



UnionPay Integrated Circuit Card Specifications **— Basic Specifications**

Part V Contactless Integrated Circuit Card Payment Specification



Version 2017

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Table of Contents

SUMMARY OF REVISIONS.....	1
1 APPLICATION SCOPE	3
2 REFERENCES	4
3 CONTACTLESS IC CARD PAYMENT	5
3.1 QUICS PROCESS.....	5
3.2 CONTACTLESS STANDARD DEBIT/CREDIT PROCESS.....	5
3.3 INTEROPERABILITY BETWEEN CONTACTLESS STANDARD DEBIT/CREDIT AND QUICS.....	5
4 QUICS OVERVIEW	7
4.1 PROCESS OVERVIEW	7
4.1.1 <i>Transaction Pre-processing.....</i>	<i>7</i>
4.1.2 <i>Card-seeking.....</i>	<i>8</i>
4.1.3 <i>Application Selection.....</i>	<i>8</i>
4.1.4 <i>Application Initialization.....</i>	<i>9</i>
4.1.5 <i>Read Application Data</i>	<i>11</i>
4.1.6 <i>Offline Data Authentication.....</i>	<i>12</i>
4.1.7 <i>Online Processing.....</i>	<i>12</i>
4.1.8 <i>Transaction Closing.....</i>	<i>12</i>
4.2 TERMINAL REQUIREMENTS	13
4.2.1 <i>General-purpose Terminal Requirements.....</i>	<i>13</i>
4.3 CARD REQUIREMENTS	14
4.3.1 <i>General-purpose Card Requirements.....</i>	<i>14</i>
4.3.2 <i>Recommended iCVN for Cards.....</i>	<i>14</i>
4.4 INTERACTION TIME.....	14
5 PRE-PROCESSING.....	16
5.1 TERMINAL DATA.....	16
5.2 TERMINAL PROCESSING	17
5.2.1 <i>Terminal Contactless Transaction Limit Check.....</i>	<i>17</i>
5.2.2 <i>Terminal Contactless Floor Limit Check.....</i>	<i>17</i>
5.2.3 <i>Status Check</i>	<i>18</i>
5.2.4 <i>Transactions with Zero Authorized Amount Check.....</i>	<i>18</i>

5.2.5	<i>Terminal CVM Required Limit Check</i>	18
6	CARD-SEEKING	20
7	APPLICATION SELECTION	23
7.1	TERMINAL DATA	23
7.2	CARD DATA	24
7.3	COMMANDS	25
7.4	TERMINAL PROCESSING	26
7.4.1	<i>Establishment of Candidate List</i>	27
7.4.2	<i>Application Determination and Selection</i>	27
7.5	CARD PROCESSING	28
7.5.1	<i>Establishment of Candidate List</i>	29
7.5.2	<i>Application Selection</i>	29
7.5.3	<i>Anti-pullout Protection</i>	31
8	APPLICATION INITIALIZATION	32
8.1	OVERVIEW	32
8.2	FLOW DIAGRAM AND COMMANDS	33
8.2.1	<i>GPO Command</i>	33
8.3	TERMINAL PROCESSING	34
8.3.1	<i>Terminal GPO Sending</i>	34
8.3.2	<i>Terminal GPO Response Processing</i>	34
8.4	UICS CARD PROCESSING	36
8.4.1	<i>Contactless Transaction Order</i>	36
8.4.2	<i>UICS Card GPO Response</i>	36
8.4.3	<i>qUICS Card Risk Management Process</i>	42
8.4.4	<i>Terminate qUICS Card GPO Processing</i>	66
9	READ APPLICATION DATA	67
9.1	TERMINAL DATA	67
9.2	CARD DATA	67
9.3	FLOW CHART AND COMMAND	67
9.4	TERMINAL PROCESSING	68
9.5	CARD PROCESSING	70

10	OFFLINE DATA AUTHENTICATION	71
10.1	TERMINAL DATA.....	71
10.2	CARD DATA.....	71
10.3	FLOW CHART.....	72
10.4	TERMINAL PROCESSING.....	72
11	TRANSACTION TERMINATION	73
11.1	CRYPTOGRAM TYPE CHECK.....	73
11.2	APPROVE OFFLINE TRANSACTION	74
11.3	TERMINAL ONLINE PROCESSING.....	74
11.4	TERMINAL OFFLINE REJECTION	75
11.5	OFFLINE DATA AUTHENTICATION FAILURE AND TERMINAL TERMINATES TRANSACTION.....	75
APPENDIX A	QUICS DATA ELEMENTS	76
A.1	NAME	76
A.2	FORMAT TAG LENGTH SOURCE	76
A.3	SHARED.....	76
A.4	REQUIREMENT.....	76
A.5	ACCESS	76
A.6	VALUE	76
APPENDIX B	QUICS AND DEBIT/CREDIT FLOW COMPARISON	88
APPENDIX C	FAST DDA (FDDA).....	91
C.1	DYNAMIC SIGNATURE GENERATION.....	91
C.2	DYNAMIC SIGNATURE VERIFICATION	93
APPENDIX D	ISSUER DEFINED DATA.....	95
D.1	ISSUER DEFINED DATA OPTION	95
D.2	ISSUER DEFINED DATA PERSONALIZATION	95
D.3	ISSUER APPLICATION DATA PERSONALIZATION EXAMPLE.....	96
D.4	ISSUER APPLICATION DATA TO GENERATE APPLICATION CRYPTOGRAM RETURN....	96
D.5	VERIFICATION CODE CALCULATION.....	97
APPENDIX E	CRYPTOGRAM VERSION	99
APPENDIX F	IMPLEMENTATION OF ONLINE-ONLY QUICS TRANSACTIONS	100
F.1	ONLINE-ONLY QUICS FLOW.....	100

F.2	QUICS PARAMETER CONFIGURATION FLOW	100
APPENDIX G IMPLEMENTATION GUIDELINES OF ODA FOR ONLINE		
	AUTHORIZATION	102
G.1	INTRODUCTION.....	102
	<i>G.1.1 Scope</i>	<i>102</i>
	<i>G.1.2 Transaction Processing Flow</i>	<i>102</i>
G.2	TECHNICAL REQUIREMENTS	103
	<i>G.2.1 Application System Architecture.....</i>	<i>103</i>
	<i>G.2.2 Transaction Stages Overview</i>	<i>104</i>
	<i>G.2.3 Technical Solution of Interaction Between Card and Terminal.....</i>	<i>104</i>
	<i>G.2.4 Technical Implementation Requirements.....</i>	<i>109</i>

Summary of Revisions

The change listed below is associated with the current version.

Change Description	Revised/ Deleted/ Added	Section
Revised the definition of the Terminal Transaction Qualifiers (tag '9F66'): enabled the Byte 1 Bit 1 to indicate if the terminal supports ODA in qUICS online authorization, and enabled the Byte 3 Bit 7 to indicate whether the terminal supports CDCVM.	Revised	7.1
To support two-tap feature of CDCVM, added the descriptions about the scenario and the GPO response processing logic of the terminal when the card returns '6986', and updated Figure 5 correspondingly.	Added	8.2, 8.3.2
Enable the Byte 1 Bit 7 of the Card Additional Function Indicator (tag 'DF61') to indicate if the ODA for online authorization is supported by the card. Added the processing logic of tag 'DF61' during the application selection, and added tag 'DF61' to template 'BF0C' in Table 6.	Added	4.1.3, 7.3, A.6
Added Table 13 to list GPO response data elements in qUICS ODA for online authorization	Added	8.4.2.2
Revised the data requirement and conditions in Table 11, Table 12 and Table 13.	Revised	8.4.2.2
Revised the description of qUICS online transaction processes.	Revised	8.4.2.4
Enabled the Byte 4 Bit 6 in Card Additional Processes (tag '9F68') to indicate if the card supports ODA in qUICS online authorization.	Revised	8.4.3
Added description to clarify cardholder verification process.	Added	8.4.3.3
Revised the requirements of reading application data and performing offline data authentication	Revised	4, 8, 9, 10
Enabled Byte 2 Bit 8 in Card Transaction Qualifiers (tag '9F6C') to indicate whether CDCVM is performed.	Added	Appendix A
Added sensitive information protection requirements in Cardholder Name (tag '5F20'), Cardholder Name Extension (tag '9F0B') and Cardholder ID number (tag '9F61') in Table A.1	Added	Appendix A
Added the definition of PAR (tag '9F24') in Table A.1 Added the definition of IIN (tag '42') in Table A.1	Added	Appendix A
Added PAR in GPO response data element list; Added PAR in recommended signature data list	Added	8.4.2.2, 8.4.2.3
Deleted the content related with dCVN	Deleted	4.3.2, Appendix B

Change Description	Revised/ Deleted/ Added	Section
Added the description for the condition of performing ODA and its requirements during application selection process. Change the condition of performing ODA to: “performing offline data authentication if AFL returned”.	Added	4.1
Revised “The offline-capable terminals shall support the fDDA” to “The terminals shall support the fDDA” in order to support ODA for both offline and online authorization.	Revised	4.2.1
Added the terminal general requirement that the terminal shall not set any bits of TVR during transaction processing	Added	4.2.1
Added requirement that the terminal shall ignore the Issuer Authorization Code-Electronic Cash (tag ‘9F74’) returned by the card in ODA for online authorization.	Added	8.4.3.14
Revised requirements of dynamic data signature generation and verification to support ODA for online authorization, signed data format ‘95’ is introduced when cards returns ARQC.	Revised	C.1
Updated the requirements of “Terminal Contactless Floor Limit”, “Terminal Contactless Transaction Limit” and “Terminal CVM Required Limit” from optional to conditional. The terminal data element presents if pre-processing supported.	Revised	A.6
Updated the requirement of “Electronic Cash Balance Upper Limit” from optional to conditional. The data element presents if low-value check or low-value and CTTA check supported.	Revised	A.6
Updated access requirement of tag 9F68, 9F6C, 9F79, 9F77, 9F6D, 9F78 that those data value can be updated via PUT DATA command	Revised	A.6
Supplemented terminal data elements to improve the readability.	Added	Appendix A
Revised the online-only parameter configuration requirement	Revised	F.2
Deleted the content related with the low-value or CTTA check, this version of specification does not support this check anymore.	Deleted	8.4.3.8
Added Appendix G to illustrate the implementation requirement for transit applications based on ODA for online authorization	Added	Appendix G

1 Application Scope

This book applies to all UPI Participants.

2 References

The following normative documents contain provisions which, through reference in this text constitute provisions of this book. For dated references, subsequent amendments to, or revisions of, any of these publications (excluding corrected contents) do not apply. However, parties to the agreements based on this book are encouraged to investigate the possibility of applying the most recent editions of the normative documents indicated below. For undated references, the latest edition of the normative document referred to applies.

UnionPay Integrated Circuit Card Specifications

ISO14443 Identifications cards – Contactless integrated circuit cards – Proximity cards

3 Contactless IC Card Payment

The payment through a contactless interface may be achieved based on the following two processes:

- Contactless Quick Payment Process (hereinafter referred to as “qUICS”);
- Contactless Standard Debit/Credit UICS Process.

This section will describe the relationship between two payment processes, and focus on qUICS.

3.1 qUICS Process

To meet the speedy requirements for transactions conducted through the contactless interface, it is necessary to adjust and optimize the standard debit/credit process. The qUICS process is an optimized standard debit/credit command and transaction process, as reflected below:

- The commands in the debit/credit process are compressed to the least possible amount so as to reduce the transaction time; the qUICS adopts command sets in the standard debit/credit process, such as SELECT, GPO, READ RECORD and GET DATA commands.
- The interaction between the card and the terminal is completed in a centralized manner. The terminal will not carry out offline data authentication, terminal risk management and terminal action analysis until the card has left the working range of the terminal. The process also allows for password operation before and after the card has left the induction range in order to minimize the stay of the card within the range.

This section will define the functional requirements of qUICS.

3.2 Contactless Standard Debit/Credit Process

As the contactless standard debit/credit process is exactly the same with the contact standard debit/credit process, except for the communication manner, it will not be described here. The transaction process described herein is only intended for the qUICS.

3.3 Interoperability between Contactless Standard Debit/Credit and qUICS

For dual-interface cards with a contact interface, the contactless debit/credit application is optional.

For contactless-only cards, the script processing (such as script processing for loading transactions) shall be conducted through the contactless standard debit/credit application. In this case, Terminal Transaction Qualifiers (tag ‘9F66’) shall indicate support for contactless standard debit/credit applications.

For contactless applications, the card shall support the qUICS process, while the contactless standard debit/credit process is optional. If the terminal and the card

both support the qUICS and contactless standard debit/credit processes, the transaction shall be completed through the contactless standard debit/credit process.

Table 1 describes the application range of contactless cards and terminals.

Table 1 The application range of contactless cards and terminals

Property of Contactless Cards Terminal Configuration	qUICS Only	qUICS and Contactless Standard Debit/Credit
qUICS Only	qUICS	qUICS
qUICS and Contactless Standard Debit/Credit	qUICS	Contactless standard debit/credit
Contactless Standard Debit/Credit Only	-	Contactless standard debit/credit

Note: For product applications of contactless standard debit/credit and qUICS in specific transactions, please refer to Auxiliary Specifications - Part IX UnionPay Integrated Circuit Card Products Applications Guide.

4 qUICS Overview

4.1 Process Overview

This section briefly describes the qUICS process in the transaction execution sequence. The conditions for the execution of a step were given in the bracket of the heading. The steps are mandatory if it is not labelled. Fig. 1 presents a qUICS process instance.

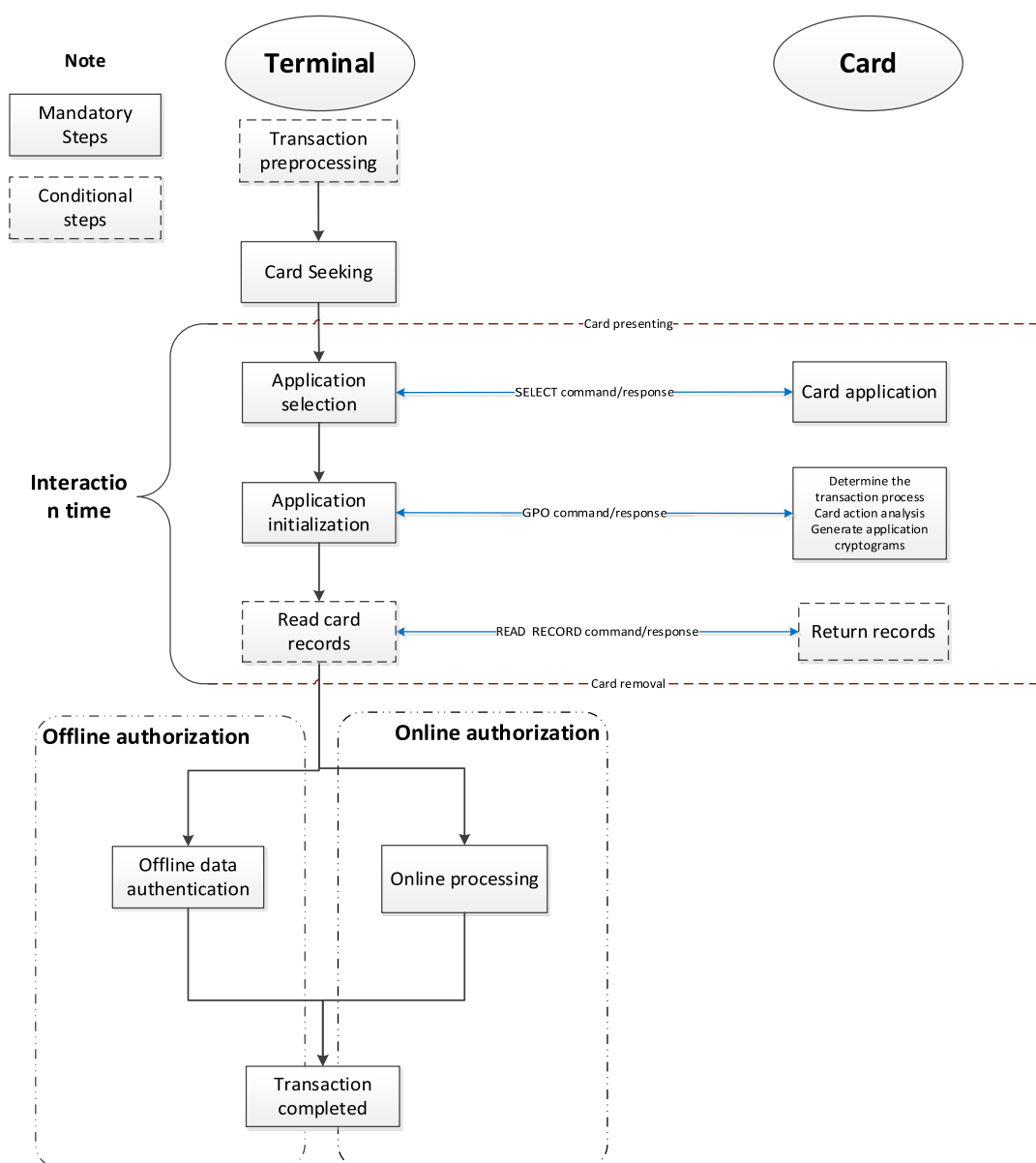


Figure 1 qUICS Process Instance

4.1.1 Transaction Pre-processing

If a terminal supports qUICS, the transaction undergoes pre-processing before the cardholder is prompted to show the card and the terminal is activated. During pre-processing, the contactless interface of the terminal is powered off, e.g. where the card is not involved at this stage. To reduce the interaction time between the

terminal and the card (e.g. where the card is present at this time), the terminal acquires the transaction amount and performs parts of the terminal risk management at this stage.

4.1.2 Card-seeking

After the terminal has completed the transaction pre-processing, the contactless interface is powered on and prompts the cardholder to present the card. At the same time, locates the card nearby. Refer to the *UICS Product Specification – Part 1* for card seeking process.

During the card seeking period, the cardholder must present the card. The interaction between the card and the terminal has already begun.

4.1.3 Application Selection

The application selection shall be conducted immediately after the terminal has finished card-seeking. The terminal will put jointly supported applications into a candidate list for a transaction to be done via the contactless interface and select final applications to complete the transaction.

The card supporting contactless applications must select the Proximity Payment System Environment (PPSE). After the PPSE is selected, the card will return all Application Identifiers (AID) of supported applications to the terminal in response to a SELECT command. The terminal will establish a candidate list and select and determine the applications together with the card. During the application determining process, the card will return the Processing Options Data Object List (PDOL) to the terminal.

The terminal supporting ODA for online authorization shall determine whether the card supports ODA for online authorization based on Card Additional Function Indicator (tag 'DF61') returned by the card, and then perform ODA for online authorization.

- If the Byte 1 Bit 7 of Card Additional Function Indicator (tag 'DF61') returned by card in a SELECT response is 1, it indicates that the card supports ODA for online authorization. The terminal can perform ODA for the transaction. The Byte 1 Bit 7 of Terminal Transaction Qualifier (tag '9F66') should be set to 0 (indicates the terminal does not support contactless UICS processing flow) by the terminal; The byte 1 bit 6 should be set to 1 (indicates the terminal supports qUICS processing flow); The Byte 1 Bit 1 should be set to 1 (indicates the terminal supports ODA for online authorization); The Byte 2 Bit 8 should be set to 1 (indicates requiring ARQC). Then, the terminal performs the qUICS flow with ODA for online authorization.
- If the Byte 1 Bit 7 of Card Additional Function Indicator (tag 'DF61') returned by card in a SELECT response is 0, which indicates that the card does not support ODA for online authorization, the terminal can process the transaction selecting the standard debit/credit transaction flow, or terminate

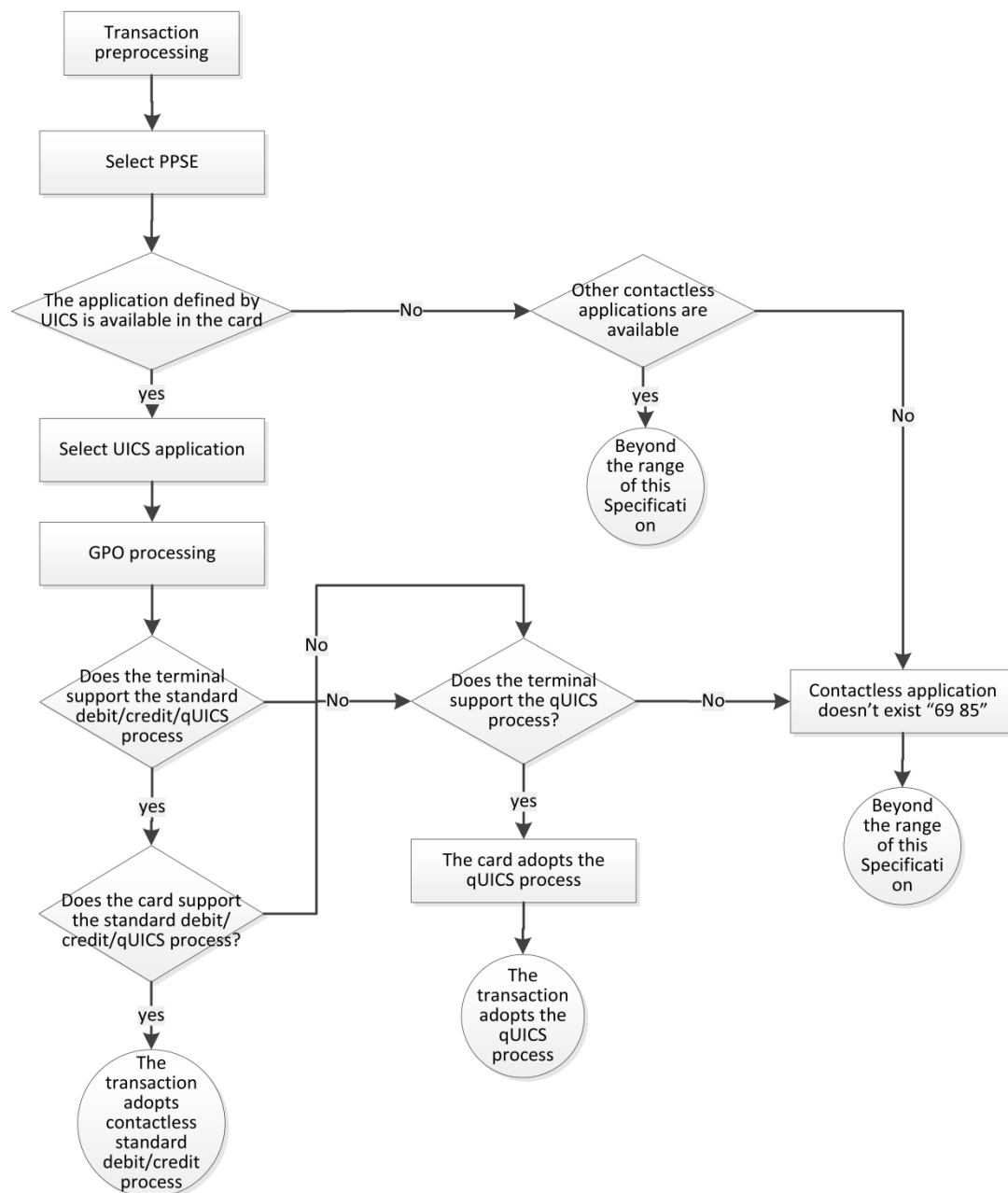
the transaction if the standard debit/credit process is not supported by the terminal.

The terminal which does not support ODA for online authorization shall ignore the returned Card Additional Function Indicator (tag 'DF61') and perform qUICS process as scheduled.

4.1.4 Application Initialization

The terminal will notify the card to start a transaction by sending a GET PROCESSING OPTIONS (GPO) command. The GPO command will include the terminal data elements required by the PDOL. The card will work out the dynamic signature data according to the fDDA requirements.

During the initialization period, the card needs to determine the payment process (qUICS or contactless standard debit/credit process) according to the terminal qualifiers and its own functions.



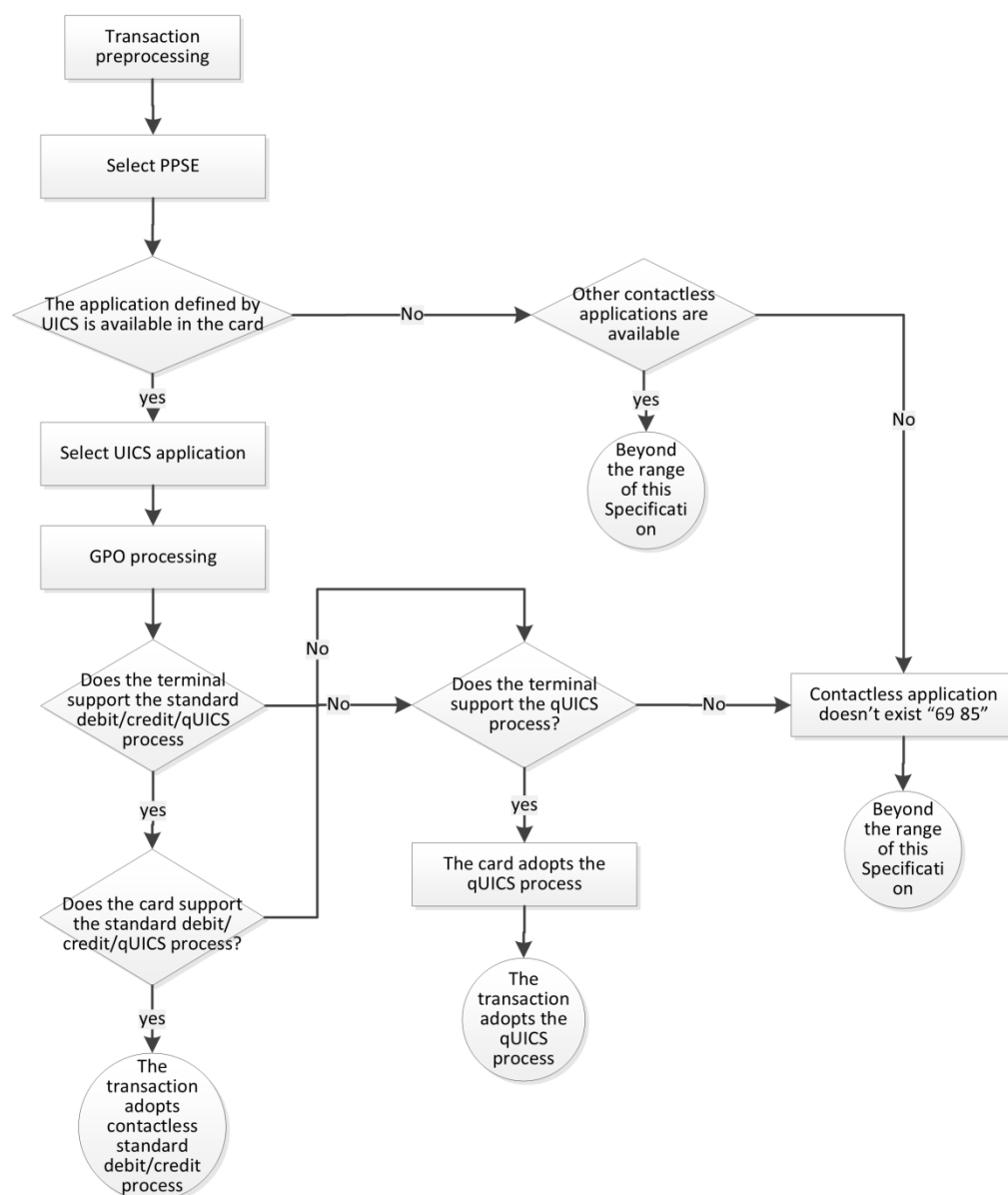


Figure 2 Processing Process Determination

In addition, the card carries out card action analysis to generate application cryptograms. If the cryptograms requires offline authorization or online authorization (with ODA), the dynamic signature data shall be generated and the Application File Locator (hereinafter referred to as “AFL”) shall be returned. The card will return all cryptograms and relevant application data to the terminal.

4.1.5 Read Application Data

If the card requires an offline authorization or online authorization (with ODA) in a return response to a GPO command, the AFL is returned simultaneously. The terminal will send a READ RECORD command to read data according to the Short File Identifier (SFI) and the record number given by the AFL.

The card will reset an anti-pullout bit while reading the last record. After the card sends out the last record, it may leave the communication area.

4.1.6 Offline Data Authentication

Both SDA and DDA are the offline data authentication method supported by this document. The Static Data Authorization (SDA) may verify whether an important application data has been illegally changed or not, but it does not provide copy protection. The Fast Dynamic Data Authorization (fDDA) not only verifies whether a card data has been illegally changed but also verifies whether the card is a valid one (not a counterfeit one with tampered data). Therefore, the terminal can apply fast shielding functions supported by the SDA when needed. If SDA is shielded, the transaction cannot be approved offline unless the card supports fDDA. If the card supports DDA, the terminal shall execute DDA;

This Specification supports two versions of fDDA. The offline data authentication in the qUICS is different from the standard debit/credit in the following ways:

- The dynamic signature is generated by a GPO command; Neither the INTERNAL AUTHENTICATE command nor the DDOL command.
- The results of SDA or fDDA are not contained in the Terminal Verification Result (TVR); they are sent to the issuer online or protected through online authorization or settlement cryptograms.

The qUICS does not require all essential data for debit/credit applications to be contained in the card or require such data to be read out if they are contained.

The qUICS may not support SDA in ODA for online authorization.

4.1.7 Online Processing

If the card requires online authorization in response to a GPO command, the online terminal performs an online processing.

The terminal will send the authentication request to the background system for the Issuer to decide whether to accept the transaction. If the Issuer decides to accept the transaction, it will deduct the authorized amount from the background debit/credit account. The Issuer's system will verify the authenticity of the card and execute a series of Issuer defined risk controls (the risk control management of the issuer's system is not described here) to generate authorization response cryptograms for accepting or rejecting the transaction to be sent to the terminal.

4.1.8 Transaction Closing

In offline authorization transactions, the terminal will inform the cardholder of the transaction result based on the application cryptograms and the offline data authentication result returned by the card.

In online authorization transactions, the terminal will inform the cardholder of the transaction result based on the authorization response cryptograms returned by the Issuer.

4.2 Terminal Requirements

4.2.1 General-purpose Terminal Requirements

- The terminals shall at least support the qUICS or support the qUICS and the contactless standard debit/credit simultaneously;
- The terminals shall support the fDDA;
- The terminals shall not set any bit of TVR during transaction processing to ensure that the TVR used to generate ARQC is consistent with TVR submitted in online authorization.
- The terminals that support multiple interfaces (contact and contactless interfaces as well as the use of magnetic stripe cards) shall have their contactless interface under the power-off state when there are no transactions and only powered after completion of the transaction pre-processing. As long as the terminal has detected a contact transaction initiated by the card or by the magnetic stripe card, the contactless interface shall be immediately powered off and remain under the power-off state during the whole transaction process;
- When the terminal supporting multiple interfaces is conducting initialization for a contact transaction or magnetic stripe card transaction, if a contactless transaction is taking place at the same time, the terminal shall terminate the contactless transaction and give up all data obtained from the card, and re-start another interface to complete the transaction.
- If the Application Authentication Cryptogram (AAC) is returned by the card to reject the transaction, the transaction shall not be conducted through any other interface.
- For the qUICS process adopted for contactless transactions, the terminal shall clearly notify the cardholder and the merchant of the following:
 - Present the card;
 - The transaction result—approved, rejected or terminated.
- For the contactless UICS standard debit/credit process adopted for contactless transactions, the terminal shall clearly notify the cardholder and the merchant of the following:
 - Present the card;
 - The prompts of “transaction is being processed” and “please don’t remove the card”;
 - The transaction result—approved, rejected or terminated.

The recommended terminal messages include:

- Present the card;

- Card reading succeeds;
- Transaction is being processed, please don't remove the card;
- Transaction approved;
- Transaction rejected;
- Present a single card [anti-collision];
- Insert or swipe the card

When giving out the prompt of presenting the card, the terminal shall display the authorized transaction amount (tag '9F02').

If the card presents an effective offline transaction amount, the terminal shall display the amount to indicate the card reading succeeds. The transaction amount may be also printed on the transaction receipt.

4.3 Card Requirements

4.3.1 General-purpose Card Requirements

- The cards shall at least support the qUICS process;
- If the contact interface is activated, the cards shall not respond to the contactless interface.
- The equivalent data of track 2 are mandatory for the qUICS;
- The offline-capable card (qUICS) shall support the DDA;

Note: To allow present chips to meet the time requirements, the cards are suggested to store and use the RSA private key in the Chinese Remainder Theorem mode.

4.3.2 Recommended iCVN for Cards

In addition to contactless risk management characteristics, qUICS is recommended to adopt iCVN (refer to *Terms and Definitions*). The Issuer shall use iCVN to encode track data in the chip cards. The iCVN is adopted to prevent the skimming of track data from the chip card, and using it to make blank plastic magnetic stripe cards. The iCVN may be used for the same purpose in contactless transactions.

4.4 Interaction Time

To complete a qUICS payment, the card is not necessarily kept within the terminal communication area throughout the entire transaction. The interaction time means the time of the card to be placed in the communication area. During this period, the terminal and the card conduct data transmission and some action analysis.

In the qUICS process, the interaction time between the terminal and the card shall not exceed 500ms. During offline authorization transactions, the interaction begins when the card first responds to the terminal during card seeking process and ends when the card sends out the last record, excluding the offline data authentication of

the terminal. During online authorization transactions, the interaction begins when the card first responds to the terminal during card seeking process and ends when the card returns response to a GPO command, excluding the online authorization.

Note: In addition to the complete qUICS process satisfying all requirements and containing all elements described in this Chapter, the online-only qUICS transactions are also defined to provide the shortest possible payment solution. Please refer to Appendix F for implementation of online-only qUICS transactions.

5 Pre-processing

To minimize the interaction time between the terminal and the card, the transaction will be pre-processed before the qUICS terminal prompts the cardholder to present the card and activate the contactless interface (power on the interface). The card is not involved at this stage.

The transaction pre-processing is not mandatory. Under some application scenarios, the pre-processing may be skipped to further enhance the transaction speed. It generally involves terminals with a fixed consumption amount, that is to say, the consumption amount has been stored in the terminals in advance, and thus does not need to be inputted again through human-machine interaction during a transaction, such as vending machines and ticket vending machines at scenic spots. For terminals that do not apply transaction pre-processing, the contactless interface may be powered on immediately.

When determining whether to apply transaction pre-processing, the Acquirer shall seriously inspect the transaction environment, analyze the transaction scenarios and consider the transaction speed and risks.

5.1 Terminal Data

Table 2 provides a list of data elements applied in the pre-processing process.

Table 2 Pre-processing--Terminal Data

Data Elements	Descriptions
Authorized Amount (‘9F02’)	A data element used in both the debit/credit and qUICS process to store the consumption amount. If it cannot be obtained, use the default value 0.
Terminal Transaction Qualifiers (‘9F66’)	A data element specific to the qUICS process, indicating the capabilities and requirements of the terminal and parameter selection of the card. It is set according to the transaction type and transaction pre-processing result, and sent to the card during the application initialization process.
Terminal Contactless Transaction Limit	An internal data element of the terminal specific to the qUICS process and not involved in the interaction with the card. This value is used to determine whether the transaction may be completed through the contactless interface. If the authorized amount is greater than or equal to the value, the transaction will be terminated; if the terminal has any other transaction interface, a prompt will be given for an interface switch.
Terminal Contactless Transaction Floor Limit	An internal data element of the terminal specific to the qUICS process and not involved in the interaction with the card. This value is used to determine whether the transaction may be completed offline. If the authorized amount is greater than the value, the terminal supporting online transactions shall request for online cryptograms; the terminal only supporting offline transactions shall terminate the transaction.
Terminal CVM Required Limit	An internal data element of the terminal specific to the qUICS process and not involved in the interaction with the card.

	This value is used to determine whether the cardholder verification shall be executed. If the authorized amount is greater than or equal to the value, the terminal will execute the cardholder verification according to its own CVM capability and the CVM requirements of the card. In this Specification, only online PIN and signature are supported.
Terminal Unpredictable Number ('9F37')	A dynamic signature used for calculating cryptograms and offline transactions. To minimize the interaction time between the terminal and the card, the terminal may generate an unpredictable number during pre-processing.
Terminal Floor Limit ('9F1B')	Data elements used in both the debit/credit process and the qUICS process. In the qUICS process, the floor limit is only applicable when Terminal Contactless Floor Limit is absent, and its application method is the same with that of the above limit.

5.2 Terminal Processing

The terminal adopts Terminal Transaction Qualifiers (tag '9F66') to indicate its contactless capability and requirements for qUICS transactions. During the pre-processing period, the terminal will re-allocate the dynamic bits in Terminal Transaction Qualifiers and set each bit according to the results of terminal risk management. The tag and length of Terminal Transaction Qualifiers (tag '9F66') are included in the PDOL, which will be sent to the terminal in response to a SELECT command. The terminal will provide a precise value according to a GPO command. Refer to Table 4 and Appendix A for details of Terminal Transaction Qualifiers (tag '9F66').

At the beginning of the transaction pre-processing, the terminal performs the following:

- Set Byte 2 Bit 7 and Byte 2 Bit 8 of Terminal Transaction Qualifiers to 0;
- The terminal obtains the value of Authorized Amount (tag '9F02').

The Acquirer may perform all or parts of the following terminal risk control checks according to requirements.

5.2.1 Terminal Contactless Transaction Limit Check

If Authorized Amount is greater than or equal to Terminal Contactless Transaction Limit, the terminal will terminate the transaction and prompt the cardholder to adopt another interface (if available);

If the terminal supports this check, set it as the first item to be checked. If the requirements for contactless transactions are met, the follow-up checks may be skipped.

5.2.2 Terminal Contactless Floor Limit Check

If Authorized Amount (tag '9F02') is greater than either Terminal Contactless Floor Limit or applicable Terminal Floor Limit (tag '9F1B') if Terminal Contactless Floor Limit is absent, then the terminal shall indicate that online application cryptogram is required. In other words, Byte 2 Bit 8 of Terminal

Transaction Qualifiers (tag '9F66') is set to 1.

5.2.3 Status Check

If the terminal configuration supports the status check and Authorized Amount is one monetary unit (this is required for status check), the terminal shall set Byte 2 Bit 8 of Terminal Transaction Qualifiers to 1 to indicate that online application cryptogram is required. The status check is a configurable option and may be executed only after power-on. This check is closed by default.

5.2.4 Transactions with Zero Authorized Amount Check

When Authorized Amount (tag '9F02') is zero, if the terminal does not supports extended applications of the qUICS, then:

- An online-capable terminal shall set Byte 2 Bit 8 of Terminal Transaction Qualifiers (tag '9F66') to 1 to indicate that online application cryptogram is required;
- An offline-only terminal terminates the transaction and prompts the cardholder to adopt another interface (if available).

5.2.5 Terminal CVM Required Limit Check

If Authorized Amount is greater than or equal to Terminal CVM Required Limit, the terminal indicates that CVM is required (Byte 2 Bit 7 of Terminal Transaction Qualifiers is set to 1) and specifies the supported CVMs: online PIN (Byte 1 Bit 3), signature (Byte 1 Bit 2) and CDCVM (Byte 3 Bit 7) are also included in Terminal Transaction Qualifiers.

- Refer to Section 8.4.3.3 for detailed descriptions of the card action corresponding to these indicators.

After a transaction pre-processing is successfully completed, if the terminal decides that the transaction will be conducted through the contactless interface, it will request for the present of a contactless card, power on the contactless terminal interface and start to seek the card. If the transaction ends here, the interface will not be powered on. Instead, it will be switched to another interface (if available).

Based on the consumption that all of the above checks are supported, the terminal processing is illustrated in Figure 3.

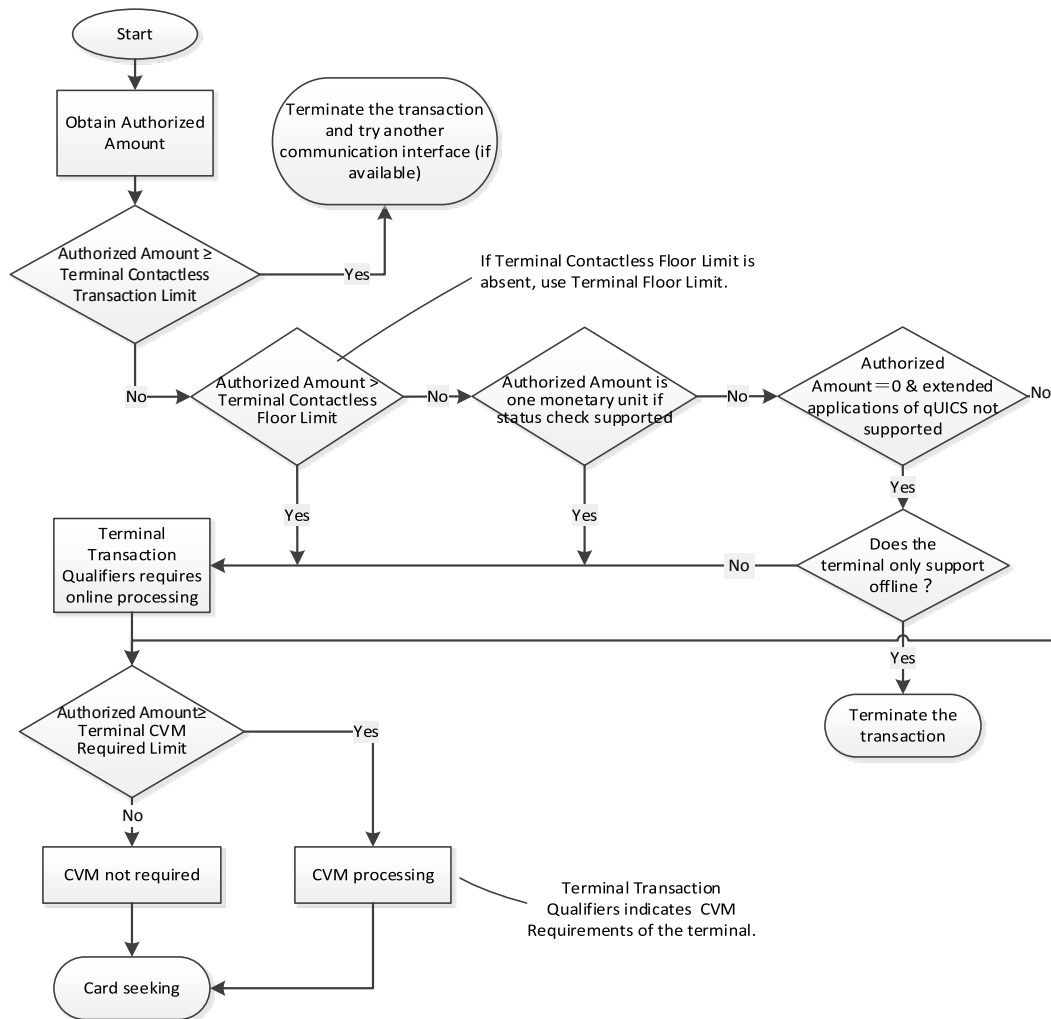


Figure 3 Terminal Pre-processing

6 Card-seeking

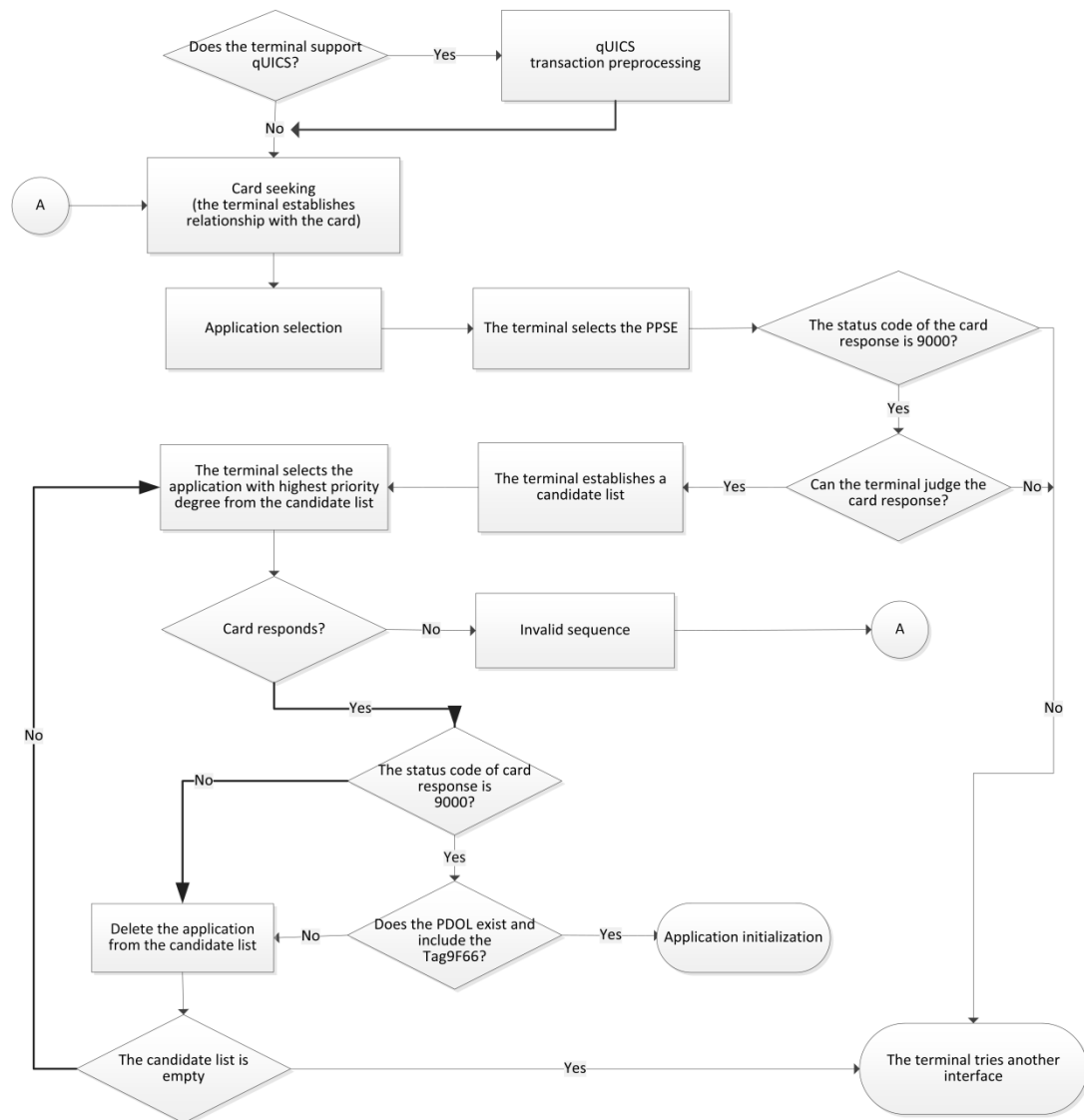
Card-seeking refers to a process during which the terminal detects whether there is any card within its communication area. The communication shall comply with the *UICS Product Specification - Part I Contactless Reader Interface Specification*.

The terminal prompts the cardholder to present the card, seeks the card and detects whether there is incompatibility between the card and the terminal. If the terminal has detected multiple contactless cards before the application selection process, it shall inform the cardholder of such situation and prompt the cardholder to present only one valid card.

When a contactless card enters the induction range of the terminal, communication initialization will be performed between the terminal and the card.

The terminal may cease the card seeking and close the contactless interface to terminate the transaction according to the merchant command or after the pre-defined timeout.

Figure 4 presents the card seeking and application selection process.



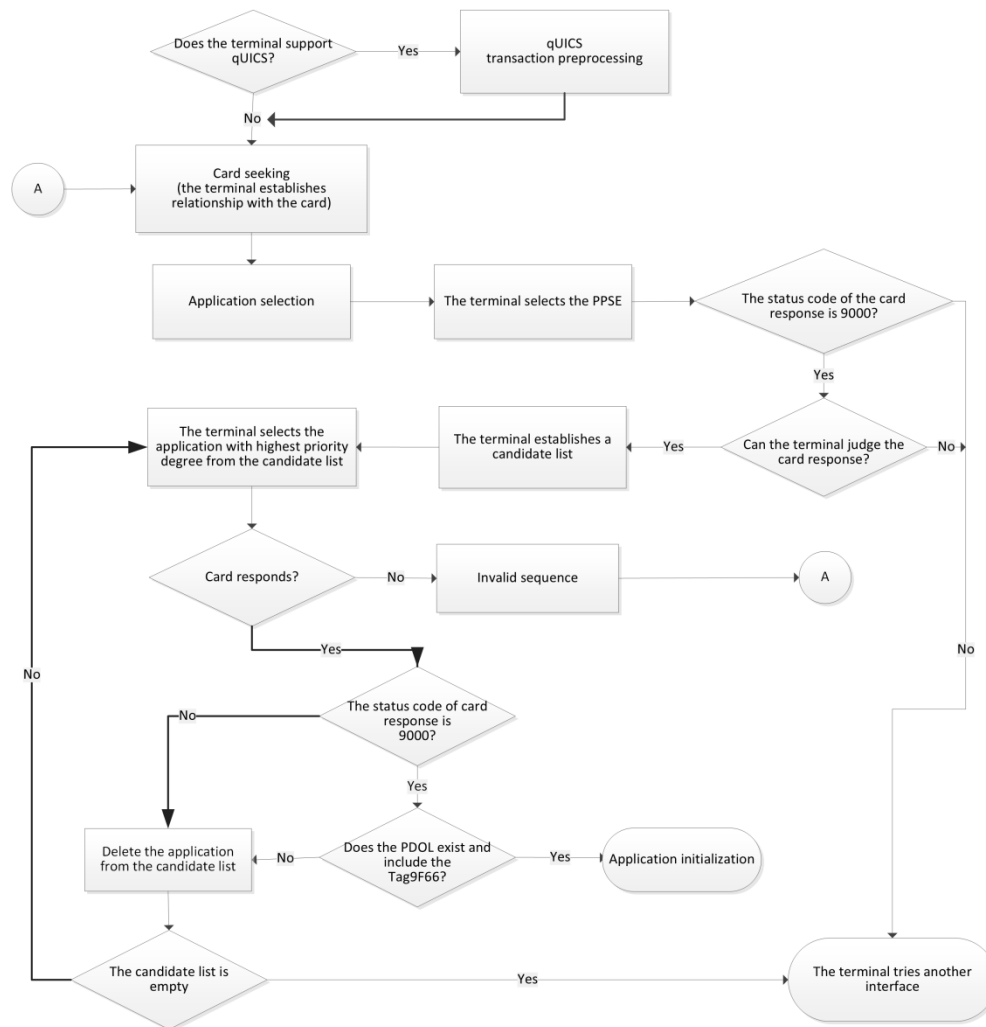


Figure 4 Card Seeking and Application Selection Process

7 Application Selection

Application selection is a process where the terminal and the card establish a mutually supported application candidate list and determine the application for the certain transaction. It may be completed in two steps:

1. The terminal establishes a application candidate list;
2. The terminal determines the application for the transaction.

7.1 Terminal Data

Table 3 Application Selection --Terminal Data

Data Elements	Descriptions
AID (tag '9F06')	A financial applications identifier supported by the terminal. After the terminal has acquired the contactless financial application list supported by the card, it will compare the list with its own AID to establish an application candidate list.
Application Selection Indicator (ASI)	An internal indicator of the terminal that is not involved in the interaction with the card. Indicating the application indicator of the terminal that completely (have the same length and value) or partially match relevant ADF names in the card (tag '4F'). Each AID has one ASI.
Terminal Transaction Qualifiers (tag '9F66')	Refer to Table 2 and Table 4 for details.

Table 4 below describes the value and its meaning of each bit in Terminal Transaction Qualifiers (tag '9F66') sent by the terminal in the GPO command. The card determines the application selection by this data set, which indicates the terminal supported functions. The setting of the terminal transaction qualifiers determines the process to be followed by a transaction (qUICS or contactless standard debit/credit process); whether the terminal supports online processing or requirements for online processing and the cardholder verification method supported by the terminal or requirements of the terminal for this method.

Byte 2 in Terminal Transaction Qualifiers (tag '9F66') shall be set by the terminal according to transaction conditions. For example, its Byte 2 Bit 8 shall be set to 1 if Authorized Amount (tag '9F02') is greater than Terminal Contactless Floor Limit , or its Byte 2 Bit 7 shall be set to 1 if Authorized Amount (tag '9F02') is greater than Terminal CVM Required Limit. Refer to 5.2 for details.

Table 4 Terminal Transaction Qualifiers (tag '9F66')

Byte	Bit	Definition
1	8	RFU

	7	1- Contactless standard debit/credit application supported 0- Contactless standard debit/credit application not supported
	6	1- qUICS supported 0- qUICS not supported
	5	1- Contact standard debit/credit application supported 0- Contact standard debit/credit application not supported
	4	1- Offline only terminal 0 - Online capable terminal
	3	1- Online PIN supported 0 - Online PIN not supported
	2	1- Signature supported 0 – Signature not supported
	1	1 - ODA for online authorization supported 0 - ODA for online authorization not supported
2	8	1- Online cryptograms required 0 - Online cryptograms not required
	7	1- CVM required 0 – CVM not required
	6-1	RFU
3	8	RFU
	7	1- CDCVM supported 0- CDCVM not supported
	6-1	RFU
4	8	1- fDDA version '01' supported
		0- only fDDA version '00' supported
	7-1	RFU

7.2 Card Data

Table 5 Application Selection — Card Data

Data Elements	Descriptions
Proximity Payment System Environment (PPSE)	The terminal may use a SELECT command to select the Directory Definition File (DDF) under the file name "2PAY.SYS.DDF01" so as to select the PPSE.
The File Control Information (FCI) of	The FCI of the PPSE will be returned by the card when the terminal selects the PPSE.

the PPSE	It includes the Issuer Discretionary Data (IDD, tag 'BF0C'). The IDD will list the Directory Entries (tag '61') of all contactless financial applications supported by the card. The terminal will establish a candidate list based on applications supported by both the card and by the terminal itself. If the card supports more than one contactless financial application, the Application Priority Indicator (tag '87') must be personalized. Refer to Table 6 for FCI of the PPSE.
PDOL	The PDOL will be returned in response to a SELECT command when the card determines the final application.

7.3 Commands

During the application selection, the terminal uses a SELECT command to select the PPSE and determines the application. The use of SELECT command should meet the requirements of B.13 of *UICS Basic Specifications – Part II*, except in the following circumstances:

- When selecting the payment environment, a SELECT command selects PPSE, instead of PSE. The FCI of the PPSE returned after the card has successfully performed the SELECT command is defined in Table 6.

Table 6 defines the PPSE formats of a single application and multiple applications. It is suggested to limit the number of personalized applications.

Table 6 FCI of the PPSE

Tag	Value		Length	Condition
'6F'	FCI Template		var.	M
	'84'	"2PAY.SYS.DDF01"	0E	M
	'A5'	FCI Proprietary Template	var.	M
	'BF0C'	FCI Issuer Discretionary Data	var.	M
	'61'	Directory Entry	var.	M
		'4F' DF Name (AID)	07-08	M
		'50' Application Label	04-10	O
		'87' Application Priority Indicator	01	C*
	'61'	Directory Entry	var.	C*
		'4F' DF Name (AID)	07-08	C
		'50' Application Label	04-10	C
		'87' Application Priority Indicator	01	C

Tag	Value		Length	Condition
	'61'	Directory Entry	var.	C*
	'4F'	DF Name (AID)	07-08	C
	'50'	Application Label	04-10	C
	'87'	Application Priority Indicator	01	C
	'DF61'	Card Additional Function Indicator	var.	C ¹

Note:

* Conditions—if more than 1 application is personalized in the card, each personalized application shall have the Application Priority Indicator. The 8-5 bit of the Application Priority Indicator should be set at '0000'.

1 Conditions—if the tag 'DF61' is personalized on the card, it shall be returned in the SELECT response by the card. However, the terminal must not terminate the transaction if it cannot obtain the tag.

7.4 Terminal Processing

The contactless applications can be selected in the following terminal environments. This section describes the selection actions from the list of multiple contactless applications. To meet the time requirements, it is recommended to list one application in the FCI. If more than one application is required, the number of applications shall be reduced to the least possible amount.

- The path through which the terminal accesses the card application shall adopt one debit/credit AID. The path cannot be accessed directly;
- The terminal will establish an application list contained in the FCI and supported by the terminal. The terminal shall determine the bits 4-1 of the application priority indicator (indicating the sequence of application selection) and select the application with the highest priority level to process a transaction.
- If there is only one application contained in the FCI and this application is supported by the terminal, the terminal shall select the application, without considering the setting of application priority indicator which might appear.
- If the status code of the card in Response to a SELECT command is not '9000' or the terminal cannot acquire the AID from the FCI due to the error format of the PPSE, the terminal shall close the contactless interface and try another interface to complete a transaction.
- If the FCI is not personalized as described in this section (for example, the application priority does not exist), and the terminal has at least one

application within the jointly supported application list, the terminal may select either of the applications.

If the card fails to respond to a SELECT command sent by the terminal, the terminal shall launch an invalid command sequence and return to the card seeking process according to requirements of section 6.

7.4.1 Establishment of Candidate List

All contactless terminals shall adopt the PPSE directory selection method based on steps below:

Step 1: The terminal uses a SELECT command to select the PPSE contactless payment system environment under the file name of “2PAY.SYS.DDF01”.

The status code returned by the card SW1 SW2=‘9000’, the terminal shall continue to the next step, otherwise, the terminal shall terminate the transaction.

Step 2: the terminal analyzes Issuer Discretionary Data in the FCI (tag ‘BF0C’). With respect to all directory entries in Issuer Discretionary Data (tag ‘61’), the terminal will acquire the DF name (tag ‘4F’) in turn.

If it matches the AID supported by the terminal, it will be added into the candidate list. According to the Application Selection Indicator (ASI), the terminal may choose to support the complete matching or partial matching.

If it does not match the AID supported by the terminal, repeat this step and check the next directory entry. If there is no next entry, the establishment of the candidate list is completed.

The terminal shall support DF file names (AID) with the maximum length of 16 bytes.

If there is no directory entry (tag ‘61’) in the Issuer Discretionary Data (‘BF0C’), the terminal shall terminate the transaction.

7.4.2 Application Determination and Selection

After a candidate list has been established, the terminal shall determine and select transaction applications.

The terminal shall determine a transaction application according to the candidate list. If any of the following cases occur during the process:

Case 1. If the candidate list is empty, i.e. the terminal and the card have no jointly supported applications, the terminal will terminate the transaction.

Case 2. If the candidate list contains one application, the terminal shall select this application.

Case 3. If the candidate list contains multiple applications:

—— The terminal shall select the application with the highest priority level according to the application priority indicator (tag ‘87’).

- If the 1-4 bit of the application priority indicator is '0000' or there is no application priority indicator (tag '87'), the application shall be considered to have the lowest priority level.
- If multiple applications share the same degree of priority, the terminal shall carry out selection based on the sequence of their directory entries in the FCI.

After the transaction applications have been determined, the terminal will send a SELECT command to the card. The command will include the ADF name of the selected applications. After receiving the card Response to a SELECT command, the terminal will determine the status code. If the status code is '9000', the terminal will perform application initialization.

If the status code of the card response is not '9000', the terminal will delete the ADF name (AID) from the candidate list and repeat the above selection process.

7.5 Card Processing

The contactless applications can be selected under following card requirements. This section describes actions of multiple contactless applications. To minimize the application selection time, it is suggested to limit the number of personalized applications in the FCI.

- The card must support the PPSE, and use the file name of "2PAY.SYS.DDF01" as the file name of the PPSE;
- The card shall support the contactless standard debit/credit process and the qUICS process in the single card application involving the debit/credit application AID.
- If more than 1 application is personalized in the FCI, the application priority indicator shall be personalized in all applications. In this section, the 8-5bit of the application priority indicator shall be set at '0000';
- The AID of the contactless financial application in the card shall be returned in the FCI in Response to a SELECT PPSE command. The complete format of the FCI is described in Table 6;
- The personalization of all contactless payment applications shall be stored in the PDOL. The PDOL shall at least include data elements with the tag '9F66' (Terminal Transaction Qualifiers) as described in Table 4;
- If the card supports a single contactless application, the length of the AID shall be at least of 7 bytes;
- If the card supports multiple contactless applications with the same UnionPay AID, it shall support 8-byte-long AIDs at least in order to extend the bytes for differentiation, examples as below:

A0 00 00 03 33 01 01 01

A0 00 00 03 33 01 01 02

7.5.1 Establishment of Candidate List

The card receives a SELECT command from the terminal to request for the selection of PPSE (File name “2PAY.SYS.DDF01”);

- If the card is blocked, or not support SELECT command, the card will respond SW1 SW2=‘6A81’;
- If there is no PPSE in the card, the card will indicate the inexistence of the file (SW1 SW2=‘6A82’) in response to a SELECT command;
- If the PPSE is blocked, the card will respond ‘6283’;
- If the PPSE is found, the card will respond ‘9000’ and return the FCI of the PPSE (refer to Table 6).

7.5.2 Application Selection

After the transaction applications have been determined the terminal will send a SELECT command to the card. The FCI responded by the card shall include the PDOL. Refer to *UICS Basic Specifications – Part 2* Table B.27 for response to an APPLICATION SELECTION command.

The qUICS does not support the CDOL, DDOL or the default DDOL. All data required for card processing are requested in the PDOL.

The card will request for the terminal transaction qualifiers to allow the contactless application to determine the card path (contactless standard debit/credit application or the qUICS). The unpredictable number, authorized amount and ATC of the card are together used for cryptogram calculation (version 01 or version 17). The unpredictable number and ATC are also used for calculating dynamic signature in offline transactions.

One card application includes a single PDOL. The PDOL includes relevant path-related tags (the qUICS or contactless standard debit/credit), and may also include tags undefined here as minimum requirements. The Issuer shall balance the advantage brought by additional data requested by the PDOL and the effect of their transmission and processing upon transaction performance.

The basic contents of the PDOL in the qUICS are dependent upon the supported cryptogram version (01 or 17) and whether the card supports offline qUICS transactions.

7.5.2.1 Online-only qUICS with the Cryptogram Version 17

Table 7 presents the basic PDOL contents in the online-only qUICS with the cryptogram version 17.

Table 7 The basic PDOL Contents in the Online-only (without ODA) qUICS

Tags in the PDOL	Name of Data Elements
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Tags in the PDOL	Name of Data Elements
'9F66'	Terminal Transaction Qualifiers
'9F02'	Authorized Amount
'9F37'	Unpredictable Number

If an online card performs the card supplementary processing, the transaction currency code (tag '5F2A') shall be also included in the PDOL.

The unpredictable number, authorized amount and ATC of the card are used together for cryptogram calculation.

7.5.2.2 Offline-capable qUICS with the Cryptogram Version 17

Table 8 presents the basic PDOL contents in the online and offline-capable qUICS with the cryptogram version 17.

Table 8 The Basic PDOL Contents in the Online and Offline-capable qUICS

Tags in the PDOL	Name of Data Elements
'9F66'	Terminal Transaction Qualifiers
'9F02'	Authorized Amount
'9F37'	Unpredictable Number
'5F2A'	Transaction Currency Code

The unpredictable number, authorized amount and ATC of the card are used together for cryptogram calculation. The unpredictable number and ATC are also used for calculating dynamic signatures in offline transactions.

7.5.2.3 qUICS with the Cryptogram Version 01

The same data tags are used in online and offline qUICS with the cryptogram version 01. The basic contents of the PDOL are summarized in Table 9.

Table 9 The Basic PDOL Contents of the qUICS with the Cryptogram Version 01

Tags in the PDOL	Name of Data Elements
'9F66'	Terminal Transaction Qualifiers
'9F02'	Authorized Amount
'9F03'	Other Amounts
'9F1A'	Terminal Country Code
'95'	Terminal Verification Result (TVR) <i>Note: Not applicable for qUICS, where all bits of TVR are set to 0.</i>

'5F2A'	Transaction Currency Code
'9A'	Transaction Date
'9C'	Transaction Type
'9F37'	Unpredictable Number

Except the terminal transaction qualifiers, the rest of data above are used for cryptogram calculation.

7.5.3 Anti-pullout Protection

If a card supports offline transactions, the transaction anti-pullout protection will be provided following the counter updating and before the completion of the present transaction. To provide this kind of protection, the card shall contain a transaction anti-pullout bit (card internal indicator). Transaction anti-pullout bit (card internal indicator) will be set after the card counter is updated, and this mark bit will be reset after the last transaction command is processed. If the mark bit has been set at the beginning of a transaction (the application has been selected), the card will know the last transaction has not been completed. The actions to be adopted by the card include but not limited to the following:

- Restore Electronic cash balance (tag '9F79') to the value of the latest successful transaction at the time of completion;
- Do not record the transaction log of the previous transaction;
- Set transaction anti-pullout bit¹.

The transaction anti-pullout protection management may be also determined by the application vender.

¹The set and reset of anti-pullout bit are determined by suppliers.

8 Application Initialization

8.1 Overview

During the application initialization period, the terminal will notify the card to start a transaction by sending a GPO command. The command includes all data required in the PDOL returned by the card when selecting applications. Refer to Figure 5 for the application initialization processing. Refer to 8.2.1 for details of the GPO command.

8.2 Flow Diagram and Commands

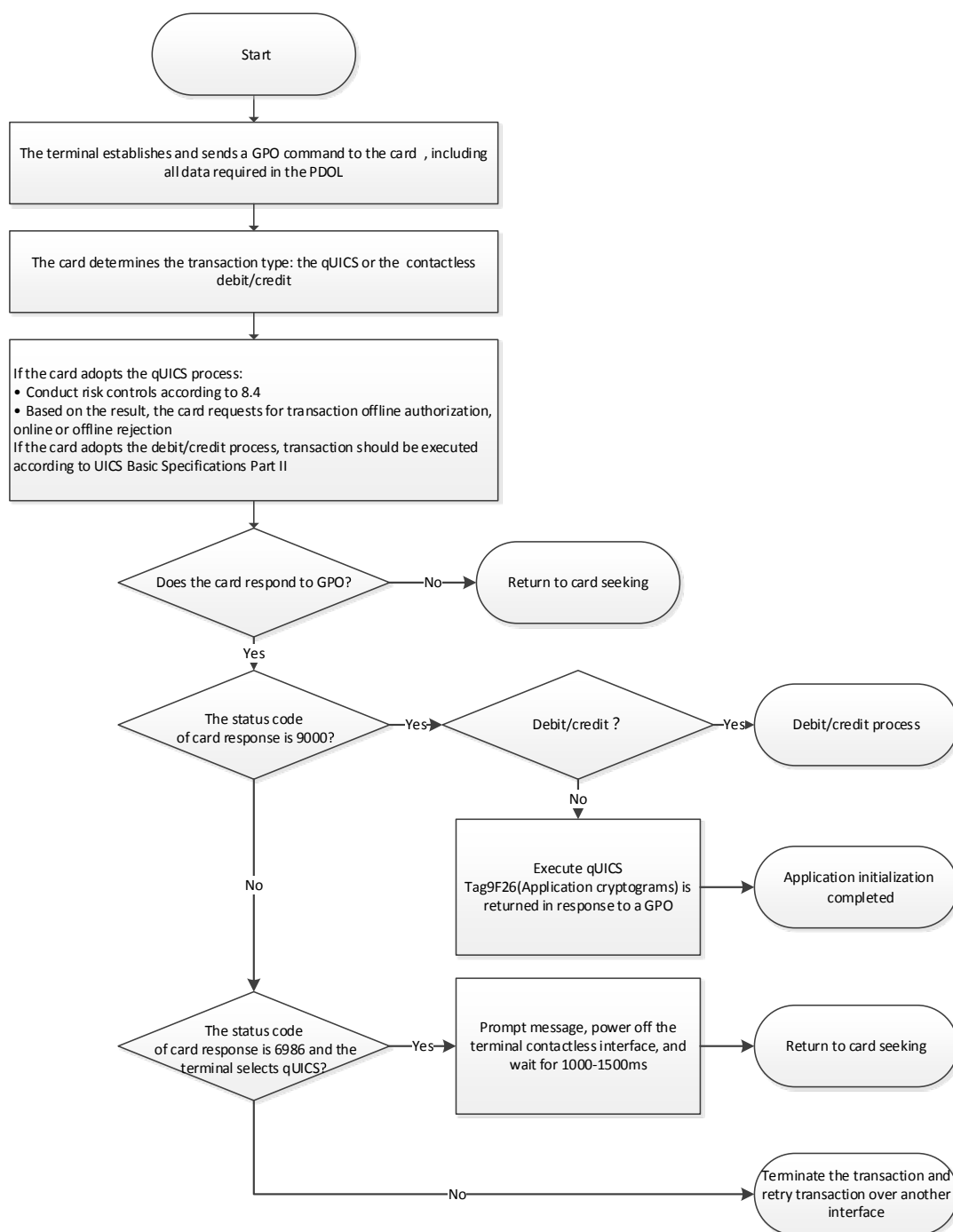


Figure 5 Application Initialization Flow

8.2.1 GPO Command

The format of GET PROCESSING OPTIONS command is shown in Table 10

Table 10 GPO Command

Code	Value
CLA	'80'

INS	'A8'
P1	'00'; other values are preserved
P2	'00'; other values are preserved
Lc	Var.
Data Field	Data concerning the Processing Options Data Object List (PDOL)
Le	'00'

8.3 Terminal Processing

8.3.1 Terminal GPO Sending

The terminal checks the response of the card to a SELECT command during application selection.

- If the PDOL does not exist in the card response or the tag '9F66' of the terminal transaction qualifiers does not exist in the PDOL, the terminal will delete the application from the candidate list and return to the application selection process.
- Otherwise, all terminals shall provide the terminal transaction qualifiers ('9F66') and other data elements in response to a GPO command according to requirements of the card in the PDOL; all terminals shall support the GPO response in Format 2, 8.4, *UICS Basic Specifications – Part 2* Appendix B;

Terminals meeting the requirements of this Specification shall support the fDDA verification of '00' version and '01' version, and show the capabilities in the terminal transaction qualifiers (set byte 4 bit 8 to 1).

8.3.2 Terminal GPO Response Processing

Case 1. No response to GPO command

If a GPO command sent by the card response terminal fails, the terminal shall return to the card seeking process, refer to section 6.

Case 2. Error code in response to GPO command

If the status code responded by the card to a GPO command is not '9000', the terminal shall terminate the contactless transaction and try another interface to complete the transaction.

If the card returns Status Word '6986' in a GPO response and the qUICS process is selected by the terminal, the terminal shall:

- Prompt the user to follow the instruction on payment device, and power off the contactless interface immediately.
- After waiting for a period of 1000ms to 1500ms, the terminal should power on the contactless interface, start the card-seeking process (Refer to Chapter 6)

and wait for the device presented in induction field. Instructions to the cardholder should be displayed during card seeking.

- The terminal seeks the mobile device according to the card-seeking requirements in Chapter 6, and restarts qUICS process after successful seeking. The terminal should use the pre-processing result from the first tap instead of performing pre-processing again.

If the card returns neither ‘9000’ nor ‘6986’ in GPO response, or if the card returns ‘6986’ in GPO response but the qUICS process is not selected on the terminal, the terminal shall terminate the contactless interface and try another interface for the transaction.

Note 1: A standard ICC will not return ‘6986’ in GPO response according to the Specifications. Only mobile payment devices that support CDCVM may return ‘6986’ to indicate that CDCVM is required but not yet performed. The transaction will only be processed and completed unless consumer has performed CDCVM on their mobile device.

Note 2: CDCVM is supported only in the qUICS process. Contactless standard debit credit process does not support CDCVM and two-tap feature.

Case 3. Successful Response to GPO command

If the card responds ‘9000’, the terminal shall determine whether to carry out the transaction through the contactless standard debit/credit process or the qUICS process based on the application interchange profile (refer to Annex A) and data elements provided by the card in response to a GPO command.

- The UICS only support the qUICS process set by the terminal as default process and does not need to inquire the AIP;
- If a terminal support the qUICS and the contactless standard debit/credit process at the same time and the byte 2 bit 8 of the AIP is set to 0, the terminal shall perform the following:
 - If the application cryptogram (tag ‘9F26’) does not appear in the GPO response, the standard debit/credit process shall be followed;
 - If the application cryptogram (tag ‘9F26’) appears in the GPO response, the qUICS process shall be followed;

If a transaction adopts the qUICS process, the following requirements must be met:

- If the mandatory data elements of the qUICS are not returned in the GPO response (refer to Table 8 and Table 9). The UICS terminal shall terminate the transaction.
- If the card returns the data elements which are mandatory for the standard debit/credit process while not required by the qUICS in the GPO response, the UICS terminal shall not reject the transaction;
- If the card does not provide the card transaction qualifiers (tag ‘9F6C’), the

signature-supporting terminal shall be considered to support signatures. If the terminal requires a CVM, the signing blank shall be printed on the receipt.

- If a terminal supports more than 1 CVM, it shall decide which CVM to be selected based on the byte 1 bit 7 and 8 of the card transaction qualifiers (tag '9F6C'). For the CVM processing details, please refer to section 8.4.3.3;
- A terminal supporting the qUICS shall read records according to the debit/credit rules of *UICS Basic Specifications – Part 1* and process the records or data elements of unknown tag codes in the PDOL.

8.4 UICS Card Processing

8.4.1 Contactless Transaction Order

Card and terminal both support the requirements of the most appropriate method, determining the order of processing chosen. qUICS supports quick online and offline transactions, and does not need card insertion into slot or placement on the card tray. Recommend making the determination in the following order:

- UICS contactless standard debit/credit procedure: If card supports contactless standard debit/credit procedure and "Terminal transaction qualifiers" byte 1 bit 7 is 1 (supports contactless debit/credit application), then card application uses contactless debit/credit procedure, terminal shall follow contactless debit/credit procedure to process the transaction;
- If card supports qUICS and "Terminal transaction qualifiers" byte 1 bit 6 is 1 (supports qUICS), then card shall use qUICS procedure, terminal shall follow qUICS to process the transaction;
- If there are no matching contactless transaction route, then card shall in response return an indicator (status word = '6985') to terminate the transaction and try using another interface;

8.4.2 UICS Card GPO Response

Card GPO response includes application interchange profile, to indicate card support for risk management features. This also includes cryptograms and related data elements, magnetic track 2 equivalent data and all mandatory data listed in table 11 for online transactions. Response data follow the format 2 encoding defined in *UICS Basic Specifications – Part 2* Appendix B.8.4, the TLV encoding that include tag and length, response data specific content follow table 11, 12 and 13 of this part.

- When ATC reach max value ('65535'), application must be permanently locked, encryption calculation is prohibited, recommend GPO command return status word '6985'.

8.4.2.1 Application File Locator (AFL)

AFL includes the currently selected file and relevant record list, there are no separators in between. Terminal shall only read AFL designated record.

— The AFL shall be returned for offline transactions and online transactions (with ODA), and shall not be returned for offline rejected transactions or online transactions (without ODA).

8.4.2.2 Application Interchange Profile

Application interchange profile indicate the application functionality supported by the card, following the encoding of Table A.1 in Appendix A. Terminal shall only try to execute IC card supported functionality. For detailed requirements see description in Table 11.

For all qUICS online transactions (without ODA), the mandatory data elements listed in Table 11 shall be included in the GPO response.

Table 11 GPO response data for qUICS online transactions (without ODA) or transaction rejection

Tag	Mandatory (M) Optional (O) Conditional (C)	Data Element Name
'82'	M	AIP
'9F36'	M	ATC
'57'	M	Magnetic track 2 equivalent value
'9F10'	M	Issuer application data (tag '9F10') Issuer self-defined data can also include available offline spending amount. Appendix D describes in detail how to include available offline spending amount
'9F26'	M	Application cryptogram
'9F27'	M	Cryptogram Information Data
'9F63'	C It's required for mobile payment devices	Product identity information The card shall return the tag '9F63' in either GPO response or READ RECORD response. The card must not return the tag in both responses. If so, the terminal may terminate the transaction.
'5F34'	C If it appears in card	Application PAN sequence number
'9F24'	C Presents if Token is used to replace PAN in the card	Payment Account Reference
'9F6C'	C If it appears in card	Card Transaction Qualifiers
'9F5D'	C If available offline spending amount display is permitted	Available Offline Spending Amount Unless tag '9F5D' has been personalized to value of 1, the card shall not return this data element in the GPO response. And Issuer shall also apply additional processing to personalize the card (byte 1 bit 1) to 1, to indicate that the amount will be calculated and included in all contactless transactions. Personalizing tag '9F5D' to 1 also indicate that GET DATA command can be used to read this data element. Content engages in calculation according to Issuer instruction and card additional processing section definitions (Low-value, low-value and CTTA)
'5F20' or	O	Cardholder name Note: Name: Cardholder names are required data elements in

'9F0B'		debit/credit applications. (If cardholder names with less than or equal to 26 bytes use '5F20', if cardholder name has greater than 26 bytes then use '9F0B')
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PAN and expiration date is retrieved from track 2 equivalent data by the terminal. For online transactions, available offline spending amount based on card configuration can be returned from two places: can be included in tag '9F10' described in Appendix D (online send to Issuer) Issuer self-defined data, or as tag element returned in GPO response (displayed or printed out by the terminal).

Data elements listed in Table 12 are either mandatory or conditional in offline transaction GPO responses.

Table 12 GPO Response Data for Offline Transaction Approvals

Tag	Mandatory (M) or Conditional (C)	Data Element Name
'82'	M	AIP
'94'	M	AFL
'9F36'	M	ATC
'9F26'	M	Application cryptogram
'9F27'	M	Cryptogram Information Data
'9F10'	M	Issuer application data (tag '9F10') Issuer self-defined data can also include available offline spending amount. Appendix D describes in detail how to include available offline spending amount
'9F63'	C It's required for mobile payment devices	Product identification information The card shall return the tag '9F63' in either GPO response or READ RECORD response. The card must not return the tag in both responses. If so, the terminal may terminate the transaction.
'57'	C If magnetic track 2 equivalent data is not static data portion pending signature	Magnetic track 2 equivalent value Unless treated as a portion of static data pending signature, track 2 equivalent data is mandatory
'5F34'	C If it appears in card	Application PAN serial number
'9F4B'	C If supports fDDA and IC card private key length is less than or equal to 1024 bits	Signed dynamic application data The card shall return the tag '9F4B' in either GPO response or READ RECORD response. The card must not return the tag in both responses.
'9F24'	C Presents if Token is used to replace PAN in the card	Payment Account Reference
'9F6C'	C If it appears in card	Card transaction qualifiers

'9F5D'	C If return of available offline spending amount is permitted and IC card private key length is less than or equal to 1024 bits	Available offline spending amount Unless tag '9F5D' has been personalized to value of 1, the card shall not return this data element in the GPO response. And issuer shall also apply additional processing to personalize the card (byte 1 bit 1) to 1, to indicate that the amount will be calculated and included in all contactless transactions. Personalizing tag '9F5D' to 1, also indicate that GET DATA command can be used to read this data element. Content engages in calculation according to Issuer instruction and card additional processing section definitions (Low-value, low-value and CTTA)
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- Any additional data, including cardholder name (tag '5F20'), is read by application using READ RECORD command;
- For offline transaction, if acting as a part of offline data authentication, application expiration date (tag '5F24'), application PAN (tag '5A') and SDA tag list (tag '9F4A') shall be included into one single record.

If a card does not return a cryptogram information data, the terminal will use the following policy to form a cryptogram information data :

- Set cryptogram information data to '00';
- Copy issuer application data '9F10' the 5th Byte, the 6-5 bit value to cryptogram information data 8-7 bit.

Data elements listed in Table 13 are mandatory, conditional or optional for GPO response in qUICS online authorizations (with ODA).

Table 13 GPO Response Data in qUICS Online Authorizations (with ODA)

Tag	Mandatory (M) Optional (O) Conditional (C)	Data Element Name
'82'	M	AIP
'94'	M	AFL
'9F36'	M	ATC
'57'	C	Magnetic track 2 equivalent data Track 2 equivalent data is required when it is part of a static data used for signature.
'9F10'	M	Issuer application data Tag '9F10' issuer self-defined data can also include available offline spending amount. Appendix D describes in detail how to include available offline spending amount.
'9F26'	M	Application cryptogram
'9F27'	M	Cryptogram Information Data
'9F63'	C It's required for mobile payment devices	Product identification information The card shall return the tag '9F63' in either GPO response or READ RECORD response. The card must not return the tag in both responses. If so, the terminal may terminate the transaction.
'5F34'	C If present in the card	Application PAN sequence number
'9F4B'	C If fDDA is supported and	Signed dynamic application data The card shall return the tag '9F4B' in either GPO response or READ RECORD response. The card must not return the tag

	length of IC card private key is less than 1024 bit	in both responses.
'9F24'	C Presents if Token is used to replace PAN in the card	Payment Account Reference
'9F5D'	C If available offline spending amount display is permitted	Available offline spending amount Unless tag '9F5D' has been personalized to value of 1, the card shall not return this data element in the GPO response. And Issuer shall also apply additional processing to personalize the card (bit 1 of byte 1) to 1, to indicate that the amount will be calculated and included in all contactless transactions. Personalizing tag '9F5D' to 1 also indicate that GET DATA command can be used to read this data element. Content engages in calculation according to Issuer instruction and card additional processing section definitions (Low-value, low-value and CTTA)
'9F6C'	C If it appears in card	Card transaction qualifiers
'5F20' or '9F0B'	O	Cardholder name

8.4.2.3 qUICS Recommended Signature Data

If a card supports qUICS then it is recommended to use the following static data elements for signatures:

- Application PAN;
- PAR (if presents)
- Application expiration date;
- AIP (if fDDA is supported);
- Application version no;
- SDA tag list (if fDDA is supported).

During personalization, the card shall set the application version number (tag '9F08') as the version for this Specification. It is strongly recommended that the application version number (tag '9F08') is included in static data for signatures, to indicate the card's actual application version. If the same card can support both qUICS and debit/credit applications (contact), then can also add the additional data elements recommended in *Part II Debit Credit Application - Member Implementation Guide for Issuer*.

8.4.2.4 qUICS card requirements

Besides the card requirements of all contactless applications, qUICS must also meet the following requirements:

- Once GPO command is received, card should immediately set issuer application data (tag '9F10') CVR part to '03000000'. CRV is byte 4-7 of the issuer application data;

CVR Byte 2, Bits 4, 3, 2, 1 are unused, remain set to 0;

CVR Byte 3, Bits 8, 4, 3, 2, 1 are unused, remain set to 0;

CVR Byte 4 is unused, all bits remain set to 0;

- After receiving a GPO command, the card will initialize cryptogram information data and set it to '00';
- Card shall add ATC value prior to calculating cryptogram and dynamic signatures;
- If the card's Available Offline Spending Amount (tag '9F5D') is personalized to 1, then card shall permit reading of this data element. Card actions shall be clarified and stored in the card internal indicator during personalization;
- For online transactions (that does not support ODA), a card should return ARQC in GPO response and the data elements to generate cryptogram listed in Table 11.
- For offline transactions, a card should return the data elements listed in Table 12 in GPO response.
- For online transactions (that supports ODA), if a card supports ODA for online authorization (Byte 4 Bit 6 is set to 1 in CAP) and a terminal supports ODA for online authorization (Byte 1 Bit 1 is set to 1 in TTQ), then the card should return the data elements listed in Table 13 in GPO response.
- If the IC card private key length is less than or equal to 1024 bits, then dynamic signature shall be generated and returned in the response to GPO;
- If IC card private key length is greater than 1024 bits, the dynamic signature shall be generated during GPO and returned in the READ RECORD command;
- If one card data element is returned in GPO response, then card shall not return this data element while reading record. So the same data element shall be only returned once during the same transaction;
- For qUICS offline approved transactions, the 70 template of the last record, which AFL indicates the terminal must read, shall not exceed 32 bytes. If the fDDA executed by the card is version '01', then recommend to place only electronic cash issuer authorization code (tag '9F74') and Card Authentication Related Data (tag '9F69') in this record, with Card Authentication Related Data only present when card executes '01' version of fDDA.
- Cards that meet this Specification version shall also support '00' and '01' version fDDA. If terminal supports '01' version fDDA (Byte 4 Bit 8 in the

Terminal Transaction Qualifiers is 1), then card must execute ‘01’ version fDDA.

Note: If IC card private key length is greater than 1024 bits, GPO response has insufficient space to return dynamic signature.

—— To ensure GPO response can be successfully sent to terminal, in the scenario that IC card private key length is equal to 1024, the AFL shall not contain more than 4 entries.

Note: If IC card private key length is shorter, there may be sufficient space to include more entries. If IC card private key length is longer, signature is transferred via record, and also will have sufficient space to transfer larger AFLs.

Note: ODA for online authorization only applies in certain scenarios (E.g. Transit) but not in ordinary contactless online transactions.

Note: If the length of the IC card private key is smaller than 1024 bit, then the dynamic signature should be generated and returned in GPO response. If the length of the IC card private key is larger than 1024 bit, then the card should generate dynamic signature in GPO command and return it in READ RECORD command.

8.4.3 qUICS Card Risk Management Process

If the Byte 1 Bit 6 in TTQ is 1, it indicates that the terminal supports qUICS procedure over the contactless interface.

Card behavior is controlled with the values of a set of indicators personalized in the Card Additional Processes (tag ‘9F68’). The contents of this data element are as shown in Table 14, and is used during the processing of the described card as below.

Table 14 Card Additional Processes (tag ‘9F68’)

Byte	Position	Notes
1	8	1 - Low-value check supported 0 - Low-value check not supported
	7	1 - Low-value and CTTA check supported 0 - Low-value and CTTA check not supported
	6	RFU
	5	1 - New card check supported 0 - New card check not supported
	4	1 - PIN try limit exceeded check supported 0 - PIN try limit exceeded check not supported
	3	1 - Offline transactions in non-matching currency allowed 0 - Offline transactions in non-matching currency not allowed
	2	1 - Card Prefers Contact debit/credit online 0 - No Card Contact debit/credit online preference
	1	1 - Return available offline spending amount 0 - Does not return available offline spending amount
2	8	0 - RFU
	7	1 - Transactions in non-matching currency not allowed 0 - Transactions in non-matching currency allowed
	6	1 - Decline transaction if new card and terminal only 0 - Does not decline traction if new card and terminal only
	5	1 - qUICS offline approved transaction, card records transaction log 0 - qUICS offline approved transaction, card does not record transaction log

	4-1	0 – RFU
3	8	1 – Online PIN supported for matching currency transactions 0 – Online PIN not supported for matching currency transactions
	7	1 – Online PIN supported for non-matching currency transactions 0 – Online PIN not supported for non-matching currency transactions
	6	1 – CVM required for non-matching currency transactions 0 – CVM not required for non-matching currency transactions
	5	1 – Signature supported 0 – Signature not supported
	4-1	0 – Reserved
4	8 - 7	Reserved for enhanced Low-value
	6	1- ODA for online authorization supported 0- ODA for online authorization not supported
	5-1	0 – Reserved

This part uses pseudo code to describe card-processing process, without indicating specific implementation details. This part details functionality and time requirements should be met, but implementation details are up to the application developer.

8.4.3.1 Set Application Currency Matching or Non-Matching

Currency is compared once with the result saved. Proceed with the following processing:

- Set currency match bit (card internal indicator) to 0;
- If the application currency code (tag ‘9F51’) is equal to transaction currency code (tag ‘5F2A’), then set currency matching bit to 1;
- If currency matching bit =0 and non-matching currency transaction is not allowed (card additional processing byte 2 bit 7 = 1), then decline transaction. For the follow-up processing steps see 8.4.3.17--decline transaction.

8.4.3.2 Terminal only supports Offline

If terminal only supports offline transactions, skip online request check.

- If terminal only supports offline (terminal transaction qualifiers, byte 1 bit 4 =1), the card must try offline processing:
 - Set offline only terminal bit (internal card indicator) to 1;
 - If the last online ATC register is 0, and if it is a new card and terminal only supports offline (card additional processing byte 2 bit 6=1), then reject transaction, for the follow-on processing steps see 8.4.3.17 -- reject transaction.

Offline PIN Try Limit Exceeded

- If terminal only supports PIN retry exceeded check (card additional processing byte 1 bit 4), then when offline PIN retry counter (tag ‘9F17’) exists and equal to 0 (no remaining PIN retries), the card shall reject transaction;

- Set CVR byte 3 bit 7 to 1 (PIN Try Limit exceeded);
- For the follow-up processing steps see 8.4.3.17--reject transaction.

Request CVM

If terminal only supports offline, and one of the following scenarios apply:

- In terminal transaction qualifiers the terminal requests CVM (byte 2 bit 7 = 1);
- Currency match bit =1, and authorized amount is greater than card CVM limit;
- Currency match bit =0, and for currency mismatch transaction card requests CVM bit = 1 (card additional processing byte 3 bit 6).

Then:

Card and Terminal Both Support Signatures

If terminal transaction qualifiers support signatures (byte 1 bit 2 = 1), and card additional processing also supports signatures (byte 3 bit 5 = 1), then set require signature in card transaction qualifiers and try offline processing:

- Set card transaction qualifiers byte 1 bit 7 to 1;
- For the follow-on processing steps see 8.4.3.5 -- offline currency check.

Either Card or Terminal Does Not Support Signature

If terminal transaction qualifiers does not support signature (byte 1 bit 2 = 0), or card additional processing does not support signature (byte 3 bit 5 = 0), then terminate contactless transaction.

- For the follow-up processing steps see 8.4.3.16--terminate contactless transaction.

For processing flow of offline-only terminal support see Figure 6.

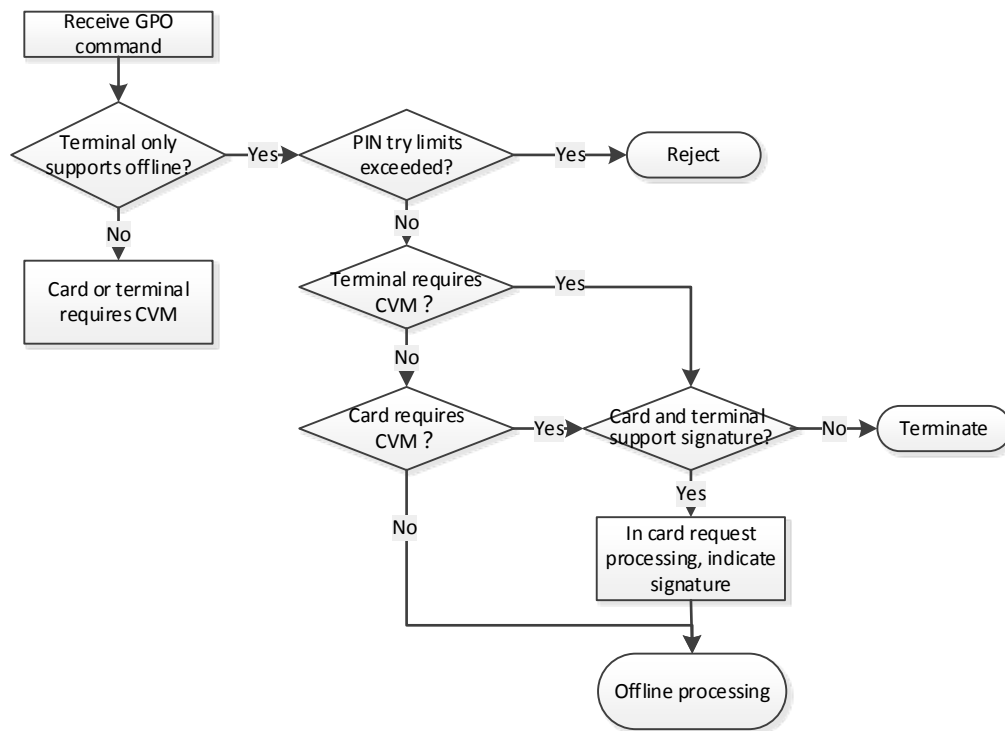


Figure 6 Terminal only support offline

8.4.3.3 Terminal or Card Requests CVM

Terminal can request CVM (always or just for transactions that exceed terminal CVM required limit), card can also request CVM. A terminal determines whether CVM needs to be performed. The CVMs supported by UICS include online PIN, CDCVM, and signature.

If CVM requested and online PIN is supported by both terminal and card, then transaction will be processed online.

If CVM requested but is not supported by both card and terminal, then transaction will be terminated.

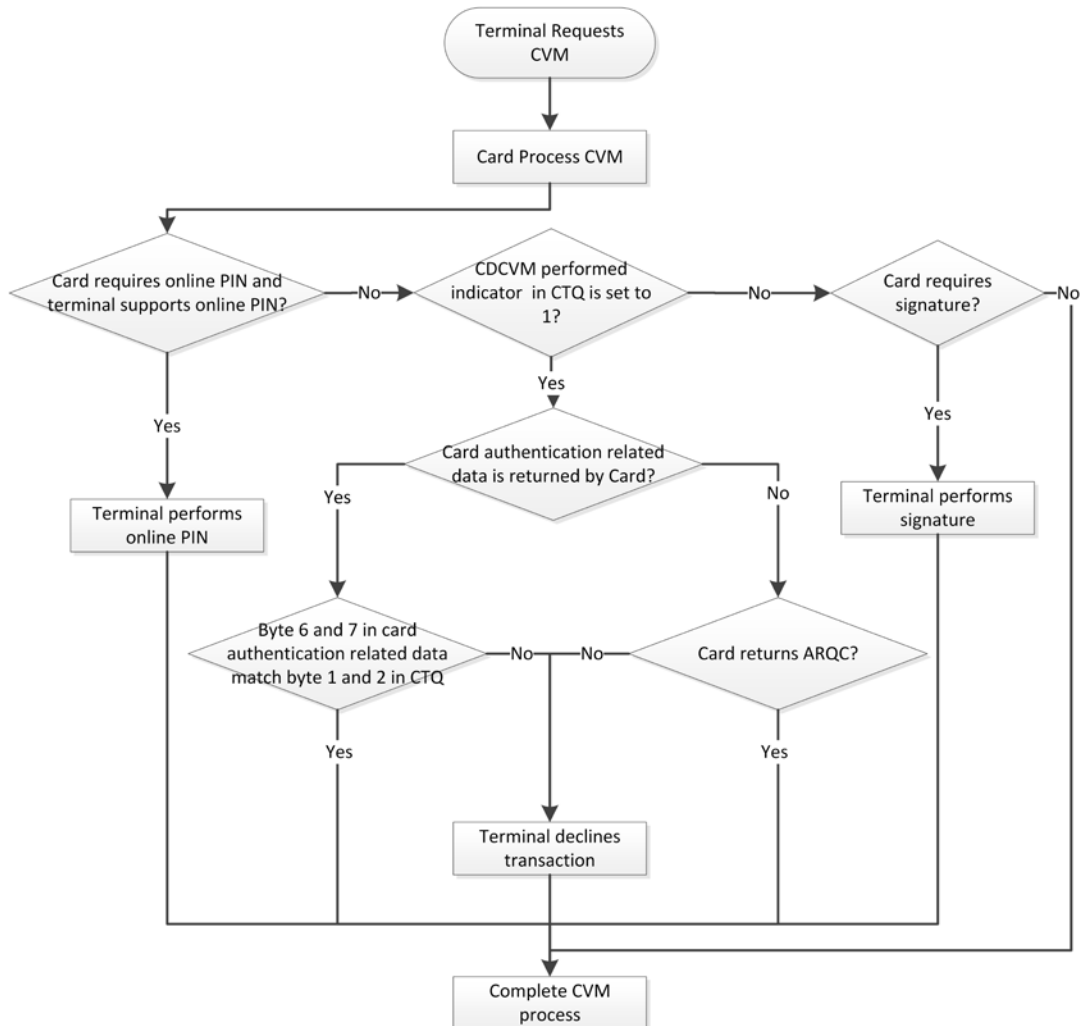
8.4.3.3.1 Terminal CVM Process

A terminal with contactless capability supporting qUICS process must support cardholder verification step. Acquirers and merchants should be able to enable and disable the CVM supported by the terminal. This document requires that the CDCVM feature must be supported and enabled; that byte 3 bit 7 in Terminal Transaction Qualifiers (tag '9F66') must be set to 1.

If a card does not return CTQ (tag '9F6C') when a terminal requires performing cardholder verification, then:

- If terminal supports signature, then performs signature as verification method.
- If terminal only supports online PIN and CDCVM, then performs online PIN.
- If terminal only supports CDCVM, then terminal declines the transaction.

If a card returns a CTQ (tag '9F6C') when a terminal requires performing cardholder verification, then the terminal should check the CTQ and perform a cardholder verification. The detailed process is illustrated in the following chart:



- If online PIN is required by CTQ (byte1 bit 8 is set to 1 in CTQ) and terminal supports online PIN, then:
 - Terminal performs online PIN and should not check the remaining bit in CTQ;
- Else, if online PIN is not required by CTQ, or online PIN is not supported by terminal
 - If CDCVM performed indicator is set to 1 in CTQ (byte 2 bit 8 is set to 1 in CTQ), then:
 - ✧ If the card also returns a card authentication related data (tag '9F69') then;
 - If the card authentication related data (tag '9F69') byte 6-7 matches the card transaction qualifiers (tag '9F6C') byte 1-2, then the CVM processing is complete and the terminal should not check the remaining bit in CTQ;
 - Else (if do not match), the terminal declines the transaction and completes the CVM processing;
 - ✧ If the card does not return a card authentication related data

(tag '9F69')² then:

- If the cryptogram type returned by card is ARQC, then terminal completes CVM processing and should not check remaining bit in CTQ;
- If the cryptogram type returned by card is not ARQC, then terminal declines transaction and completes CVM processing;

○ If online PIN is not required and CDCVM is not performed, then:

- ✧ If signature is required by CTQ (byte 1 bit 7 is set to 1 in CTQ), and terminal supports signature, then signature is performed as cardholder verification method.

- Else (none of the above), no common supported CVM is found by terminal, then CVM processing is not performed;

If terminal requests CVM and CVM is not performed, then terminal should decline the transaction.

8.4.3.3.2 Card CVM Process

CVM Not Required

—— If the CVM request bit in Terminal Transaction Qualifiers is 0, and either of the following condition is true:

- Currency match bit =1, and authorized amount is less than or equal to card CVM limit;
- Currency match bit =0, and for currency mismatch transaction card requests CVM bit = 0 (card additional processing byte 3 bit 6).

Then card continues the risk management process, and proceeds as Section 8.4.3.4.

Terminate CVM Not Required

Request CVM

—— If terminal transaction qualifiers' CVM request bit (byte 2 bit 7) is 1, or if terminal transaction qualifiers' CVM request bit (byte 2 bit 7) is 0, and any one of the following scenarios is true:

- Currency match bit =1, and authorized amount is greater than card CVM limit;
- Currency match bit =0, and for currency mismatch transaction card requests CVM bit = 1 (card additional processing byte 3 bit 6).

Continue following the steps below.

Card and Terminal Both Support Online PIN

—— If terminal transaction qualifiers (byte 1 bit 3) supports online PIN, and any one of the scenarios below is true:

² For qUICS online transactions, card authentication related data is not ordinarily included in GPO response. Therefore, terminals cannot ensure that card transaction qualifiers are not tampered by other application program. However, issuers can verify whether CDCVM is successfully performed by checking card verification result (CVR) byte 2 bit 3.

- Currency match bit = 1, and for matched currency, online PIN support bit = 1 (card additional processing byte 3 bit 8);
 - Currency match bit = 0, and for mismatched currency, online PIN support bit = 1 (card additional processing byte 3 bit 7);
- Card and terminal both support online PIN;
- Card sets card transaction qualifiers (tag ‘9F6C’) byte 1 bit 8 to 1, and requests online processing;
 - If available offline spending amount bit returned =1, then card must use card additional processing designated offline Low-value option (low-value, low-value and CTTA) to calculate available offline spending amount. If no option is specified, then card must set available offline spending amount to zero.

Follow 8.4.3.15 steps to continue processing -- complete online processing.

Card and Terminal Both Support CDCVM

- If CDCVM supported by reader (TTQ byte 3 bit 7 is 1):
- Indicate CDCVM Performed (set CTQ byte 2 bit 8 to 1).
 - Card sets card transaction qualifiers (tag ‘9F6C’) byte 1 bit 8 to 1, and requests online processing.

Card and Terminal Both Support Signature

- If terminal transaction qualifiers (byte 1 bit 2) supports signature and card additional processing also support signature with bit = 1 (byte 3 bit 5):
- Card sets card transaction qualifiers signature request bit to 1, and then continues card risk management processing;
 - Follow 8.4.3.4 to continue processing -- check online processing request.

No Common CVM

- Card shall terminate transaction:

For the follow-up processing steps see 8.4.3.16--terminate contactless transaction.

Terminate CVM Request

For card CVM processing flow see Figure 7.

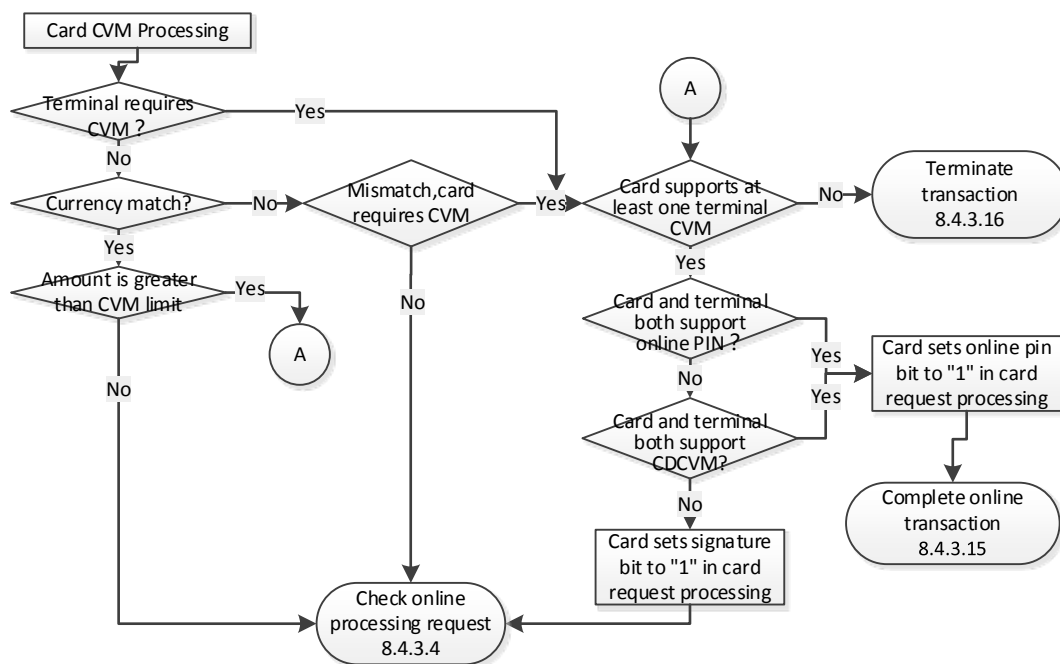


Figure 7 Card CVM process

8.4.3.4 Check Online Processing Request

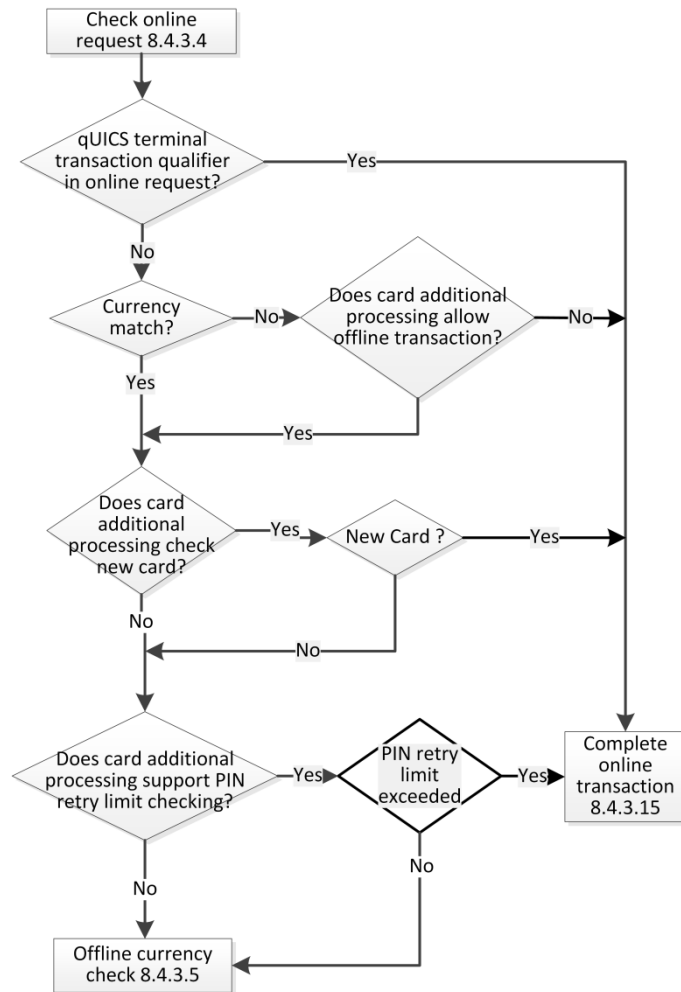
Card and terminal can request online processing based on transaction conditions. If prior check did not specify requirement for online processing, or terminate contactless transaction, execute this check to determine whether there exists other conditions leading to online processing.

- If terminal requests online processing (terminal transaction qualifiers byte 2 bit 8 =1), then card may also request processing;
 - If available offline spending amount bit returned =1, then card must use card additional processing designated offline Low-value option (Low-value, Low-value and CTTA) to calculate available offline spending amount. If no option is specified, then card must set available offline spending amount to zero.
 - Follow 8.4.3.15 for follow-on processing -- complete online transaction.
- If mismatched currency offline transaction is not permitted (card additional processing byte 1 bit 3 = 0) and currency match = 0, then card shall request online processing;
 - If available offline spending amount bit returned =1, then card must use card additional processing designated offline Low-value option (Low-value, Low-value and CTTA) to calculate available offline

spending amount. If no option is specified, then card must set available offline spending amount to zero.

- Follow 8.4.3.15 for follow-on processing -- complete online transaction.
- If new card check is supported (card additional processing byte 1 bit 5 = 1) and last online ATC register is 0 (new card did not complete online processing), then card requests online processing;
- If available offline spending amount bit returned =1, then card must use card additional processing designated offline Low-value option (Low-value, Low-value and CTTA
-) to calculate available offline spending amount. If no option is specified, then card must set available offline spending amount to zero.
- Set CVR byte 3 bit 5 to 1(new card);
 - Follow Figure 14 for follow-on processing -- complete online transaction.
- If PIN retry exceeded check is supported (card additional processing byte 1 bit 4 = 1), and offline PIN retry counter (tag '9F17') exists and equal to 0 (no remaining PIN retries), then card shall request online processing;
- If available offline spending amount bit returned =1, then card must use card additional processing designated offline Low-value option (Low-value, Low-value and CTTA) to calculate available offline spending amount. If no option is specified, then card must set available offline spending amount to zero;
 - Set CVR byte 3 bit 7 to 1 (PIN Try Limit exceeded);
 - Follow 8.4.3.15 for follow-on processing -- complete online transaction.

See Figure 8 for check online processing request.



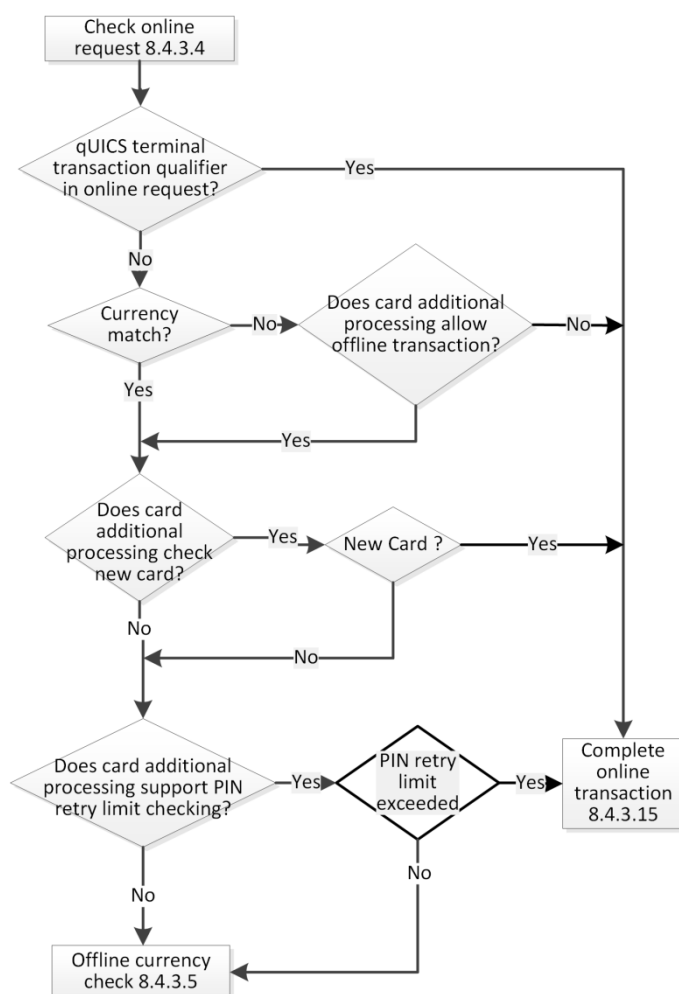


Figure 8 Check online processing request

8.4.3.5 Offline Currency Check

When transaction currency matches application currency, execute offline purchase check. If currency doesn't match, skip these checks and execute currency mismatch processing.

Check whether processing is match or mismatch currency, and whether corresponding checks of the offline purchase check category is supported.

Low-value check, Low-value and CTTA check check are the three qUICS kinds of methods to check offline purchase. *UICS Basic Specifications – Part 6* defined electronic cash relevant data (electronic cash balance, electronic cash balance upper limit, and electronic cash single transaction limit) is used to execute Low-value processing, but the functional requirements for processing these relevant tags are detailed in the three methods below.

—— If currency match bit = 0:

For follow-on steps see 8.4.3.13 - offline currency mismatch.

Otherwise currency match flag is 1, then card and terminal currencies match. Check which offline purchase check option is supported. If no option is supported, for

terminals that only support offline reject transaction, for terminals that support online proceed with online processing.

8.4.3.6 Low-value Check for Matching Currency Transactions

This check is realized through the Low-value upper limit on the card (electronic cash balance upper limit). Contactless transaction offline purchase total available capital is electronic cash balance. Executing this option can provide available offline spending amount equal to electronic cash balance.

- If Low-value balance check is supported (card additional processing byte 1 bit 8 = 1), then electronic cash balance is the total available purchase amount, then execute low-value check.

For follow-up steps see 8.4.3.10-- low-value check.

8.4.3.7 Low-value and CTTA Check for Matching Currency Transactions

This part checks CTTA whether exceeds cumulative offline transaction amount upper limit (CTTAUL) or if CTTAUL does not exist whether cumulative offline transaction amount limit (CTTAL) is exceeded. If CTTA available capital -- CTTAUL (if doesn't exist use CTTAL) minus CTTA is available, then similarly check whether transaction amount exceeds electronic cash single transaction limit. Only when low-value and CTTA checks pass could offline transaction proceed.

For this option, can use offline purchase amount equal to available CTTA capital.

- If low-value and CTTA check is supported (card additional processing byte 1 bit 7 = 1), then capital shall be usable in both low-value and CTTA. CTTA available capital is the total offline purchase amount available, and execute low-value and CTTA check.

For follow-up steps see 8.4.3.11-- low-value and CTTA check.

8.4.3.8 This Section has been Deleted

This version of specification does not support low-value or CTTA check anymore.

8.4.3.9 No Offline Option Supported

No offline purchase check is directed.

- If terminal only supports offline (terminal transaction qualifiers, byte 1 bit 4 =1), the card must reject transaction;

For follow-up steps see 8.4.3.17-- reject transaction;

- If terminal supports online (terminal transaction qualifiers, byte 1 bit 4 =0), the card must request online processing.

If available offline spending amount bit returned =1, then card must use card additional processing designated offline Low-value option (Low-value, low-value and CTTA) to calculate available offline spending amount. If no option is specified, then card must set available offline spending amount to zero.

For follow-up steps see 8.4.3.15 -- complete online transaction.

For offline currency check processing flow see Figure 9.

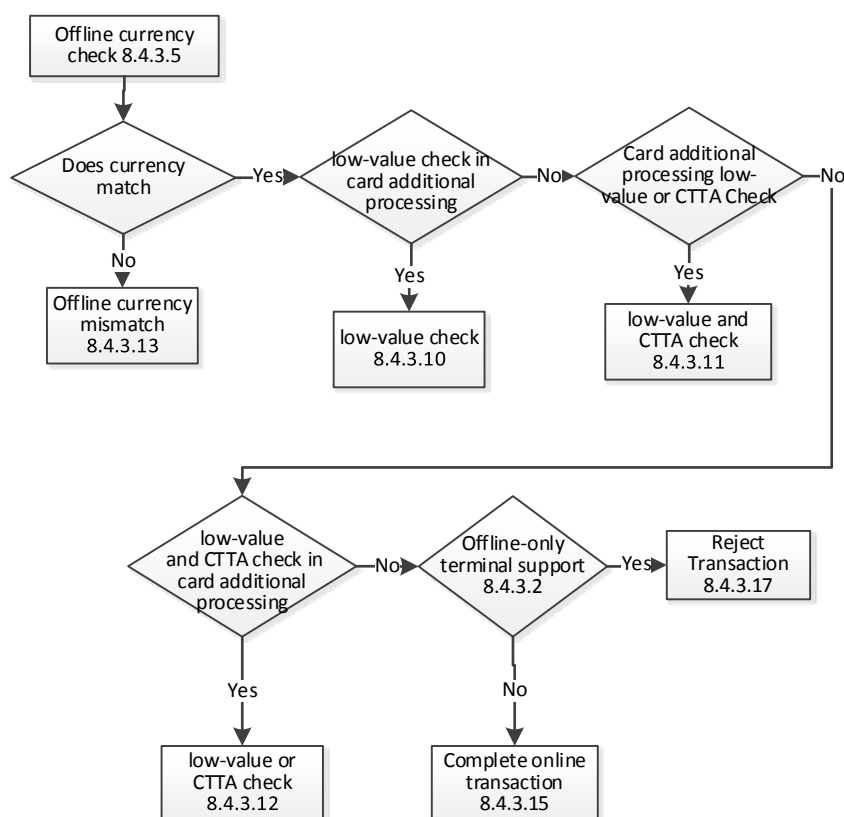


Figure 9 Offline Currency Check

8.4.3.10 Low-value Check

Check whether transaction can be processed offline.

If authorized amount (tag ‘9F02’) is less than or equal to electronic cash single transaction limit, and electronic cash balance has sufficient offline purchase available amount for the transaction, then the transaction undergoes offline processing.

Otherwise (if authorized amount is greater than the electronic cash single transaction limit or there is insufficient offline purchase available amount):

- If terminal has online processing capability, then card requests online processing;
- If terminal has no online processing capability, then card requests rejection.

Terminal Can Get Online

When terminal has online capability (terminal transaction qualifiers, byte 1 bit 4 = 0), the following requirements are appropriate.

- If authorized amount (tag ‘9F02’) is greater than the electronic cash single transaction limit (if exists, tag ‘9F78’), then card should prepare returning available offline spending amount (if supported), and request online

processing;

- If returning available offline spending amount is permitted (card additional processing byte 1 bit 1 = 1) and currency match bit = 1, then card shall set available offline spending amount (tag '9F5D') to be the electronic cash balance value, and return the available offline spending amount in the GPO response;
 - Set CVR byte 3 bit 6 to 1 (Velocity check counters exceeded);
 - For follow-up steps see 8.4.3.15-- complete online transaction.
- If authorized amount (tag '9F02') is greater than the electronic cash balance minus electronic cash reset threshold (if exists, tag '9F6D'), then card should prepare returning available offline spending amount (if obtaining is supported), and request online processing;
- If returning available offline spending amount is permitted (card additional processing byte 1 bit 1 = 1) and currency match bit = 1, then card shall set available offline spending amount (tag '9F5D') to be the electronic cash balance value, and return the available offline spending amount in the GPO response;
 - Set CVR byte 3 bit 6 to 1 (Velocity check counters exceeded);
 - For follow-up steps see 8.4.3.15-- complete online transaction.

Terminal only support Offline

When terminal has only offline support (terminal transaction qualifiers, byte 1 bit 4 = 1), the following requirements are appropriate.

- If authorized amount is greater than the electronic cash balance or greater than the electronic cash single transaction balance (if exists) then card should prepare to return available offline spending amount (if obtaining is supported), and also reject transaction.
- If returning available offline spending amount is permitted (card additional processing byte 1 bit 1 = 1), then card shall set available offline spending amount (tag '9F5D') to be the electronic cash balance value, and return the available offline spending amount in the GPO response;
 - Set CVR byte 3 bit 6 to 1 (Velocity check counters exceeded);
 - For follow-up steps see 8.4.3.17-- reject transaction;

Transaction Offline Completion Approved

- If the steps above all do not match, then card shall complete the processing flow below:
- Save current electronic cash balance value;

- Set transaction anti-pullout bit (card internal indicator) to indicate that the indicator is being updated. This indicator is only reset to 0 prior to the last read record command response. See 7.5.3 -- anti-pullout protection;
- Calculate new electronic cash balance, equal to electronic cash balance minus authorized amount (tag '9F02');
- If returning available offline spending amount is permitted (card additional processing byte 1 bit 1 = 1), then card shall set available offline spending amount (tag '9F5D') to be the electronic cash balance value, and return the available offline spending amount in the GPO response;
- Request offline approval in cryptogram information data and CVR;
- For follow-up process see 8.4.3.14-- complete offline transaction.

For low-value check process see Figure 10.

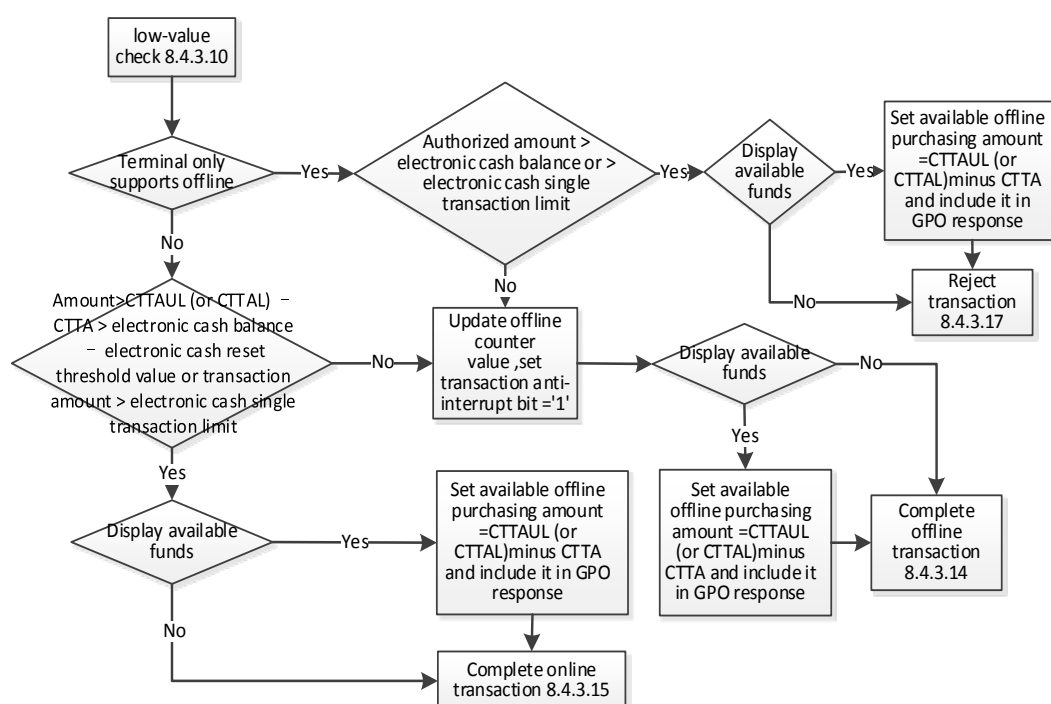


Figure 10 Low-value check

8.4.3.11 Low-value and CTTA Check

The goal of this check is to determine whether transaction can be processed offline.

If authorized amount (tag '9F02') is less than or equal to electronic cash single transaction limit, and electronic cash balance and CTTA has sufficient offline funds for the transaction, then the transaction undergoes offline processing.

Otherwise [if authorized amount (tag '9F02') is greater than the electronic cash single transaction limit or there is insufficient offline purchase available amount]:

— If terminal has online processing capability, then card requests online

processing;

- If terminal has no online processing capability, then card requests rejection.

Terminal Can Get Online

When terminal has online capability (terminal transaction qualifiers, byte 1 bit 4 = 0), the following requirements are appropriate.

- If authorized amount (tag '9F02') is greater than the electronic cash single transaction limit (if exists, tag '9F78'), then card should prepare returning available offline spending amount (if supported), and request online processing;
 - If returning available offline spending amount is permitted (card additional processing byte 1 bit 1 = 1), then card shall calculate available offline spending amount (tag '9F5D'), equal to CTTAUL (or CTTAL if CTTAUL does not exist) minus CTTA, then return this value in the GPO response;
 - Set CVR byte 3 bit 6 to 1 (Velocity check counters exceeded);
 - For follow-up steps see 8.4.3.15-- complete online transaction.
- If authorized amount (tag '9F02') is greater than the electronic cash balance (tag '9F79') minus electronic cash reset threshold (if exists, tag '9F6D'), then card should prepare returning available offline spending amount (if obtaining is supported), and request online processing.
 - If returning available offline spending amount is permitted (card additional processing byte 1 bit 1 = 1), then card shall calculate available offline spending amount (tag '9F5D'), equal to CTTAUL (or CTTAL if CTTAUL does not exist) minus CTTA, then return this value in the GPO response;
 - Set CVR byte 3 bit 6 to 1 (Velocity check counters exceeded);
 - For follow-up steps see 8.4.3.15-- complete online transaction.
- If authorized amount (tag '9F02') plus CTTA is greater than CTTAUL/CTTAL (if exists, tag '9F54'), then card should prepare returning available offline spending amount (if obtaining is supported), and request online processing.
 - If returning available offline spending amount is permitted (card additional processing byte 1 bit 1 = 1), then card shall calculate available offline spending amount (tag '9F5D'), equal to CTTAUL (or CTTAL if CTTAUL does not exist) minus CTTA, then return this value in the GPO response;
 - Set CVR byte 3 bit 6 to 1 (Velocity check counters exceeded);
 - For follow-up steps see 8.4.3.15-- complete online transaction.

Terminal only support Offline

When terminal has only offline support (terminal transaction qualifiers, byte 1 bit 4 = 1), the following requirements are appropriate.

- If authorized amount (tag '9F02') is greater than the electronic cash balance, or authorized amount is greater than the electronic cash single transaction limit, or authorized amount plus CTTA is greater than CTTAUL (or CTTAL if CTTAUL does not exist), then card shall prepare to return available offline spending amount (if supported), and reject transaction.
 - If returning available offline spending amount is permitted (card additional processing byte 1 bit 1 = 1), then card shall calculate available offline spending amount (tag '9F5D'), equal to CTTAUL (or CTTAL if CTTAUL does not exist) minus CTTA, then return this value in the GPO response;
 - Set CVR byte 3 bit 6 to 1 (Velocity check counters exceeded);
 - For follow-up steps see 8.4.3.17-- reject transaction;

Transaction Offline Completion Approved

- If none of the procedures above match, then card shall:
 - Store current CTTA value;
 - Save current electronic cash balance value;
 - Set transaction anti-pullout bit (card internal indicator) to indicate that the indicator is being updated. This indicator is only reset to 0 prior to the last read record command response. See 7.5.3—anti-pullout protection;
 - Calculate new CTTA equal to CTTA plus authorized amount (tag '9F02');
 - Calculate new electronic cash balance, equal to electronic cash balance minus authorized amount (tag '9F02');
 - If returning available offline spending amount is permitted (card additional processing byte 1 bit 1 = 1), then card shall calculate available offline spending amount (tag '9F5D'), equal to CTTAUL (or CTTAL if CTTAUL does not exist) minus CTTA, then return this value in the GPO response;
 - Request offline approval;
 - For follow-up steps see 8.4.3.14-- complete offline transaction.

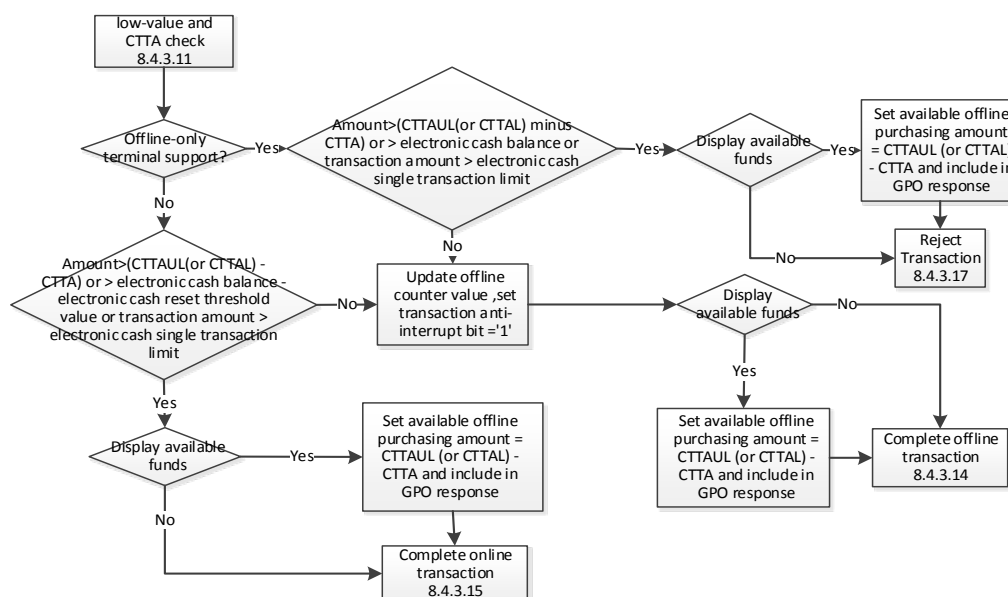


Figure 11 Low-value and CTTA check

For low-value and CTTA check process see Figure 11.

This check is to determine whether transaction can be processed offline.

If authorized amount (tag '9F02') is less than or equal to single transaction amount, and electronic cash balance or CTTA contain sufficient offline funds, then transaction can be processed offline.

Otherwise (if authorized amount (tag '9F02') is greater than the electronic cash single transaction limit or there is insufficient offline purchase available amount):

- If terminal has online processing capability, then card requests online processing;
- If terminal has no online processing capability, then card requests rejection.

For this option, can use offline purchase amount equal to the sum of CTTA available balance and electronic cash balance.

Terminal Can Get Online

The following is only appropriate for terminals that can get online (terminal transaction qualifiers byte 1 bit 4 = 0).

If terminal offline only bit (card internal indicator) = 0:

- If authorized amount (tag '9F02') is greater than the electronic cash single transaction limit (if exists, tag '9F78'), then card shall prepare returning available offline spending amount (if supported), and request online processing;
 - If returning available offline spending amount is permitted (card additional processing byte 1 bit 1 = 1), then card shall set available offline spending amount (tag '9F5D') equal to electronic cash balance

plus CTTAUL (or CTTAL if CTTAUL does not exist), minus CTTA, then return this value in the GPO response;

- Set CVR byte 3 bit 6 to 1 (Velocity check counters exceeded).
- See 8.4.3.15 for follow-on processing -- complete online transaction.

— If authorized amount (tag '9F02') is greater than the electronic cash balance (tag '9F79'), and authorized amount (tag '9F02') plus CTTA (no tag) is greater than CTTAUL/CTTAL (tag '9F54'), then card shall prepare returning available offline spending amount (if supported), and request online processing.

- If returning available offline spending amount is permitted (card additional processing byte 1 bit 1 = 1), then card shall set available offline spending amount (tag '9F5D') equal to electronic cash balance plus CTTAUL (or CTTAL if CTTAUL does not exist), minus CTTA, then return this value in the GPO response;
- Set CVR byte 3 bit 6 to 1 (Velocity check counters exceeded).
- See 8.4.3.15 for follow-on processing -- complete online transaction.

Terminal only support Offline

The following is only appropriate for terminals that cannot get online (terminal transaction qualifiers byte 1 bit 4 = 1).

— If authorized amount (tag '9F02') is greater than the electronic cash single transaction limit, or authorized amount is greater than the electronic cash balance and authorized amount plus CTTA is greater than CTTAUL/CTTAL, then card shall prepare to return available offline spending amount (if supported), and reject transaction.

- If returning available offline spending amount is permitted (card additional processing byte 1 bit 1 = 1), then card shall set available offline spending amount (tag '9F5D') equal to electronic cash balance plus CTTAUL (or CTTAL if CTTAUL does not exist), minus CTTA, then return this value in the GPO response;
- Set CVR byte 3 bit 6 to 1 (Velocity check counters exceeded).
- To continue see 8.4.3.17 -- reject transaction.

Transaction Offline Completion Approved

— If none of the procedures above apply, then card shall complete the following processing:

- Set transaction anti-pullout bit (card internal indicator) to indicate that the indicator is being updated. This indicator will be reset to 0 prior to the last READ RECORD response. See 7.5.3 -- anti-pullout protection;

Electronic Cash Funds Available

- If authorized amount (tag '9F02') is not greater than the electronic cash balance (tag '9F79'), then save the electronic cash balance value, and calculate new electronic cash balance = electronic cash balance - authorized amount (tag '9F02').

Electronic Cash Funds Not Available So Use CTTA Funds

- If authorized amount (tag '9F02') is greater than the electronic cash balance (tag '9F79'), then save CTTA, and calculate new CTTA = CTTA plus authorized amount.
- If returning available offline spending amount is permitted (card additional processing byte 1 bit 1 = 1), then card shall set available offline spending amount (tag '9F5D') equal to electronic cash balance plus CTTAUL (or CTTAL if CTTAUL does not exist), minus CTTA, then return this value in the GPO response;
- Request offline approval.
- See 8.4.3.14 for follow-on processing -- complete online transaction.

8.4.3.12 Offline Currency Mismatch

If application currency does not match transaction currency, need to check whether the upper limit for these transactions are exceeded. Section 7.7.5 outlines currency check, if currency is mismatched please see this Section.

—— If consecutive transaction counter (international - currency) is less than the consecutive offline transaction limit (international - currency) (tag '9F53'), then card shall:

- Store the consecutive transaction counter's current value (international);
- Set transaction anti-pullout bit (card internal indicator) to indicate that the indicator is being updated. This indicator will be reset to 0 prior to the last READ RECORD response. See 7.5.3-- anti-pullout protection;
- Consecutive transaction counter (international - currency) adds 1;
- Request offline approval;
- See 8.4.3.14 for follow-on processing -- complete online transaction.

—— If the conditions above are not met, and offline only terminal bit = 0, then card shall request online processing;

- Set CVR byte 3 bit 6 to '1' (Velocity check counters exceeded);
- See 8.4.3.15 for follow-on processing -- complete online transaction.

—— If the conditions above are not met, and offline only terminal bit = '1', then card shall request reject transaction;

- Set CVR byte 3 bit 6 to '1' (Velocity check counters exceeded);
- To continue see 8.4.3.17 -- reject transaction.

For offline currency mismatch check processing flow see Figure 12.

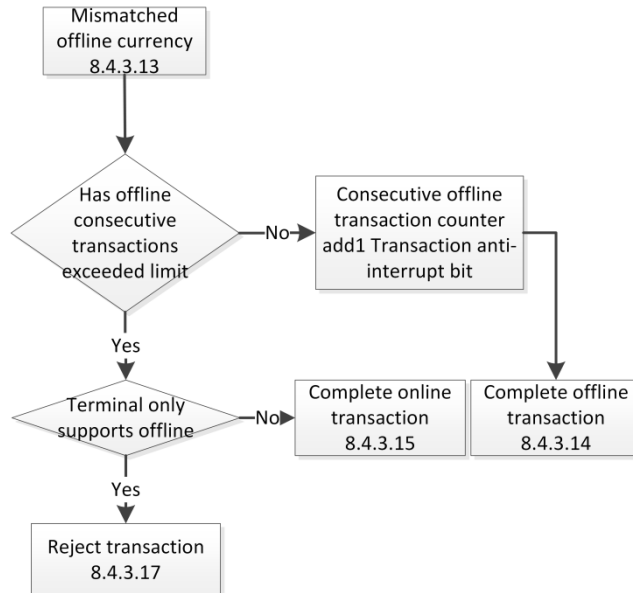


Figure 12 Mismatched offline currency

8.4.3.13 Complete Offline Transaction

Transaction can be completed offline. Provide additional data pointer and approval cryptogram available for terminal reading in the GPO response.

— Card shall:

- ATC adds 1;
- Generate dynamic application data signature (SDAD - tag '9F4B');
 - a) If terminal supports '01' version fDDA (terminal transaction qualifiers byte 4 bit 8 = '1'), then generate a 4 byte random number, then follow Appendix C to execute version '01' fDDA;
 - b) If terminal does not support '01' version fDDA (terminal transaction qualifiers byte 4 bit 8 is '0'), then follow Appendix C and execute '00' version fDDA;
- Return fDDA needed data SFI and record number AFL in the GPO response.

Note: Not all debit/credit mandatory data (such as CDOLs) exists in AFL labeled records.

During the process of ODA for online authorization, the card offline data authentication shall be consistent with the existing offline data authentication in the qUICS offline transaction, including the card using the same set of AFL for transaction processing. The terminal shall ignore the Issuer Authorization Code-Electronic Cash (tag '9F74') returned by the card, and shall not process this

case as the Electronic-cash transaction.

- Card shall set the cryptogram information data ‘9F27’ bit 8-6 and CVR byte 2 bit 6-5 to ‘01’, to indicate an offline approval cryptogram (TC), follow *Part II Debit Credit Application - Member Implementation Guide for Issuer* Appendix E cryptogram version 01 generate application cryptogram (TC). Cryptogram 17 is generated in the same way as cryptogram 01, but use different card and terminal data elements as cryptogram input (see Appendix E of this part).

Note: CDOLs are not used in qUICS, cryptogram is generated using PDOL requested data.

- Card shall follow Section 8.4.2 in building GPO response;
- To continue see Section 8.4.4 -- terminate qUICS card GPO processing.

For complete offline transaction processing flow see Figure 13.

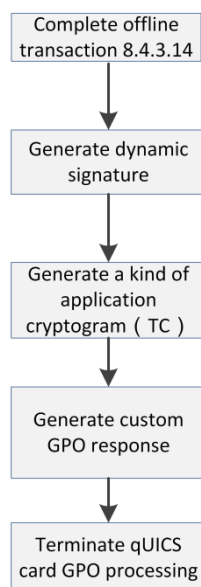


Figure 13 Complete offline transaction

8.4.3.14 Complete Online Transaction

Card or terminal requesting transaction needs online authorization. Prior to completing online transaction confirm other checks does not require termination or rejection of transaction.

Card Requires Contact Type Debit/Credit Online

If card requires contact type debit/credit online (card additional processing, byte 1 bit 2) and terminal supports contact debit/credit (terminal transaction qualifiers byte 1 bit 5), then card shall request transaction termination; if terminal does not support contact type debit/credit, continue to complete online transaction.

To continue see 8.4.3.16 -- terminate contactless transaction.

Continue and Complete Online Transaction

- ATC adds 1;
- Based on *Part II Debit Credit Application - Member Implementation Guide for Issuer* Appendix E cryptogram version 01, card generates application cryptogram (ARQC). Cryptogram version 17 and cryptogram version 01 generation methods are the same, but the card and terminal data elements used as cryptogram input are different (see Appendix E of this part);

Note: CDOLs are not used in qUICS, cryptogram is generated using PDOL requested data;

- Card shall set the cryptogram information data '9F27' bit 8-7 and CVR bit '10', and then indicate an ARQC. Then based on the description in Section 8.4.2, include cryptogram and relevant data in the GPO response (note that for online transaction, AFL is not returned);
- When Issuer personalizes the card, if it is required to provide available offline spending amount in the issuer application data (tag '9F10') issuer defined data, then card shall include it to facilitate online authorization, see Appendix D;
- If return of available offline spending amount is permitted (card additional processing, byte 1 bit 1 = '1') and currency match bit = '1', then card shall include these data in the GPO response;
- To continue see 8.4.4 -- terminate qUICS card GPO processing.

For complete online processing flow see Figure 14.

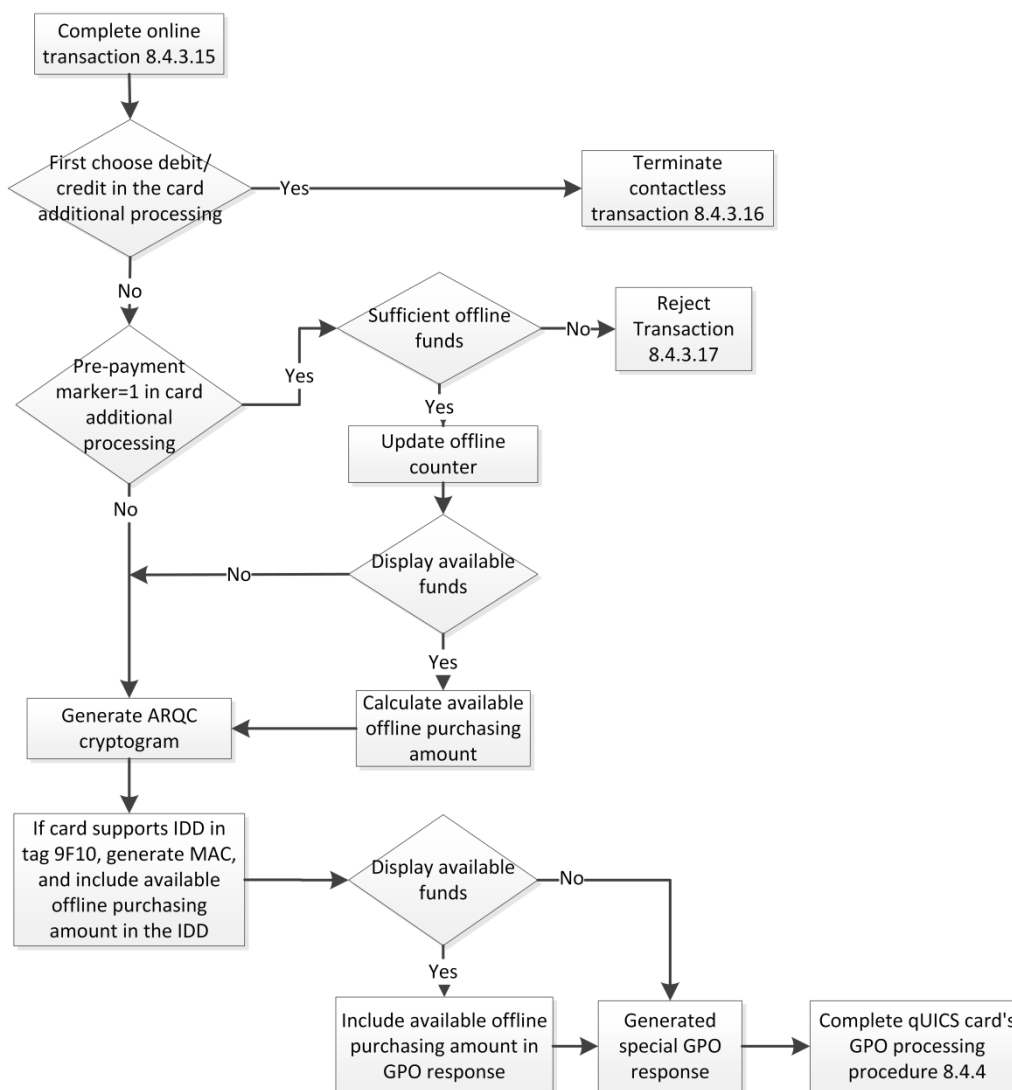


Figure 14 Complete online transaction

8.4.3.15 Terminate Contactless Transaction

Card has requested termination of contactless transaction.

—— Return of error code in GPO response:

SW1 SW2=x'6985'

8.4.3.16 Reject Transaction

Regardless of whether the terminal is an offline only terminal and offline transaction can't be completed due to offline transaction upper limit being exceeded, or card uses pre-paid option and transaction has insufficient funds, transaction must be rejected.

—— If returned available offline purchase amount bit = '1', then card shall include available offline spending amount in the GPO response;

—— Card shall set the cryptogram information data '9F27' bit 8-7 and CVR bit '00', indicate a AAC cryptogram in the CVR, generate AAC cryptogram,

and (based on Section 8.4.2) include CVR and cryptogram and related data in the GPO response;

- ATC adds 1;
- Cryptogram is generated based on Appendix E cryptogram version 01. Cryptogram version 17 and cryptogram version 01 generation methods are the same, but the card and terminal data elements used as cryptogram input are different (see Appendix E of this part).
- To continue see 8.4.4 -- terminate qUICS card GPO processing.

8.4.4 Terminate qUICS Card GPO Processing

Card follows description of Section 8.4.2 , format GPO command response and return to terminal.

9 Read Application Data

9.1 Terminal Data

This step does not involve terminal data elements.

9.2 Card Data

Table 15 List out the data that was returned by the card in the prior step of application initialization processing and to be used in read application data processing.

Table 15 Read application data -- card data

Date Element	Description
Application File Locator (AFL)	In application initialization processing, the data returned by the card to the terminal includes a set of record entries to be read, with each entry including: • File short file identifier (SFI) • First and last record number that needs to be read • Used to save the record numbers of SDA and DDA data. Begin calculation from the first record number read in the file.

Table 16 lists out application elementary file records data read from the card.

Table 16 Read application data -- card file

Date Element	Description
Application Elementary File (AEF)	Card data file, includes data used for application processing. An AEF includes a series of records that are labelled with record numbers. Each AEF uses an unique SFI. Terminal uses READ RECORD command to read record content, command includes SFI and record number
Short File Identifier (SFI)	Used to uniquely identify application data files. Listed out in AFL, can be used by terminal to identify file to be read

9.3 Flow Chart and Command

READ RECORD

Terminal sends a READ RECORD command to the card for each record to be read. This command includes the SFI that identifies the file and a record number that identifies the record in the file.

Card provides the requested record in the response to READ RECORD.

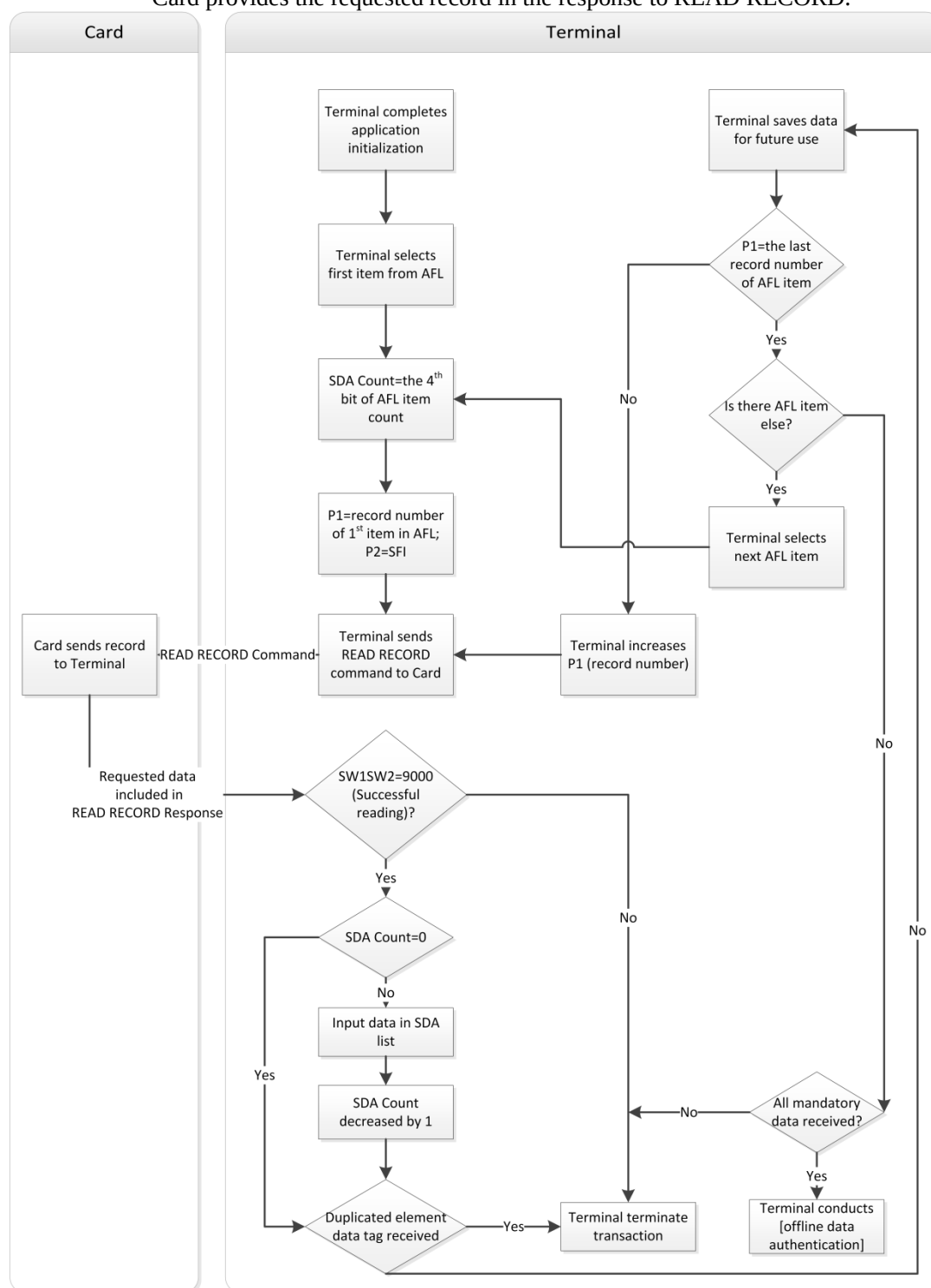


Figure 15 Read application data flow chart

9.4 Terminal Processing

Terminal uses AFL to determine which transaction data records needs to be read from the card, each AFL item (4 bytes) represents the consecutive records in a file on the card. For each AFL item (4 bytes), beginning from record 1, terminal sends the card a READ RECORD command for each record in order to read record data, until the last record is reached. This continues until all AFL items are processed.

The recognizable transaction data read shall be stored on the terminal for transaction use. If reads tags with correct TLV format but unspecified definition, the terminal shall store it for future use, and shall not terminate transaction on this basis. For records where AFL indicates as used for offline data authentication, then add the data into the offline authentication data list for use during offline data authentication.

If one of the following situations occur during read data processing, terminal shall terminate the transaction:

- Card returns the same tag two or more times within one or more records;
- Card returns in a record a tag that has already been returned by the card in the GPO response;
- Card is missing mandatory data;
- Data format error;
- READ RECORD command return status word is not '9000'.

Existence of one or more of the following conditions should not be considered cause to terminate transaction:

- Card returns card holder name (tag '5F20') but the tag length does not meet *UICS Basic Specifications - Part II*;
- Card returns extended card holder name (tag '9F0B') but the tag length does not meet *UICS Basic Specifications - Part II*;
- Card returns card holder name (tag '5F20') as well as extended card holder name (tag '9F0B').

For terminal processing flow chart see Figure 15.

When terminal obtains card data through the READ RECORD command, after obtaining the card expiration data, shall immediately carry out expiration check. If terminal current date is after the expiration date given by card, then terminal shall treat the card as expired. If card is expired, then terminal shall determine card transaction qualifiers byte 1 bit 4, if this is 1, then terminal shall transition into online process, and prompt the card holder "card has passed expiration date, transaction going online", if this position is 0, then terminal shall reject transaction and prompt the card holder with "card has passed expiration date, transaction is declined". Therefore because the last record indicated by AFL is not read, the card's transaction anti-pullout bit will not be reset, so before the next transaction, the card shall recover the various counters and electronic cash balance to the values prior to this transaction.

During personalization, card expiration date shall not be in the last record.

9.5 Card Processing

For offline transaction, the transaction will continue. Terminal sends READ RECORD command for each record in the AFL. When the card successfully returns the last record, transaction anti-pullout bit is reset, to indicate that the terminal has completed transaction with card.

- Card shall be able to know the last record has been read;
- Prior to the last READ RECORD command, the card shall set the transaction anti-pullout bit (card internal indicator);

Note: Card would not know whether the terminal successfully received the last READ RECORD command response. This means that interruption may still occur, and once it occurs, would abnormally impact offline available balance. The time window for this type of situation to occur has been reduced to the minimum. If offline data authentication fails, the terminal can still reject the transaction, but this is very rare for real cards.

- Prior to responding to the last READ RECORD command, the card shall check card additional processes (tag '9F68') byte 2 bit 5, if this position is 1, then card shall make a transactional journal record, see *UICS Basic Specifications - Part 2* Section 16 for how to record transactional journal.
- After the card records the last READ RECORD command, prior to responding to the last READ RECORD command, the card shall update the 9F69 value into the last record for storage, and guarantee that this operation and the other card update operations is one atomic operation. For example updates of electronic cash balance, contactless transactional log recording, etc. shall be a single atomic operation with the update of 9F69.

In order to boost transaction-operating speed, the terminal shall follow the order in the AFL to read card records.

10 Offline Data Authentication

10.1 Terminal Data

For SDA related data on the terminal see Table 17.

Table 17 Offline data authentication - terminal data

Date Element	Description
Public Key Index (PKI)	Each CA public key used in offline data authentication is uniquely identified together by PKI and registered application provider id (RID) in the application id (AID)
CA public key	Public keys stored in the terminal used for verification of issuer public key certificates
Registered application provider id (RID)	Indicates the public key list for specified payment organizations in the terminal and jointly identifies CA public keys with PKI

10.2 Card Data

Used by terminal whether to execute SDA or DDA card data as listed in Table 18.

Table 18 Offline data authentication - card data

Date Element	Description
Application interchange profile (AIP)	Includes indicators: - Card supports static data authentication SDA - Card supports dynamic data authentication fDDA
CA public key index (PKI)	Each public key in static data authentication used for offline data authentication are jointly identified by CA public key index (PKI) and registered application provider id
Issuer public key certificate	Issuer public key certificate that includes the issuer public key signed using CA private keys
Issuer RSA public key index	Used to recover RSA signed static application data and IC card public key certificate
Issuer RSA public key remainder	Includes part of issuer RSA public key that is not included in the issuer public key certificate
Registered application provider id (RID)	Part of AID (first 5 bytes), used to identify payment organization. RID and public key index jointly determines the public key needed for transaction
Signed static application data (SAD)	Signature calculated using issuer private key, includes the hash value of major card data
Static data authentication tag list	Includes the tag list of data used in offline data authentication, this data element is optional, but if it appears, is only permitted to include AIP (tag '82'), if it includes other data, then SDA fails
Signed authenticated	Card data used for verifying signed static application data (SAD), includes record data designated in the AFL for use in offline data

static data	authentication, as well as designated data in the SDA tag list. If SDA tag list exists, it should only contain AIP (tag '82'), terminal checks SDA tag list to see whether only AIP tag exists
IC card dynamic data	Issuer designated data that is included in signed dynamic application data.
IC card dynamic number	Part of IC card dynamic data, card generated value that changes over time.
IC card public key certificate	Includes IC card public key that is signed using issuer private key, placed in card during card personalization. Certificate contains static application data signed and encrypted using issuer private key
IC card RSA public key index	Used to recover RSA signed dynamic application data, value is 3 or 65537
IC card RSA public key remainder	IC card RSA public key part (if exists) that is not included in the IC public key certificate
Signed dynamic application data	Data generated by the card upon receiving INTERNAL AUTHENTICATE command

10.3 Flow Chart

Offline data authentication uses fDDA. For fDDA flow see Appendix C.

10.4 Terminal Processing

IC card public key certificate used in DDA includes card static data hash value. This part recommends that qUICS and debit/credit applications process utilize the same static data. If signed debit/credit static data is different from signed qUICS static data, then shall need to support two card public key certificates, which will add implementation complexity.

Terminal shall read all static data elements included in IC card public key certificate, in order to complete DDA. For shared static data, Issuer shall weigh the pros and cons of including special data elements in debit/credit static data elements and the increased interaction time for qUICS transactions.

For qUICS, see recommended signature data in 7.2.5.

— Offline data authentication will fail if any of the case is true:

- AIP indicates that fDDA is not support;
- or fDDA supported, but fDDA required data is incomplete.

11 Transaction Termination

When terminal receives the correct GPO command response from the card, it will check the cryptogram information data '9F27' to determine the cryptogram type provided by card. Based on cryptogram type, determine transaction reject, online processing, or offline approval.

11.1 Cryptogram Type Check

- If returned ARQC (cryptogram information data (tag '9F27') bit 8-7 = '10'), then terminal shall send transaction online;

To continue see 11.3 -- terminal online processing.

- If returned AAC (cryptogram information data (tag '9F27') bit 8-7 = '00'), then terminal shall reject transaction;

To continue see 11.4 -- terminal offline rejection.

- If returned TC (cryptogram information data (tag '9F27') bit 8-7 = '01'), then terminal shall check terminal abnormal file (if exists), if application PAN appears in terminal abnormal file, then terminal shall reject transaction offline;

To continue see 11.4 -- terminal offline rejection.

- Terminal shall process AFL based on *Part II Debit Credit Application - Member Implementation Guide for Issuer*, sending a READ RECORD command for each record in AFL;
- If card response is READ RECORD command failure, then terminal shall discard current transaction data and return check processing;
- Once all indicated records have been read, terminal shall prompt card holder and merchant to remove card, but transaction is still processing;
- If AIP indicates DDA supported, then terminal shall leverage UnionPay IC card Specifications – Basic Specification Part 4 Appendix C fDDA definitions to verify DDA dynamic signature.
- If fDDA fails, or offline data authentication is not executed, then terminal shall check card transaction properties:

If card transaction qualifiers byte 1 bit 6 = '1', terminals with online capability shall notify cardholder that transaction is in process, and generate an online message to acquirer, then use the card provided TC to send transaction online. To continue see Section 11.3 -- terminal online processing.

Or else, If card transaction qualifiers byte 1 bit 5 = '1', terminals that support contact type debit/credit applications (terminal transaction property '9F66' byte 1 bit 5 = '1') shall terminate the transaction and request

cardholder to use contact type debit/credit interface. To continue see Section 11.5.

If none of the above conditions are met, the terminal shall reject transaction, and shall not try to use other interface for transaction. To continue see Section 11.4 -- terminal offline rejection.

- If TC returned and fDDA is executed and pass, then terminal shall approve the transaction. To continue see Section 11.2 -- approve offline transaction.

11.2 Approve Offline Transaction

- Terminal shall power off the contactless interface;
- Terminal shall prompt cardholder and merchant that transaction has been approved;
- If card (in card transaction qualifiers) or terminal requests a CVM (signature), then terminal shall print signature line on the receipt;
- If card provides available offline spending amount, and terminal can display or print, then terminal shall display or print it out;
- Terminal shall use the cryptogram provided by GPO response and relevant data to clear the transaction. For details see Appendix E for data needed in cryptogram version 17.

11.3 Terminal Online Processing

- Terminal shall power off the contactless interface;
- Terminal shall prompt cardholder and merchant that card can be removed, and the transaction is requesting authorization;
- Terminal should check the card transaction qualifier '9F6C' to determine CVM for this transaction;
- Terminal shall send an online authorization request message to the acquirer, with the message including online cryptogram (ARQC) provided by the card in GPO response and other required data;
- In the event of full issuer migration, the terminal shall be capable of providing online message that contains basic IC card transaction data. See Appendix E for the most elementary data provided by online message when cryptogram version 17 is supported;
- Terminal shall approve or reject transaction based on issuer response;
- Terminal shall prompt cardholder and merchant that transaction has been approved or rejected;
- If online transaction could not be completed, terminal shall reject transaction and prompt the cardholder and merchant that the transaction has been rejected;

- If transaction is approved, then terminal shall clear the transaction, and include cryptogram (ARQC) provided by the card GPO response and relevant data. For details see Appendix E for data needed in cryptogram version 17.
- Terminal shall not reject transaction due to issuer returned scripts.

11.4 Terminal Offline Rejection

- Terminal shall power off the contactless interface;
- Terminal shall reject transaction and prompt cardholder and merchant that transaction has been rejected;
- If available offline spending amount is provided, and terminal can display or print, then terminal shall display or print it out;
- Terminal shall not try another interface to carry out the transaction.

11.5 Offline Data Authentication Failure and Terminal Terminates Transaction

- Terminal shall power off the contactless interface;
- Read card shall terminate contactless transaction and prompt user to use contact type interface.

Appendix A qUICS Data Elements

This part lists out the data elements specific for the qUICS flow.

A.1 Name

Name of this data element;

A.2 Format Tag Length Source

- Data element format follows *Part II Debit Credit Application - Member Implementation Guide for Issuer A.1*;
- Data element tag is hexadecimal unique encoding that represents the data element; Tag usage shall follow *Part II Debit Credit Application - Member Implementation Guide for Issuer A.3*;
- Data element length value is represented in decimal.
- Data element is from the card or the terminal

A.3 Shared

Y: indicates that this data element is also being used in the standard debit/credit flow, both standard debit/credit flow completed transaction and qUICS flow completed transactions shall impact this data element;

N: indicates that this data element is only used in qUICS.

A.4 Requirement

Requirement column indicates whether the data element is mandatory (M), required (R), conditional (C), optional (O), or not applicable (N/A) in qUICS, and indicates whether the data element is card data or terminal data.

A.5 Access

Obtain column indicates whether this data element can be read by terminal or returned through command. If yes, give the read command.

A.6 Value

The value of each bit of the data element. If there are no special notes, value is set as reserved (byte or bit), and set to 0.

Table A.1 Data Elements

Name	Format Tag Length Source	Shared	Requirement	Description	Access	Value
Available Offline Spending Amount	F: n 12 T: 9F5D L: 6 S: Card	N	C	A calculation field used to approve the terminal printing or displaying the offline transaction amount available in the card. Unless this tag is personalized to 1, otherwise the card will not this tag to be included in terminal read records or in GPO response. Personalization of this data would not impact its inclusion in issuer defined data.	GET DATA GPO READ RECORD	If personalized value is greater than 0, obtaining this data element with GET DATA operation would be approved; If this data element is personalized to 1 and card application processing (byte 1 bit 1) has value 1, then this data element is included in GPO, and permitted to be read with READ RECORD; If IC card private key length is greater than 1024, then this data element is read through READ RECORD and not through GPO.
Card Additional Processes	F: b 32 T: 9F68 L: 4 S: Card	N	C If card is not online-only qUICS, then must possess this data element	Points to card processing requirements and parameter selection.	GET DATA	This data can be updated via PUT DATA command For details see Table 10 card additional processing
Card CVM Limit	F: n 12 T: 9F6B L: 6 S: Card	N	C	If it appears this indicates that when card and terminal currency types match and a contactless transaction exceeds this value, then card must provide CVM. The cardholder verification defined in this part is online PIN and signature.	GET DATA	Note: This tag can be modified by PUT DATA command.

Name	Format Tag Length Source	Shared	Requirement	Description	Access	Value
Card Transaction Qualifiers	F: b 16 T: 9F6C L: 2 S: Card	N	C If Issuer supports cardholder authentication or Issuer has some requirements for card transaction qualifiers	Used in this part to show the device which CVM, card capabilities and Issuer requirements are required by the card.	GPO	This data can be updated via PUT DATA command Byte 1 Bit 8 1 = requires online PIN Bit 7 1 = requires signature Bit 6 1 = if offline data authentication fails and terminal can go online then request online Bit 5 1 = if offline data authentication fails and terminal supports standard debit/credit flow then terminate Bit 4 1 = if application has expired, then online transaction Bit 3-1 = reserved Byte 2 Bit 8 1= CDCVM performed (Bit 8 is not applicable for traditional IC card transactions, and is set to 0) Bit 7-1 = reserved
Application Interchange Profile (AIP)	F: b 16 T: 82 L: 2 S: Card	N, separate from AIP used for standard debit/credit flow	M	Indicates the application designated functionality supported by the card.	GPO	Byte 1 Bit 8 RFU Bit 7 1= support SDA Bit 6 1= support DDA Bit 5 1 = support cardholder verification Bit 4 1 = support terminal risk management Bit 3 1 = support Issuer authentication Bit 2 1 = RFU Bit 1 1= support CDA Byte 2 Bit 8 = 0 ¹ Bit 7~1 RFU

Name	Format Tag Length Source	Shared	Requirement	Description	Access	Value
Cryptogram Information Data (CID)	F: b 8 T: 9F27 L: 1 S: Card	N	M	Indicates cryptogram type returned by card and next action of terminal.	GPO	Bit 8-7: 00=AAC 01=TC 10=ARQC 11=AAR (Not supported in UICS) Bit 6-5: RFU (00) Bit 4: 1= need notification Bit 3-1 (reason/notification/authorization reference code): 000 = no information 001 = the service is not allowed 010 = PIN try limit exceeded 011 = Issuer authentication failed xxx = RFU This data will be initialized to 0
Last Online ATC Register ³	F: b 16 T: 9F13 L: 2 S: Card	Y	C If executed new card check	Last online submission of transaction ATC value.	GET DATA	
Terminal Contactless Floor Limit	F: n 12 T: – L: 6 S: Terminal	N	C If pre-processing supported	If contactless transaction's transaction amount is greater than or equal to this value, then terminal requests online authorization.	N/A	

³ This bit was formerly used for MSD id, so some cards may already have bit set as 1.

Name	Format Tag Length Source	Shared	Requirement	Description	Access	Value
Terminal Contactless Transaction Limit	F: n 12 T: – L: 6 S: Terminal	N	C If pre-processing supported	If contactless transaction value is greater than or equal to this value, then terminate transaction, permit retrying this transaction through another interface.	N/A	
Terminal CVM Required Limit	F: n 12 T: – L: 6 S: Terminal	N	C If pre-processing supported	If contactless transaction is greater than or equal to this value, terminal requests a cardholder verification method (CVM). Online PIN, signature and CDCVM are the cardholder verification methods (CVM) defined in this part.	N/A	
Terminal Transaction Qualifiers	F: b 32 T: 9F66 L: 4 S: Terminal	N	M	Indicates terminal functionality, requirements, and card parameter selection.	N/A	Refer to Table 4 for values of Terminal Transaction Qualifiers.
Electronic Cash Balance	F: n 12 T: 9F79 L: 6 S: Card	Y (debit/credit based Low-value flow usage)	M	If authorized amount exceeds electronic cash balance, then all transactions shall be authorized online or rejected offline.	GET DATA	This data can be updated via PUT DATA command Shall not be returned via READ RECORD command

Name	Format Tag Length Source	Shared	Requirement	Description	Access	Value
Electronic Cash Balance Upper Limit	F: n 12 T: 9F77 L: 6 S: Card	Y (debit/credit based Low-value flow usage)	C If low-value check or low-value and CTTA check supported	If authorized amount plus electronic cash balance exceeds this constraint, card requests online processing.	GET DATA	This data can be updated via PUT DATA command Shall not be returned via READ RECORD command
Electronic Cash Reset Threshold	F: n 12 T: 9F6D L: 6 S: Card	Y (credit based Low-value flow usage)	O	If authorized amount is greater than electronic cash balance minus this threshold, then card requests online processing.	GET DATA	This data can be updated via PUT DATA command Shall not be returned via READ RECORD command
Electronic Cash Single Transaction Limit	F: n 12 T: 9F78 L: 6 S: Card	Y (debit/credit based Low-value flow usage)	O		GET DATA	This data can be updated via PUT DATA command Shall not be returned via READ RECORD command
Electronic Cash Issuer Authorization Code	F: an 6 T: 9F74 L: 6 S: Card	Y (debit/credit based Low-value flow usage)	O	Electronic cash transaction or qUICS offline approved transaction, card shall return this data element.	READ RECORD	
Application Version Number	F: b16 T: 9F08 L: 2 S: Card	Y	M	Version number assigned to application by payment system. Same as defined in UICS – Basic Specifications Part 5.	READ RECORD	Defined by payment system

Name	Format Tag Length Source	Shared	Requirement	Description	Access	Value
Card Authentication Related Data	F: b T: 9F69 L: var 8-16 S: Card	N	C If supports fDDA version '01' or above.	If card executes '01' or higher version of fDDA, then this data shall be returned in the last record; otherwise this data shall not appear in records.	READ RECORD	Byte 1: fDDA version number (in this specification version is '01') Byte 2-5: card unpredictable number Byte 6-7: card transaction qualifiers Byte 8: RFU (00), specific usage method is not defined in this part. Note: in this version of this specification, card authentication related data use 8 bytes, and is personalized into the card.
Cumulative Total Transaction Amount Limit (CTTAL)	F:n12 T:9F54 L:6 S: Card	Y	C If executing cumulative amount frequency check.	Cumulative offline transaction amount max constraint. If exceeded by transaction then request online.	GET DATA	
Cumulative Total Transaction Amount Upper Limit (CTTAUL)	F: n 12 T:9F5C L: 6 S: Card	Y	C If executing cumulative amount frequency check.		GET DATA	
Issuer Application Data	F: b T: 9F10 L: var. up to 32 S: Card	Y	M	Issuer application data.	GPO	
Application Currency Code	F: n 3 T: 9F51 L: 2 S: Card	Y	M	Issuer application data	GET DATA	

Name	Format Tag Length Source	Shared	Requirement	Description	Access	Value
Application Transaction Counter	F: b 16 T: 9F36 L: 2 S: Card	Y	M		GET DATA GPO	
Cardholder Name	F: ans2-26 T: 5F20 L: 2:20 S: Card	Y	O	Cardholder name, consistent with ISO/IEC 7813.	GPO READ RECORD	The Issuer shall not write the cardholder's authentic and integral name into the application, and shall partially omit cardholder name, replace it with other titles (e.g., Mr. X/Mrs. Y) or specific format characters (e.g., space), to ensure that authentic cardholder information is not transmitted in contactless transactions. Notes: The value of this data object can be Chinese characters, and it may cause transaction rejections on a limited number of international terminals outside Mainland China.
Cardholder Name Extension	F: ans27-45 T: 9F0B L: 27-45 S: Card	Y	O	It indicates the whole Cardholder name when greater than 26 characters, using the same coding convention as in ISO7813.	GPO READ RECORD	The Issuer shall not write the cardholder's authentic and integral name into the application, and shall partially omit cardholder name, replace it with other titles (e.g., Mr. X/Mrs. Y) or specific format characters (e.g., space), to ensure that authentic cardholder information is not transmitted in contactless transactions. Notes: The value of this data object can be Chinese characters, and it may cause transaction rejections on a limited number of international terminals outside Mainland China.

Name	Format Tag Length Source	Shared	Requirement	Description	Access	Value
Cardholder ID Number	F: an40 T: 9F61 L: 1-40 S: Card	Y	O	Cardholder ID Number	READ RECORD	The Issuer shall not write the cardholder's authentic and integral ID number into the application. The value can be personalized with partially omitted cardholder ID information or specific format character (e.g. space) under business requirements.
Payment Account Reference (PAR)	F: an T: 9F24 L: 29 S: Card	Y	C If the card uses token	It is used to associate different Tokens deriving from the same bankcard. The same PAN and its Tokens shall have one identical PAR. For more information about PAR application, please refer to <i>UnionPay Payment Tokenization Implementation Guide</i> .	READ RECORD or GPO	For details, please refer to <i>EMV Payment Tokenisation Specification Technical Framework</i>
Issuer Identification Number (IIN)	F: n6 T: 42 L: 3 Source: Card	Y	O	The number that identifies the major industry and the card issuer and that forms the first part of the Primary Account Number (PAN). It is referred to as BIN in this document. In template 'BF0C' or '73'. It is returned to the terminal in SELECT or READ RECORD response.	READ RECORD or SELECT	
Card Additional Function Indicator	F: b8 T: DF61 L: var. Source: Card	N	M	It is used to identify the card's capability to support additional functions, and can be read by the Terminal. It is personalized in template 'BF0C' and supports GET DATA and PUT DATA commands.	SELECT or GET DATA	Byte 1: Bit 5: 1= Credit-mode low-value payment supported Bit 7: 1= ODA for online authorization supported

Name	Format Tag Length Source	Shared	Requirement	Description	Access	Value
Card Verification Results (CVR)	F: b32 T: – L: 4. S: Card	Y	M	UnionPay dedicated data. It indicates the exception conditions that occurred in the current and previous transactions. It is returned to the terminal in the Issuer Application Data.	GPO	Byte 1: Length indicator (03) Byte 2: bits 8–7: 00 = Not applicable for qUICS 01 = Not applicable for qUICS 10 = 2nd GENERATE AC not requested. 11 = RFU bits 6–5: 00 = GPO returns AAC; 01 = GPO returns TC; 10 = GPO returns ARQC; 11 = RFU; bit 4: 1 = Not applicable for qUICS bit 3: 1 = CDCVM performed successfully; bit 2: 1 = Not applicable for qUICS bit 1: 1 = Not applicable for qUICS Byte 3: bit 8: 1 = Not applicable for qUICS bit 7: 1 = PIN Try Limit exceeded; bit 6: 1 = Velocity check counters exceeded; bit 5: 1 = Not applicable for qUICS bit 4: 1 = Issuer Authentication failed in the last online transaction ; bit 3: 1 = Not applicable for qUICS bit 2: 1 = Not applicable for qUICS bit 1: 1 = Not applicable for qUICS Byte 4: bits 8–5: Issuer Script Command counter bit 4: 1 = Issuer Script processing failed on last transaction; bit 3: 1 = Not applicable for qUICS bit 2: 1 = Not applicable for qUICS bit 1: Not applicable for qUICS Bytes 2-4 are reset to zero during application initialization..

Name	Format Tag Length Source	Shared	Requirement	Description	Access	Value
Amount, Authorized (Numeric)	F: n12 T: 9F02 L: 6 S: Terminal	N/A	R	Authorized amount of the transaction (excluding adjustment)	N/A	
Amount, Other (Numeric)	F: n12 T: 9F03 L: 6 S: Terminal	N/A	C If cash-back supported	Secondary amount related with the transaction, indicating the amount of cash-back	N/A	
Application Identifier (AID) ID	F: b40-128 T: 9F06 L: 5-16 S: Terminal	N/A	R	It indicates an application according to the definition of <i>ISO/IEC 7816-5:1994/Amd 1:1996</i>	N/A	
Merchant name and location	F: ans T: 9F4E L: var. S: Terminal	N/A	R	It indicates merchant name and location	N/A	
Terminal country code	F: n 3 T: 9F1A L: 2 S: Terminal	N/A	R	Terminal country code specified in <i>ISO 3166-1:1997</i>	N/A	
Terminal floor limit	F: b 32 T: 9F1B L: 4 S: Terminal	N/A	C If the terminal is capable of offline transaction processing.	It indicates the floor limit in the terminal related with the AID that causes transactions to be transmitted online.	N/A	

Name	Format Tag Length Source	Shared	Requirement	Description	Access	Value
Terminal Verification Result (TVR)	F: b 40 T: 95 L: 5 S: Terminal	N/A	R	A set of indicators that records the processing results of each debit/credit function that terminal performs.	N/A	Bytes 1-5 are not used in qUICS, filled with all zeros.
Transaction currency code	F: n 3 T: 5F2A L: 2 S: Terminal	N/A	R	Transaction currency code specified in <i>ISO 4217:2001</i>	N/A	
Transaction date	F: n6 YYMMDD T: 9A L: 3 S: Terminal	N/A	R	Local date when transaction authorized	N/A	
Transaction Type	F: n 2 T: 9C L: 1 S: Terminal	N/A	R	Financial transaction types indicated by the first 2 bits of processing code according to definition of <i>ISO 8583:1987</i>	N/A	
Unpredictable Number	F: b 32 T: 9F37 L: 4 S: Terminal	N/A	R	Dynamic unique data that the terminal provides to the card for application cryptogram generation.	N/A	

Appendix B qUICS and Debit/Credit Flow Comparison

This appendix compares qUICS and contact standard debit/credit application flows.

qUICS requirements are different from that of contact standard debit/credit application flow. With respect to application selection, the former uses PPSE and the latter uses PSE. If PPSE is chosen, contactless application list is returned in Select command response. In the contact standard debit/credit application flow, after PSE is chosen, will use Read Record command to obtain the contact application list from the card.

Contact debit/credit flow PSE use is not required (directory selection method). In the credit/debit flow, AID list method is required, while in qUICS, this method is not recommended.

For qUICS, PDOL should ideally exist and provide terminal data elements - terminal transaction qualifiers; this data will indicate terminal support for contact type debit/credit applications, contactless debit/credit applications, qUICS or all three.

qUICS does not follow debit/credit application processing rules, and does not need to support debit/credit application required data and requirements. GPO command is used to provide terminal with cryptogram, cryptogram information data, and dynamic signatures.

If qUICS supports fDDA, then fDDA related data also must be read from chip. Table B.1 makes a detailed listing of qUICS and debit/credit flow processing

Table B.1 qUICS and contact type standard debit/credit flow comparison

qUICS Terminal		Contact Type Standard Debit/Credit Flow Device	
Command	Description	Command	Description
SELECT	Mandatory: choose PPSE (2PAY.SYS.DDF01), no options. Response for selecting PPSE includes all contactless application AIDs (and some additional information). Requires PDOL and includes tag '9F66' minimum value and terminal data tag used for encryption. Flow is described in Figure 3 of this part.	SELECT	Touch type standard debit/credit. Mandatory: select AID Option: select PSE(1PAY.SYS.DDF01) For directory file read record, directory file is related to AIDs provided by card PSE and application data. PDOL is optional. Flow is described in UICS – Basic Specifications Part 3.
Obtain processing option GPO	Terminal sends data value tag '9F66', indicating support for contactless debit/credit application or qUICS, sending terminal data used for encryption and other card required data for completing	Obtain processing option GPO	If transaction conditions are met, card sends response to AFL and AIP. If there is PDOL, terminal

	transaction. Offline For offline transactions, as GPO response, card returns cryptogram, card cryptogram data, other transaction data and dynamic signature, an AFL including offline data authentication (DDA/SDA) is also returned. Online For online transaction, there is no AFL return As response to GPO, card returns cryptogram, card cryptogram data.		provides card with data requested in PDOL, and card may use other logic to determine what AFL or AIP to return.
READ RECORD	Offline: If GPO returned cryptogram is not AAC, terminal read AFL indicated records. AFL can also point out which record has been signed for use in offline data authentication. Terminal checks whether the card has reached expiration date, if expiration date has not been reached then execute offline data authentication (SDA/DDA), if failed then reject this transaction. If offline data authentication pass and card has not expired, then use the GPO returned cryptogram to complete transaction. Offline data authentication is compatible with UICS – Basic Specifications Part 2, unless the dynamic signature is generated in GPO or if the card does not need to save the field after reading the last record. Online: After card moves away, terminal sends cryptogram provided by the card. Cryptogram replies with a GPO approval or rejection, online, and Issuer. AFL is not returned, and no other records can be read.	READ RECORD	Device uses AFL to determine which record to read and reads these records. AFL also points out which records will be signed. If mandatory data element is lost, transaction will be terminated.
N/A	N/A	INTERNAL AUTHENTICATE	Device checks AIP to determine which risk management characteristic is supported by the card. If AIP requires DDA support, internal authentication command is sent to the card. Execute DDA according to UICS.

			Set UICS specified indicator.
N/A	N/A	N/A	Processing constraint
N/A	N/A	N/A	Cardholder verification.
N/A	N/A	Obtain random number GET CHALLENGE	Optional offline encryption PIN.
N/A	N/A	VERIFY	Optional offline PIN verification (plaintext or cryptogram).
N/A	N/A	N/A	Terminal risk management
N/A	N/A	Generate application cryptogram (1st time)	Offline approval or rejection or request online processing.
N/A	N/A	External authentication	If online processing and issuer authentication.
N/A	N/A	Generate application cryptogram (2 nd)	Approve or reject
N/A	N/A	Issuer script command	Device sends issuer script command to the card.

Appendix C Fast DDA (fDDA)

In contactless payment environment, transaction speed is a business requirement. DDA as a dynamic data authentication method is used offline to protect against fake cards.

Besides the majority of contact chip applications that use unpredictable number (terminal) for signature, fDDA also signs other transaction dynamic data. Authorized amount, transaction currency, and unpredictable number (card) are all used for signing during fDDA.

Card uses PDOL to obtain for fDDA from terminal. In the GPO command, card receives data requested from card reader. These terminal data elements and card data jointly generate dynamic signature.

The AFL returned by GPO points to records including certificate and other fDDA related data. Once the last record is read by a card reader, the card no longer needs to remain in the field. The card reader then verifies the dynamic signature returned by the card. If signature verification fails, transaction will be rejected offline depending on the card transaction qualifiers, request online authorization or termination.

To accommodate possible new fDDA algorithms and inputs, newly defined card data element fDDA version (part of tag '9F69') is used to identify the fDDA version used by the card. fDDA version number is returned by the card, card reader uses it to determine the fDDA algorithm to be executed. *Part II Debit Credit Application - Member Implementation Guide for Issuer* defined fDDA algorithm is defined as '00' version fDDA by this version. This Specification will define a new fDDA algorithm and set its version as '01'.

Card applications that meet this Specification version also supports '00' and '01' versions of fDDA, specific versions used can be determined by terminal capability (indicated in terminal transaction qualifiers).

Card readers that meet this Specification version shall also support '00' and '01' versions of fDDA. They also support '01' version fDDA to the card (terminal transaction qualifier byte 4 bit 8 is '1') in GPO command.

For version '01' fDDA, cards use unpredictable number (terminal), authorized amount, and transaction currency code obtained from card reader GPO command, concatenated with card ATC and card authentication related data, for dynamic signature calculation.

C.1 Dynamic signature generation

Data concatenation and dynamic signature generation meets this Specification part 4 3.3.5.1 step 2, excluding the following content:

- For qUICS online authorization requests (ARQC), the Signed Data Format input to the DDA hash algorithm must be the value '95' for ODA for online

authorization. The Signed Data Format of '95' distinguishes the dynamic signature returned for an online authorization request from the dynamic signature returned for an offline authorization request (where Signed Data Format is the value '05').

- For offline approval requests (TC), the Signed Data Format input to the DDA hash algorithm must be the value '05'.
- If a contactless-enabled terminal supports ODA for online authorization, the terminal should support Signed Data Format request from the dynamic signature returned for an offline authorization. In a traditional financial POS environment which does not support ODA for online authorization, the DDA will fail when the Signed Data Format is not mat i

Terminal dynamic data elements are not specified in DDOL (DDOL for qUICS is unrecognized data). Dynamic data in this Specification part 4 3.3.5.1 table 13 or part 17 3.2.4 table 7 are made up by concatenating data elements in order as specified by table B.1. If any required data element is missing, then fDDA fails.

Prior to including related data to card authentication in the terminal dynamic data, the card generates and fills unpredictable number (card) and card transaction qualifiers into card authentication related data.

Note: If card transaction qualifier has not been personalized, then use value '0' as replacement, for use in card authentication related data.

IC card dynamic data includes table C.2 content.

Table C.1 Dynamic terminal data used as input for DDA hash algorithm

Tag	Data Element	Length	Data Source	Version '00'	Version '01'
9F37	Unpredictable number	4 bytes	terminal	√	√
9F02	Authorized amount	6 bytes	terminal		√
5F2A	Transaction currency code	2 bytes	terminal		√
9F69	Card authentication related data	variable	card		√

Table C.2 Dynamic card data used as input for DDA hash algorithm

Tag	Data Element	Length	Data Source	Version '00'	Version '01'
9F36	Application Transaction Counter (ATC)	2 bytes	card	√	√

C.2 Dynamic signature verification

To verify fDDA dynamic signature, card readers can recover CA public key, issuer public key, and IC card public key. For this process see part 4 3.3.

Dynamic signature verification process matches this Specification in part 4 3.3, but excludes the following content:

- Terminal determines the fDDA signature algorithm to be used based on the card authentication related data (tag '9F69') returned by the card; if not returned, then treat as using '00' version fDDA signature algorithm;
- Terminal dynamic data elements that are inputs to the hash algorithm are not specified in the DDOL (DDOL is unrecognized data to qUICS), but consist rather of data elements concatenated in order as specified by table B.1. Terminal can interpret table B.1 specified tags as '01' version fDDA's default DDOL.

Note: Card authentication related data is variable length data. Card reader use the entire card authentication related data returned by the card for dynamic signature authentication.

In the following circumstances, fDDA fails when:

- Application Interchange Profile (AIP) indicates that cards do not support DDA (AIP byte 1 bit 6 is 0);
- fDDA is supported but required data for fDDA is missing;
- Card requested fDDA version is not supported by the card reader. '00' version fDDA and '01' version fDDA are fDDA versions supported by this part;
- If terminal supports '01' version fDDA (terminal transaction qualifier byte 4 bit 8 is 1) and card returns application version number (tag '9F08'), then it indicates that the card meets this Specification version or later, but returns a '00' version fDDA signature.

See Figure C.1 for fast DDA (fDDA) qUICS example.

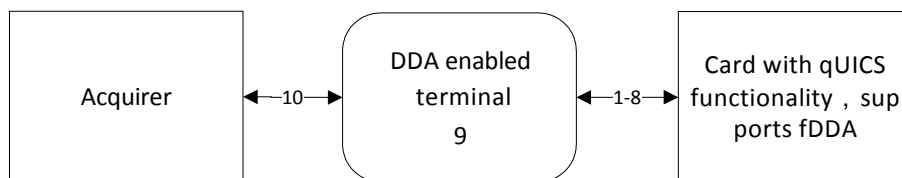


Figure C.1 fast DDA (fDDA) qUICS example

- 1) Terminal selects PPSE;
- 2) Card returns unique debit/credit AID;
- 3) Terminal selects debit/credit AID;
- 4) Card returns request:

- Terminal transaction qualifiers (tag '9F66');
 - Unpredictable number (tag '9F37');
 - Authorized amount (tag '9F02')
 - Transaction currency code (tag '5F2A')
 - Other PDOL specified tag.
- 5) Terminal sends GPO, provides:
- Tag '9F66' indicates qUICS supported only;
 - Tag '9F37' unpredictable number;
 - Tag '9F02' authorized amount;
 - Tag '5F2A' transaction currency code;
 - Other PDOL specified tag.
- 6) Card response:
- Transaction certificate (TC);
 - Dynamic signature;
 - AFL list records related to offline data authentication (fDDA);
 - Other non-fDDA related data.
- 7) Terminal reads AFL designated records;
- 8) Card provides certificate and data to authenticate static data signatures, while card authentication related data are added to the last record to be returned (if card has been personalized to support '01' version fDDA);
- At this time the card can leave the communication zone.
- 9) Terminal authentication dynamic signature;
- 10) If fDDA authentication passes, terminal provides settlement data.
- Transaction certificate (TC);
 - Relevant data.

If fDDA authentication fails, transaction is rejected, terminated, or sent as online request based on issuer configuration.

Appendix D Issuer Defined Data

D.1 Issuer Defined Data Option

In order for Issuers to closely track funds on the host end, an option allows adding special data into Issuer defined data portion of the issuer application data (tag '9F10'). For debit/credit transactions, this data is provided to the terminal through the Generate AC response, and sent online to the Issuer. For qUICS transactions, this data is provided to the terminal through the GPO command response, and sent online to the Issuer.

Cumulative total transaction amount limit (CTTAL) added on CTTA, electronic cash balance, available offline purchasing amount, and personalize static data with no more than 15 bytes are the 5 data options that can be optionally sent online by the Issuer. Issuer can select any of these 5 options to be sent online. Also if this data exists, verification code would be added to commands sent to ensure data integrity.

D.2 Issuer Defined Data Personalization

If issuer defined data (IDD) exists, it is returned after the defined data in the issuer application data (tag '9F10').

For qUICS transactions, this is returned when card application responds to GPO command with online authentication request (ARQC).

However, issuer defined data (IDD), based on choices made at personalization as described in table D.1, and varies.

Table D.1 Issuer defined data (IDD)

Issuer Discretionary Data Options	Length (byte)	IDD ID	Amount Field	MAC Byte Size
Electronic Cash Balance	10	0x01	Tag '9F79' value (lower 5 bytes)	4
Cumulative total transaction amount (CTTA)	10	0x02	Value, this data has no tag (lower 5 bytes)	4
Electronic Cash Balance and CTTA	15	0x03	Value (10 bytes, '9F79' value is in the first position)	4
CTTA and CTTAL	15	0x04	Value (10 bytes, CTTA value is in the first position)	4
Available Offline Spending amount	10	0x05	Tag '9F5D' value (lower 5 bytes)	4
Contactless Extended	10	0x06	Reserved	4

Reserved ⁴				
Mobile payment reserved ⁵	Var.	0x07	Mobile payment reserved	4
Static	1 to 15	N/A	Issuer designated fixed data	None

Issuer defined data (IDD) ID value selects the return data type in the issuer defined data field. By default, issuer defined data would not be returned. If Issuer wants to receive issuer defined data, then the above corresponding data length and id bytes (after the debit/credit application self-defined data) must be added in the personalization value of issuer application data (tag '9F10').

For example, 0x0A02 means that in the generate transaction cryptogram command response, there will be 10 bytes of returned issuer defined data, including data type id (0x02), cumulative transaction total, and verification code. Return electronic cash balance option, only after application is personalized for electronic cash would be valid.

D.3 Issuer Application Data Personalization Example

Debit/credit self-defined data (mandatory)

Length: 0x 07

Value: 0x 011003000000001 (assume cryptogram version number is 10)

Issuer defined data

Length: 0x 0A (expected return value length in the response to Gen AC command)

Value: 0x 02 (ID value for CTTA request)

The TLV value for the case above is:

9F10 0A

07 011003000000001

0A 02

Applications on the card use personalized issuer defined data length and ID (0x0A02), when generating application cryptogram return online cryptogram request for the first time, enable internal code, thereby providing an indicator for cumulative transaction total in the issuer defined data.

D.4 Issuer Application Data to Generate Application Cryptogram Return

Debit/credit self-defined data

Length: 0x 07

⁴ IDD ID 0x06 please refer to: *UICS -Product Specifications - Part II Extended Purchase Specification Based on Contactless Low-value Payment Application*

⁵ IDD ID 0x07 refer to UnionPay mobile payment technical specification.

Value: 0x 011003000000001 (example)

Issuer defined data

Length: 0x 0A

Value: 0x 02 (ID) cumulative transaction amount (5 bytes)

Verification code: 4 bytes, for description see D.5

D.5 Verification Code Calculation

Data that undergoes verification calculation includes 2 byte application transaction counter, plus one or two 5 byte amount field and padding characters 0x00, specific data composition rules are as shown in table D.2.

For issuer defined data ID 0x01, 8 byte data, including application transaction counter, electronic cash balance, and 1 byte padding.

For issuer defined data ID 0x02, 8 byte data, including application transaction counter, CTTA amount, and 1 byte padding.

For issuer defined data ID 0x03, 16 byte data, including application transaction counter, electronic cash balance, CTTA, and 4 byte padding.

For issuer defined data ID 0x04, 16 byte data, including application transaction counter, CTTA, CTTAL, and 4 byte padding.

For issuer defined data ID 0x05, 8 byte data, including application transaction counter, available offline purchasing amount, and 1 byte padding.

The 4 byte verification code is calculated using process key derived from MAC UDK. For key derivation method and MAC calculation method see *UICS Basic Specifications - Part 4*.

Table D.2 MAC calculation

IDD ID Option	Data Block Length	Element	
0x01	8 bytes	ATC	2 bytes
		Electronic Cash Balance	Lower 5 bytes
		Padding	1 byte
0x02	8 bytes	ATC	2 bytes
		CTTA amount	Lower 5 bytes
		Padding	1 byte
0x03	16 bytes	ATC	2 bytes
		Electronic Cash Balance	Lower 5 bytes
		CTTA	Lower 5 bytes
		Padding	4 bytes

IDD ID Option	Data Block Length	Element	
0x04	16 bytes	ATC	2 bytes
		CTTA	Lower 5 bytes
		CTTAL	Lower 5 bytes
		Padding	4 bytes
0x05	8 bytes	ATC	2 bytes
		Available offline purchase amount	Lower 5 bytes
		Padding	1 byte

Appendix E Cryptogram Version

Cryptogram version 17 uses the same algorithm and parameters as cryptogram version 01, the only difference is that it does not support all the data required by cryptogram version 01. Table E.1 lists the data required by cryptogram version 17 in the required order.

Table E.1 Data elements included in cryptogram version 17

Tag	Data Element	Data From Terminal	Input From Card
'9F02'	Authorized amount*	✓	
'9F37'	Unpredictable number	✓	
'9F36'	Application Transaction Counter (ATC)		✓
'9F10'	Issuer application data (byte 5) According to <i>Part II Debit Credit Application - Member Implementation Guide for Issuer</i>, byte 5 is the second data byte in CVR, with CVR fixed length as x '03' Only byte 5 is used in cryptogram calculations, but the first 8 bytes of issuer application data appears in the message. For qUICS online transactions, issuer defined data (IDD) may be included Byte 1 - '07' Byte 2 - DKI Byte 3 - Cryptogram version number Byte 4-7 - CVR Byte 8 - algorithm id Byte 9 - Length of IDD Byte 10-23 - IDD		✓

For qUICS, terminal messages to Acquirer include these data. Acquirer packages these data into message field 55.

Application cryptogram and Table E.1 data will appear in the messages from terminal to Acquirer, as well as the authentication-clearing message from Acquirer to exchange center.

Appendix F Implementation of Online-Only qUICS Transactions

This Specification outlines two ways to implement UICS to meet the online-only transaction requirements of some special markets.

F.1 Online-only qUICS Flow

Online-only qUICS flow is a simplified, online-only card side qUICS route, which can simplify card implementation and accelerate transaction speed. Whether the card supports online-only flow is transparent to the terminal.

Online-only qUICS flow basically follows qUICS flow, but risk control during application initialization stage is relatively simplified. Online-only qUICS flow has the following characteristics:

- Online-only qUICS flow does not support offline processing and card does not personalize GPO offline response.
- Card executes the following risk management:
 - If application is locked, the card terminates transaction and returns SW1 SW2 = '6985'.
 - If terminal requires execution of cardholder check (terminal transaction qualifiers byte 2 bit 7 is 1), then card will choose CVM based on the Cardholder verification method jointly supported by the terminal and card. If both terminal and card support two methods (online PIN and signature) then execute the following: 1) online PIN; 2) signature.
- Card responds to GPO command according to UnionPay IC card Specification Part II specified formats, and includes mandatory data listed in Table 11;
- Card qUICS route supports cryptogram version 17;
- PDOL content for online-only qUICS meets the requirements in Table 11.

F.2 qUICS Parameter Configuration Flow

Using qUICS flow, Issuer and Acquirer can initiate online-only transactions through parameter configuration. Because qUICS flow is still used, personalization process is not simplified and executes require risk control steps, increase in transaction speed is relatively minor. There is no need to re-personalize existing cards to realize online transaction.

Based on transaction risk checking, Issuer and Acquirer can realize online-only transactions through the following parameter configuration:

- Issuer can set electronic cash balance to 0;
- Issuer can set electronic cash transaction limit to 0;
- For the card compliant to this version of specification, Issuer can set “low-value check” and “low-value and CTTA check” bit of Card additional

processes to 0;

- Acquirer can set offline contactless terminal transaction floor limit to 0.

Appendix G Implementation Guidelines of ODA for Online Authorization

This appendix illustrates the general implementation requirement for transit applications based on ODA for online authorization, and focuses on the interaction between the card/consumer device and the terminal. For the transaction processing requirement from the terminal to payment network via industry-specific back-office system and the product business rules, please refer to the transit product implementation guide and UPI Operating Regulations for details.

G.1 Introduction

G.1.1 Scope

G.1.1.1 Terminal and Channel Scope

Applicable to specifically modified contactless terminals which support ODA for online authorization.

Applicable to the transactions where the card or consumer device has physical interaction with the terminal.

G.1.1.2 Card Scope

Applicable to cards which support ODA for online authorization in qUICS process and cards which support contactless standard debit/credit process, including IC cards, hybrid cards (IC cards with magnetic stripe), and consumer devices. Both debit cards and credit cards should support ODA for online authorization.

G.1.1.3 Transaction Scope

Applicable to industry-specific or merchant-specific online transactions, such as self-service purchase transactions, which read chip information in the contactless mode, and adopt the qUICS or standard debit/credit process.

G.1.1.4 Merchant Scope

Applicable to only two kinds of Merchants: 1. Merchants which accept low-value payments with the requirement for high interaction speed; 2. Merchants which are not capable of constantly initiating real-time online transactions. Typical Merchants are subways, buses and parking lots. In addition, Merchants shall meet certain qualification requirements to support ODA for online authorization.

G.1.1.5 Transaction Scenario Scope

Applicable to face-to-face card-present purchase transactions, as well as self-service card-present transactions initiated by Cardholders from self-service terminals.

G.1.2 Transaction Processing Flow

Transaction processing flow is as follows:

- The terminal initiates the transaction, or the Cardholder operates on the self-service terminal to confirm transaction information;

- The cardholder presents the card within the induction range of the contactless terminal;
- After passing ODA and other related checks, the terminal displays a success message, and the Merchant provide the Cardholder with commodities or services;
- The terminal shall save the related transaction data which will be used for deferred online authorization, be connected with back-office system, and upload the transaction data to the back-office system within a certain time period;
- The back-office system shall complete the transaction data processing within the pre-defined time limit and upload the data to the GSCS system of UPI. The merchant back-office system shall perform checks and processing on these data according to related rules, and then initiate online transactions;
- In the cases which meet the black list rule, such as the presentment failure, the Merchant adds the corresponding PAN to the black list and updates the list in every terminal. The terminal shall deny the transactions initiated from the card in the black list.

G.2 Technical Requirements

G.2.1 Application System Architecture



Figure G.1 Application System Architecture

The industry-specific terminal shall:

- Interact with the card according to this Specification;
- Collect and retain transaction data and transaction elements for online messages;
- Perform the checks accordingly as required by the Merchant;
- Communicate with the industry-specific back-office system and transmit data.

The industry-specific back-office system shall:

- Communicate with the industry-specific terminal and transmit data;
- Summarize statistics and conduct related data processing;
- Communicate with the GSCS system and transmit data.

The Acquirer shall:

- Perform the transaction compliance check (to prevent duplicate transactions and overdue transactions);

- Initiate the deferred presentment transaction.

The GSCS system shall:

- Conduct processing of switch and settlement

The Issuer shall:

- Conduct processing of transaction authorization

G.2.2 Transaction Stages Overview

G.2.2.1 Stage 1: Interaction Between the Card and the Terminal

Industry-specific terminals shall perform offline data authentication to confirm the legitimacy of the card, and retain transaction data after a successful verification and other passed related checks.

G.2.2.2 Stage 2: Deferred Online Authorization

Industry-specific terminals shall upload the retained transaction data to the industry-specific back-office system within a certain time period, from which the transaction is submitted to the GSCS system to complete the online authorization.

G.2.3 Technical Solution of Interaction Between Card and Terminal

G.2.3.1 Overview

The interaction process between the card and the terminal varies according to different card characteristics. There are two categories of cards:

The first is existing cards which are already issued which do not support qUICS ODA for online authorization. For those existing cards which support contactless standard debit/credit applications, card authentication and online authorization request can also be completed by adopting the standard debit/credit process via contactless interface. For those existing cards which neither support ODA for online authorization nor support contactless standard debit/credit applications, the transaction shall be terminated.

The second is newly issued cards which support ODA for online authorization, including IC cards and consumer devices. Cards interact with terminals through the qUICS process, and complete card authentication and the online authorization request.

G.2.3.2 Existing Card

G.2.3.2.1 Existing Card Identification

The terminal identifies the existing card using Byte 1 Bit 7 of Card Additional Function Indicator (tag 'DF61') in the template 'BF0C' returned in the SELECT response. If Byte 1 Bit 7 of the tag 'DF61' returned by the card is not enabled or the tag is not returned, it indicates an existing card.

G.2.3.2.2 Interaction Process between Existing Cards and Terminals

If the existing card does not support the contactless standard debit/credit process, the transaction ends. If the existing card supports the contactless debit/credit process, the transaction continues and the interaction process between this existing card and the terminal is as follows:

- The terminal sends the SELECT AID/SELECT PPSE command to select the application;
- The card returns FCI information in response to the SELECT command, and Byte 1 Bit 7 of the tag 'DF61' is 0 or the tag 'DF61' is not returned;
- The terminal sets Byte 1 Bit 7 of Terminal Transaction Qualifier (tag '9F66') to 1, which indicates support for contactless standard debit/credit process by the terminal. Then, it enters into the contactless standard debit/credit process and performs offline data authentication;
- Based on the ODA result, the terminal determined the legitimacy of the card. If ODA fails, which indicates a counterfeit card, the transaction is declined;
- If ODA is successfully performed, the card will be recognized as a genuine one. Then the terminal checks if the card returned the ARQC as required, and declines the transaction if ARQC is not returned;
- When the terminal confirms that ARQC is returned by the card, it can then perform new card check (optional);
- The terminal compares the PAN with the black list, and declines the transaction if the PAN is in the black list;
- Terminal can perform other checks as required by the industry-specific merchant, and decline the transaction if the check fails;
- After passing each check, the card authentication is successfully completed, and the terminal retains online transaction data and related transaction elements.

Note: Since the support of contactless debit/credit application process is not mandatory required for terminals, the interaction process above is optional to be implemented.

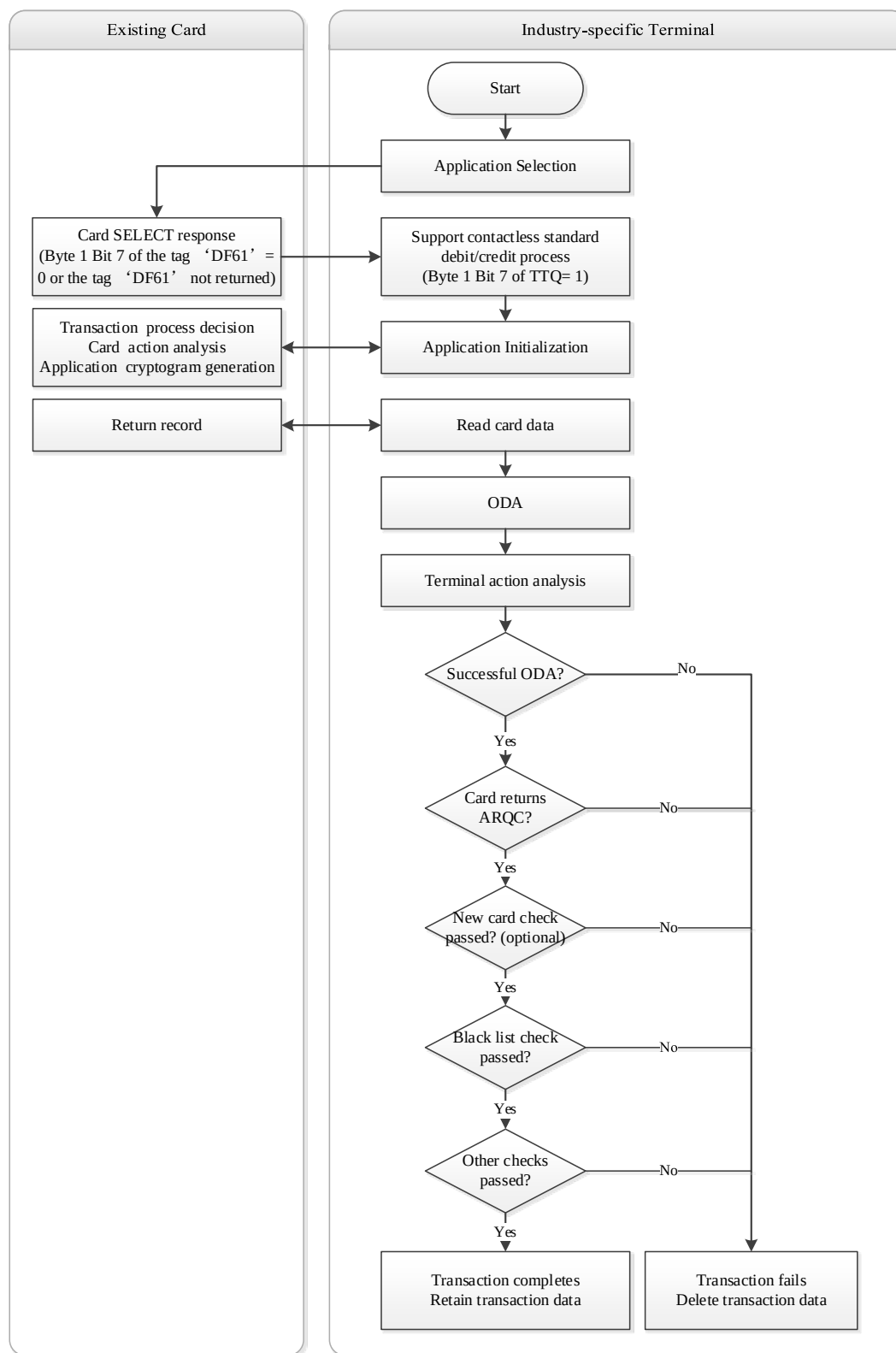


Figure G.2 Interaction Process between Existing Cards and Terminals

G.2.3.3 New Card

G.2.3.3.1 New Card Identification

The terminal identifies the new card using Byte 1 Bit 7 of Card Additional Function Indicator (tag 'DF61') in the template 'BF0C' returned in the SELECT response. If Byte 1 Bit 7 of the tag 'DF61' returned by the card is enabled, it indicates a new card.

G.2.3.3.2 Interaction Process Between New Cards and Terminals

The interaction process between a new card and the terminal is as follows:

- The terminal sends the SELECT AID/SELECT PPSE command to select the application;
- The card returns FCI information in response to the SELECT command, and Byte 1 Bit 7 of the tag 'DF61' is 1;
- The terminal sets different bits in Terminal Transaction Qualifier (tag '9F66'): sets Byte 1 Bit 6 to 1, which indicates that the terminal supports qUICS process; sets Byte 1 Bit 7 to 0, which indicates that the terminal does not support contactless standard debit/credit process; sets Byte 1 Bit 1 to 1, which indicates ODA for online authorization is supported; set Byte 2 Bit 8 to 1, which indicates online cryptogram is required. Then, the terminal enters into the qUICS process;
- The terminal reads the card data using the AFL returned by the card and READ RECORD command, and then performs ODA;
- Based on the ODA result, the terminal determined the legitimacy of the card. If ODA fails, which indicates a counterfeit card, the transaction is declined;
- If ODA is successfully performed, the card will be recognized as a genuine one. Then the terminal checks if the card returned the ARQC as required, and declines the transaction if ARQC is not returned;
- When the terminal confirms that ARQC is returned by the card, it can then perform new card check (optional);
- The terminal compares the PAN with the black list, and declines the transaction if the PAN is in the black list;
- Terminal can perform other checks as required by the industry-specific merchant, and decline the transaction if the check fails;
- After passing each check, the card authentication is successfully completed, and the terminal retains online transaction data and related transaction elements.

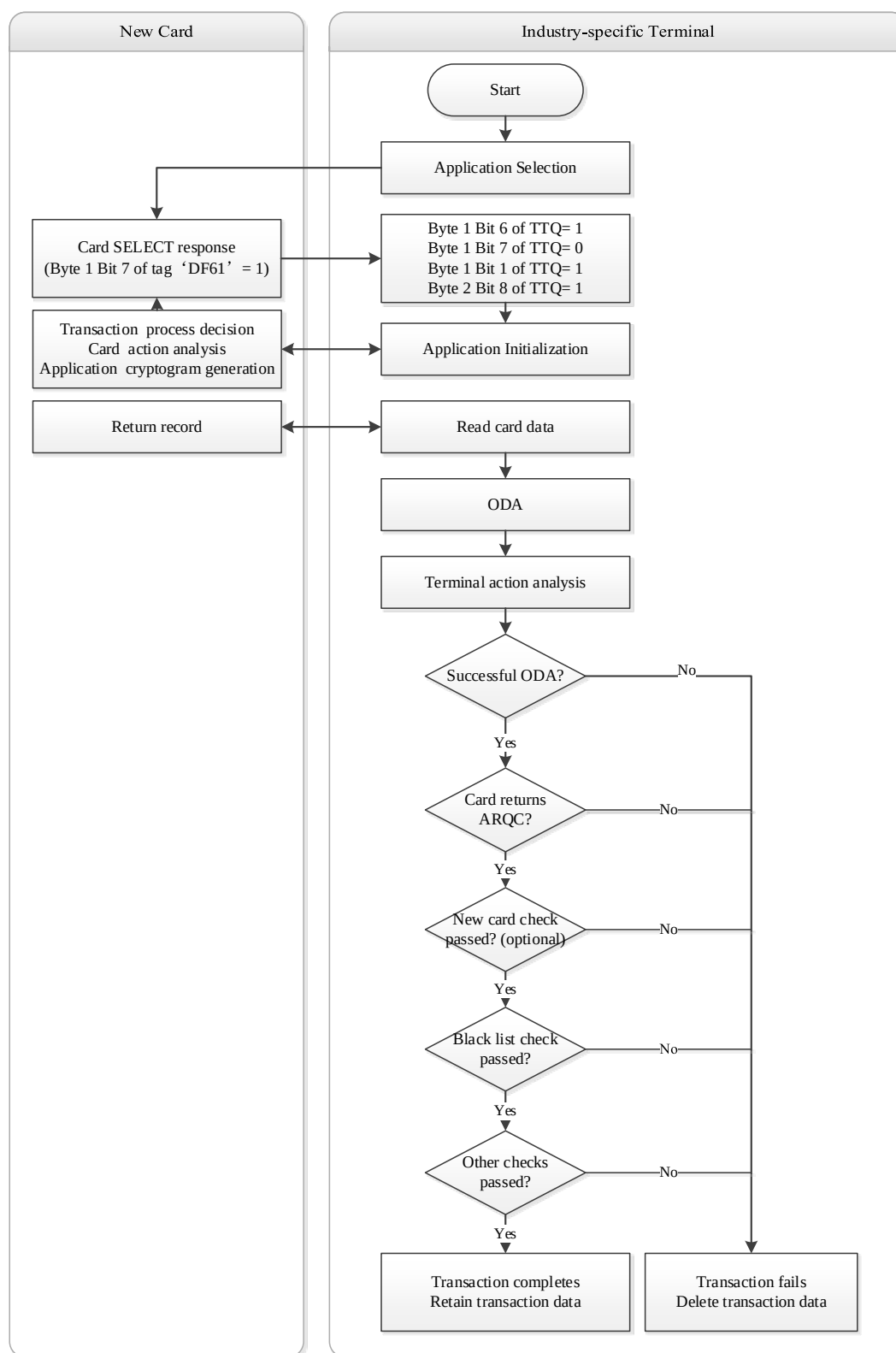


Figure G.3 Interaction Process between New Cards and Terminals

G.2.4 Technical Implementation Requirements

G.2.4.1 Enabled Card Indicator of ODA for Online Authorization in the Tag 'DF61'

Byte 1 Bit 7 of Card Additional Function Indicator (tag 'DF61') in template 'BF0C' is enabled to indicate support for ODA for online authorization. Cards, which support ODA for online authorization, set this bit to 1. For detailed definition of the data element, please refer to Appendix A.

Note 1: Offline data authentication during ODA for online authorization is the same as that during the qUICS process, including the same set of AFL used by the card for transactions.

Note 2: During card personalization, Byte 1 Bit 7 of Card Additional Function Indicator (tag 'DF61') shall be consistent with Byte 4 Bit 6 of Card Additional Processes (tag '9F68').

G.2.4.2 Card Additional Processes

Byte 4 Bit 6 of CAP (tag '9F68') is the indicator of support for ODA during the qUICS online authorization process. For new cards with support for ODA for online authorization, this bit in the tag '9F68' shall be set to 1.

G.2.4.3 Returned Card Data Elements Requirements

For qUICS online transactions with ODA, where ODA for online authorization is supported by both the card and the terminal with Byte 4 Bit 6 of CAP (tag '9F68') set to 1 and Byte 1 Bit 1 of TTQ (tag '9F66') set to 1, the card shall return the data elements listed in Table 13 in the GPO response.

G.2.4.4 Signature and Authentication Requirements

Please refer to Appendix C Fast DDA for signature and authentication requirements in ODA for online authorization.

G.2.4.5 Terminal Modification Implementation

The terminal or its back-office system shall be capable of processing online transactions, and the terminal shall support the qUICS process with Byte 1 Bit 4 of TTQ (tag '9F66') set to 0 and Byte 1 Bit 6 of TTQ (tag '9F66') set to 1, which indicates that offline transactions not supported and the qUICS process supported by the terminal respectively. If the terminal does not support E-cash transactions, Terminal Contactless Floor Limit shall be set to 0. If the terminal does not support contactless standard debit/credit process, and thus not support existing cards, Byte 1 Bit 7 of TTQ (tag '9F66') shall be set to 0. In addition, Byte 2 Bit 8 of TTQ (tag '9F66') shall be set to 1 to indicate the requirement for online cryptogram.

The terminal shall be capable of processing qUICS online transactions with ODA, and shall be able to enable and disable this function by setting the bits in terminal data elements. If this function is enabled, Byte 1 Bit 1 of TTQ (tag '9F66') shall be set to 1.

